



**SSC-CGL**

**TIER-I(English Medium)**

*Toppers Handwritten Notes*

**REASONING  
BOOK-4**

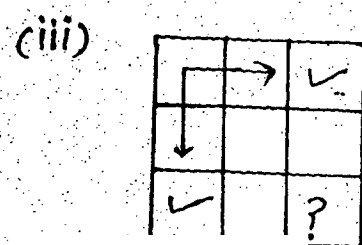
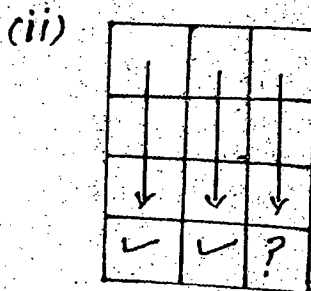
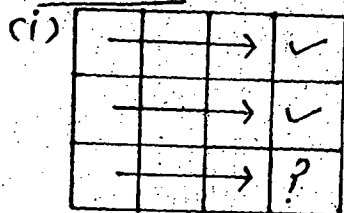


# CONTENT

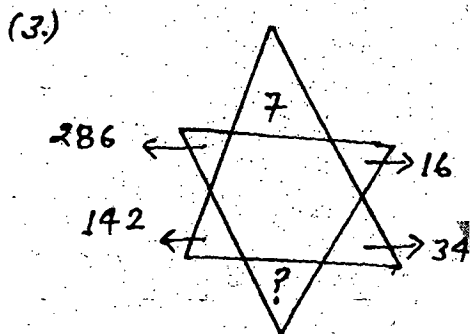
| TOPIC NAME                          | PAGE NO |
|-------------------------------------|---------|
| Missing Number                      | 4-13    |
| Repeated Series                     | 14-17   |
| Analogy                             | 18-25   |
| Venn Diagram,Blood Relation         | 26-37   |
| Logical Arrangement Of Words        | 38-41   |
| Coding Decoding,Sitting Arrangement | 42-59   |
| Ranking Arrangement                 | 60-67   |
| Dice                                | 68-73   |
| Cubes & Cuboid                      | 74-81   |
| Calendar                            | 82-93   |
| Number Series                       | 94-101  |
| Coded Equation                      | 102-105 |
| Direction Test                      | 106-115 |
| Clock                               | 116-127 |
| Counting Of Figures                 | 128-133 |
| Problems Based On Alphabet          | 134-143 |
| Sequence Test                       | 144-147 |
| Syllogism                           | 148-155 |
| Water,Mirror Images                 | 156-160 |

# \* MISSING NUMBER \*

Logic :-



(1) 2, 4, 6, 8, ?  
= 10



$$\Rightarrow 7 \times 2 + 2 = 16$$

$$16 \times 2 + 2 = 34$$

$$34 \times 2 + 2 = \boxed{70}$$

$$70 \times 2 + 2 = 142$$

$$142 \times 2 + 2 = 286$$

(5)

|   |   |    |
|---|---|----|
| 3 | 2 | 13 |
| 5 | 2 | 29 |
| 8 | 4 | ?  |

(a) 78

$$\Rightarrow (3)^2 + (2)^2 = 13$$

(b) 79

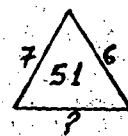
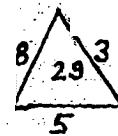
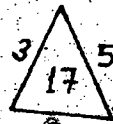
$$(5)^2 + (2)^2 = 29$$

(c) 80

$$(8)^2 + (4)^2 = \boxed{80}$$

(d) 81

(2)



$$\Rightarrow 3 \times 5 + 2 = 17$$

$$8 \times 3 + 5 = 29$$

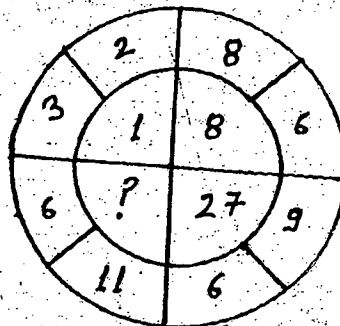
$$7 \times 6 + ? = 51$$

$$x = 51 - 42$$

$$x = \boxed{9}$$

(4)

(CGL-2015)



$$\Rightarrow 3 - 2 = 1 \Rightarrow (1)^3 = 1$$

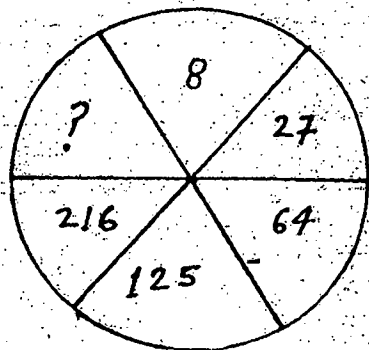
$$8 - 6 = 2 \Rightarrow (2)^3 = 8$$

$$9 - 6 = 3 \Rightarrow (3)^3 = 27$$

$$11 - 6 = 5 \Rightarrow (5)^3 = \boxed{125}$$



(6.)



(a) 343

(CGL-2014)

(b) 341

(c) 2

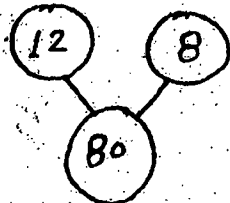
(d) 1

(e) a or d (✓)

$$\Rightarrow 1 \Rightarrow (1)^3 = \boxed{1}, \quad 2^3 = 8, \quad 3^3 = 27, \quad 4^3 = 64, \quad 5^3 = 125, \quad 6^3 = 216, \quad 7^3 = \boxed{343}$$

TYPE-1 :

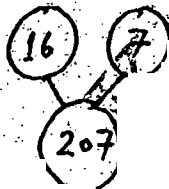
(7.)



$$\Rightarrow 12 = (12)^2 = 144$$

$$8 = (8)^2 = 64$$

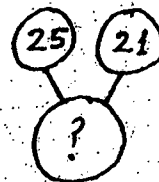
$$\underline{\quad 80 \quad}$$



$$(16)^2 = 256$$

$$7^2 = 49$$

$$\underline{\quad 207 \quad}$$

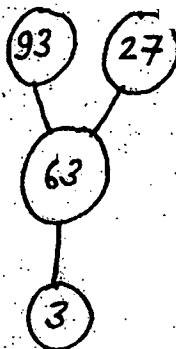


$$(25)^2 = 625$$

$$(21)^2 = 441$$

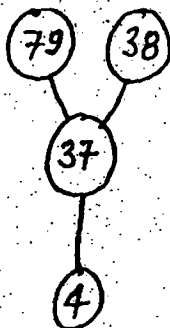
$$\underline{\quad 184 \quad}$$

(8.)



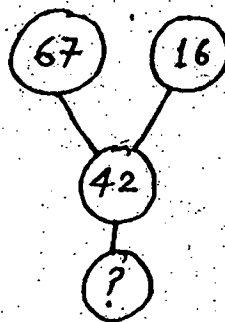
$$\Rightarrow 93 - 27 - 3$$

$$= 63$$



$$79 - 38 - 4$$

$$= 37$$

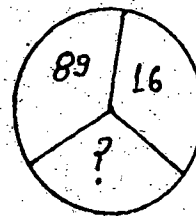
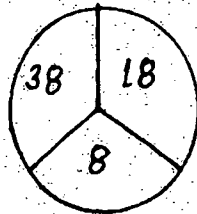
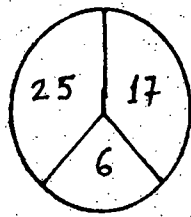


$$67 - 16 - x = 42$$

$$\Rightarrow 42 = 51 - x$$

$$x = \boxed{9}$$

(9.)



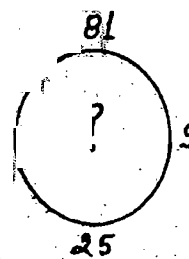
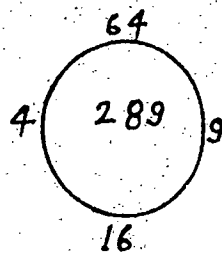
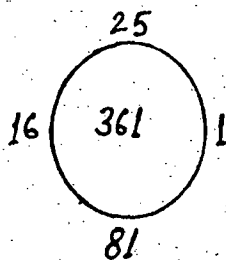
(CPO-2013)

$$\Rightarrow 25 + 17 = \frac{42}{7} = 6$$

$$38 + 18 = \frac{56}{7} = 8$$

$$89 + 16 = \frac{105}{7} = \boxed{15}$$

(10.)



(CPO-2011)

$$\Rightarrow \sqrt{16} = 4$$

$$\sqrt{25} = 5$$

$$\sqrt{1} = 1$$

$$\sqrt{81} = 9$$

$$\Rightarrow 4 + 5 + 1 + 9 = 19$$

$$\text{Now, } (19)^2 = 361$$

$$\sqrt{4} = 2$$

$$\sqrt{64} = 8$$

$$\sqrt{9} = 3$$

$$\sqrt{16} = 4$$

$$\Rightarrow 2 + 8 + 3 + 4 = 17$$

$$(17)^2 = 289$$

$$\sqrt{1} = 1$$

$$\sqrt{81} = 9$$

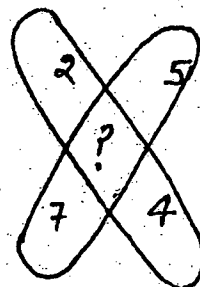
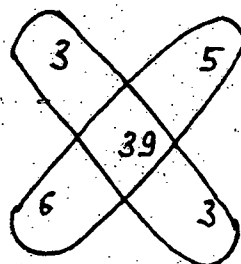
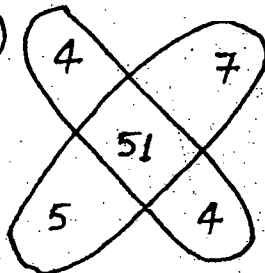
$$\sqrt{9} = 3$$

$$\sqrt{25} = 5$$

$$\Rightarrow 1 + 9 + 3 + 5 = 18$$

$$(18)^2 = \boxed{324}$$

(11.)



(SSC-LDC-2011)

$$\Rightarrow 7 \times 5 = 35$$

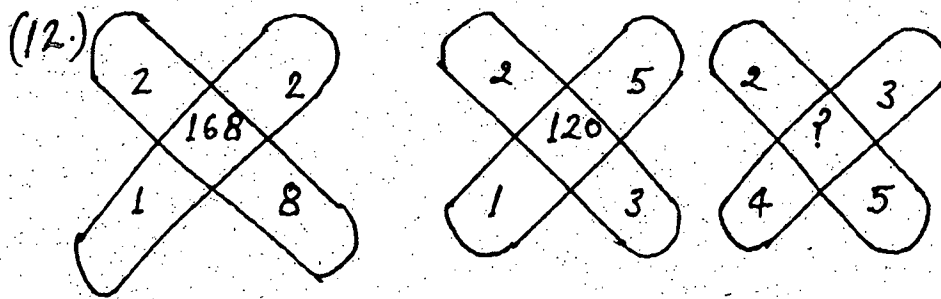
$$\begin{array}{r} 4 \times 4 = 16 \\ + \\ \hline 51 \end{array}$$

$$5 \times 6 = 30$$

$$\begin{array}{r} 3 \times 3 = 9 \\ + \\ \hline 39 \end{array}$$

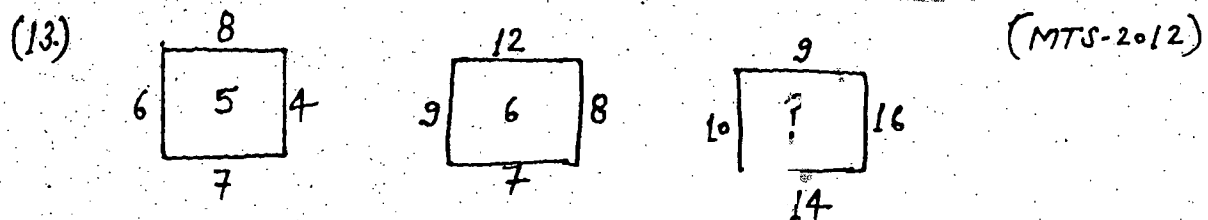
$$5 \times 7 = 35$$

$$\begin{array}{r} 4 \times 2 = 8 \\ + \\ \hline \boxed{43} \end{array}$$



$$\Rightarrow 2+2+8+1=13 \quad | \quad 2+5+3+1=11 \quad | \quad 2+3+5+4=14$$

$$(13)^2 = 169-1 = 168 \quad | \quad (11)^2 = 121-1 = 120 \quad | \quad (14)^2 = 196-1 = \boxed{195}$$



$$\Rightarrow 6+8+4+7=25 \quad | \quad 9+12+8+7=36 \quad | \quad 10+9+16+14=49$$

$$\sqrt{25} = 5 \quad | \quad \sqrt{36} = 6 \quad | \quad \sqrt{49} = \boxed{7}$$

OR

(FCI-2011)

$$\begin{array}{c} 2 \textcircled{5} \\ \downarrow \\ 5 \end{array}$$

$$\begin{array}{c} 3 \textcircled{6} \\ \downarrow \\ 6 \end{array}$$

$$\begin{array}{c} 4 \textcircled{9} \\ \downarrow \\ \boxed{9} \rightarrow \text{Unit Place} \end{array}$$

OR

$$8-4 = 4$$

$$12-8 = 4$$

$$16-9 = 7 \quad (\text{CGL-2016})$$

$$\begin{array}{r} 7-6 = 1 \\ + \\ \hline 5 \end{array}$$

$$\begin{array}{r} 9-7 = 2 \\ + \\ \hline 6 \end{array}$$

$$\begin{array}{r} 14-10 = 4 \\ + \\ \hline \boxed{11} \end{array}$$

OR

$$(6-4)^2 = 4$$

$$(9-8)^2 = 1$$

$$(16-10)^2 = 36$$

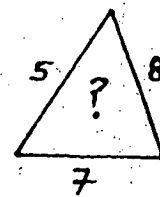
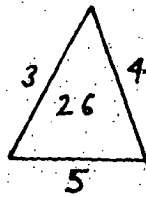
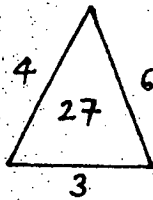
(CGL-2016)

$$\begin{array}{r} (7-6)^2 = 1 \\ + \\ \hline 5 \end{array}$$

$$\begin{array}{r} (12-9)^2 = 9 \\ + \\ \hline 6 \end{array}$$

$$\begin{array}{r} (14-10)^2 = 16 \\ + \\ \hline \boxed{41} \end{array}$$

(14.)



(CGL-2008)

(a) 41

(b) 38

(c) 43 ✓

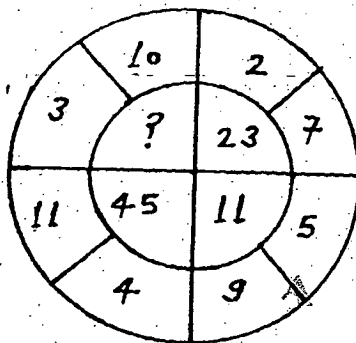
(d) 40

$$\Rightarrow 4 + 6 + 3 = 13 \rightarrow 13 \times 2 + 1 = 27$$

$$3 + 4 + 5 = 12 \rightarrow 12 \times 2 + 2 = 26$$

$$5 + 8 + 7 = 20 \rightarrow 20 \times 2 + 3 = \boxed{43}$$

(15.)



(a) 34

(b) 46 ✓

(c) 91

(d) 21

$$\Rightarrow 2 - 2 = 0 \rightarrow (5)^2 = 25$$

$$= 25 - 2 = 23$$

$$9 - 5 = 4 \rightarrow (4)^2 = 16$$

$$= 16 - 5 = 11$$

$$11 - 4 = 7 \rightarrow (7)^2 = 49$$

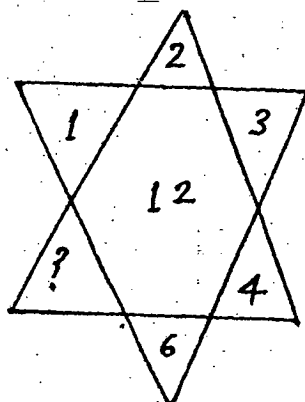
$$= 49 - 4 = 45$$

$$\Rightarrow 10 - 3 = 7 \rightarrow (7)^2 = 49$$

$$= 49 - 3 = \boxed{46}$$

(CGL-2015)

(16.)



$$\Rightarrow 2 \times 6 = 12$$

$$3 \times 4 = 12$$

$$1 \times ? = 12$$

$$? = \boxed{12}$$

(CPO-2015)

(A) 8

(B) 1

(C) 12 ✓

(D) 6

# TYPE-2 :

(17.)

|    |    |    |
|----|----|----|
| 14 | 16 | 18 |
| 26 | 32 | 18 |
| 10 | 12 | ?  |

$$\Rightarrow \begin{aligned} (14+26) \div 4 &= 10 \\ (16+32) \div 4 &= 12 \\ (18+18) \div 4 &= \boxed{9} \end{aligned}$$

(18.)

|    |    |    |
|----|----|----|
| 42 | 44 | 38 |
| 28 | 55 | 23 |
| 39 | ?  | 37 |

(a) 11 (b) 33 (c) 66 (d) 22 (✓)

$$\Rightarrow \begin{aligned} 42 - 38 &= 4 \times 11 = 44 \\ 28 - 23 &= 5 \times 11 = 55 \\ 39 - 37 &= 2 \times 11 = \boxed{22} \end{aligned}$$

(19.)

|     |    |    |
|-----|----|----|
| 72  | 24 | 6  |
| 96  | 18 | 12 |
| 108 | ?  | 18 |

$$\Rightarrow \begin{aligned} 72 \div 6 &= 12 \times 2 = 24 \quad (\text{CGL-2011}) \\ 96 \div 12 &= 8 \times 2 = 16 \\ 108 \div 18 &= 6 \times 2 = \boxed{12} \end{aligned}$$

(20.)

|     |    |   |
|-----|----|---|
| 1   | 2  | 3 |
| 11  | 7  | 5 |
| 120 | 45 | ? |

$$\Rightarrow \begin{aligned} (11)^2 - (1)^2 &= 120 \quad (\text{CPO-2013}) \\ (7)^2 - (2)^2 &= 45 \\ (5)^2 - (3)^2 &= \boxed{16} \end{aligned}$$

(21.)

|    |     |    |    |
|----|-----|----|----|
| 31 | 17  | 58 | 87 |
| 68 | 19  | 61 | 56 |
| 91 | 22  | 70 | 50 |
| 10 | 142 | 11 | ?  |

$$\Rightarrow \begin{aligned} 31 + 68 + 91 + 10 &= 200 \quad (\text{CGL-2013}) \\ 17 + 19 + 22 + 142 &= 200 \\ 58 + 61 + 70 + 11 &= 200 \\ \therefore 87 + 56 + 50 + x &= 200 \\ x &= 200 - 193 = \boxed{7} \end{aligned}$$

(22.)

|     |   |     |
|-----|---|-----|
| 729 | 3 | 216 |
| 343 | ? | 64  |

$$\Rightarrow \begin{aligned} 729 &= 9^3 \rightarrow 9 \\ 216 &= (6)^3 \rightarrow 6 \\ 343 &= (7)^3 \rightarrow 7 \\ 64 &= (4)^3 \rightarrow 4 \end{aligned} \rightarrow 9 - 6 = 3 \rightarrow 7 - 4 = \boxed{3}$$

(a) 1 (b) 2 (c) 4 (d) 3 (✓)

(23.)

|     |   |     |
|-----|---|-----|
| 963 | 2 | 844 |
| 464 | ? | 903 |

(a) 2 (✓) (b) 3 (c) 1 (d) 4 (MTS-2014)

$$\Rightarrow (9+6+3) - (8+4+4) = 18-16 = 2$$

$$(4+6+4) - (9+0+3) = 14-12 = \boxed{2}$$

(24.)

|    |    |    |    |
|----|----|----|----|
| 7  | 9  | 5  | 11 |
| 4  | 15 | 12 | 7  |
| 13 | 8  | 11 | ?  |

$\Rightarrow 7+9-5 = 11$  (CGL-2012)

$$4+15-12 = 7$$

$$13+8-11 = \boxed{10}$$

(25.)

|           |   |           |    |          |    |
|-----------|---|-----------|----|----------|----|
| 7         | 8 | 10        | 11 | 13       | 16 |
| <u>29</u> | 9 | <u>38</u> | 12 | <u>?</u> | 11 |

(CGL-2016)

$$\Rightarrow 7+8+9 = 24+5 = 29$$

$$10+11+12 = 33+5 = 38$$

$$13+16+11 = 40+5 = \boxed{45}$$

(26.)

|   |   |   |
|---|---|---|
| A | D | H |
| F | I | M |
| ? | N | R |

$\Rightarrow$

|           |    |    |    |    |
|-----------|----|----|----|----|
| 1         | +3 | 4  | +4 | 8  |
| 6         | +3 | 9  | +4 | 13 |
| <u>11</u> |    | 14 | +4 | 18 |

$\downarrow$  K (According to Place Value)

(CGL-2014)

(27.)

|   |   |   |
|---|---|---|
| F | I | O |
| A | J | K |
| P | M | ? |

$\Rightarrow$  (According to place Value)-

|    |     |                                                 |
|----|-----|-------------------------------------------------|
| F  | I   | O                                               |
| 6  | +9  | $\rightarrow 15$                                |
| A  | J   | K                                               |
| 1  | +10 | $\rightarrow 11$                                |
| P  | M   |                                                 |
| 16 | +13 | $\rightarrow \textcircled{29} \rightarrow 26+3$ |

$\downarrow$  C

(CGL-2014)

(28.)

|   |   |   |
|---|---|---|
| Z | ? | S |
| R | 0 | ? |
| ? | 6 | C |

$\nearrow$  W  
 $\rightarrow$  K  
 $\nwarrow$  J

$26 \xrightarrow{-3} 23 \xrightarrow{-4} 19$   
 $18 \xrightarrow{-3} 15 \xrightarrow{-4} 11$   
 $10 \xrightarrow{-3} 7 \xrightarrow{-4} 3$

$\boxed{W}$   
 $\boxed{K}$   
 $\boxed{J}$

(29.)

|    |    |   |
|----|----|---|
| 2  | 2  | 2 |
| 3  | 2  | 1 |
| 2  | 3  | 4 |
| 25 | 64 | ? |

$\downarrow$   
 Row of Power  
 $\rightarrow \boxed{81}$

$2+3 = (5)^2 = 25$   
 $2+2 = (4)^2 = 16$   
 $2+1 = (3)^2 = 9$

$2+3 = (5)^2 = 25$   
 $2+2 = (4)^2 = 16$   
 $2+1 = (3)^2 = 9$

$2+3 = (5)^2 = 25$   
 $2+2 = (4)^2 = 16$   
 $2+1 = (3)^2 = 9$

(CGL-2015)

(30.)

|   |   |    |    |    |    |
|---|---|----|----|----|----|
| 1 | 2 | 3  | 2  | 10 | 12 |
| 2 | 5 | 12 | 10 | 16 | 13 |
| 1 | 2 | 1  | ?  | 10 | 24 |

$\Rightarrow$

$1+2+1 = 4 = (2)^2$   
 $2+5+2 = 9 = (3)^2$   
 $3+12+1 = 16 = (4)^2$   
 $2+10+13 = 25 = (5)^2$   
 $10+16+10 = 36 = (6)^2$   
 $12+13+24 = 49 = (7)^2$

$\boxed{25}$

(IBPS-2014)

(31.)

|    |    |    |
|----|----|----|
| 6  | 11 | 25 |
| 8  | 6  | 16 |
| 12 | 5  | ?  |

$\div 2$   
 $\times 2$

$(a) 12$   
 $(b) 16$   
 $(c) 22$   
 $(d) 18$

$\Rightarrow (6 \div 2) + (11 \times 2) = 25$   
 $(8 \div 2) + (6 \times 2) = 16$   
 $(12 \div 2) + (5 \times 2) = 16$

$\boxed{16}$

(IB-2013)

(32.)

|    |    |    |
|----|----|----|
| 51 | 11 | 61 |
| 64 | 30 | 32 |
| 35 | ?  | 43 |

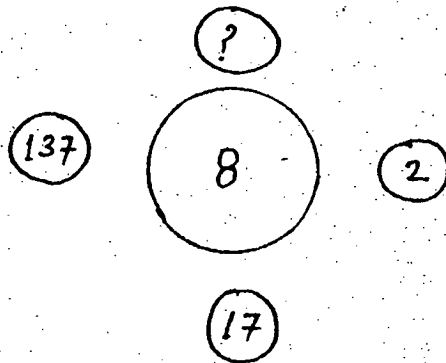
$$\Rightarrow 5 \times 1 + 6 \times 1 = 11$$

(CGL-2011)

$$6 \times 4 + 3 \times 2 = 30$$

$$3 \times 5 + 4 \times 3 = \boxed{27}$$

(33.)



$$\Rightarrow 2 \times 8 = 16 + 1 = 17$$

$$17 \times 8 = 136 + 1 = 137$$

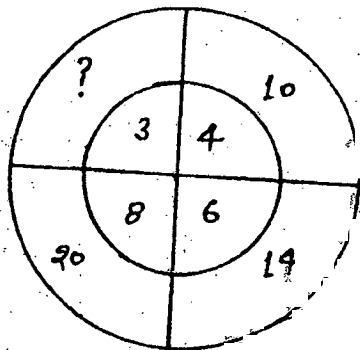
$$137 \times 8 = 1096 + 1 = \boxed{1097}$$

(a) 97

(b) 907

(c) 1097 (✓) (d) 9107

(34.)



(A) 18

(B) 19 (✓)

(C) 12

(D) 24

(CGL-2012)

$$\Rightarrow 3 \times 2 + 4 = 10$$

$$4 \times 2 + 6 = 14$$

$$6 \times 2 + 8 = 20$$

$$8 \times 2 + 3 = \boxed{19}$$

(35.)

|   |    |   |
|---|----|---|
|   | 8  |   |
| 6 | 54 | 7 |
|   | 4  |   |

|   |    |   |
|---|----|---|
|   | 12 |   |
| 8 | 51 | 4 |
|   | 7  |   |

|   |    |   |
|---|----|---|
|   | 14 |   |
| 9 | ?  | 5 |
|   | 9  |   |

(LDC-2015)

(A) 68 (✓)

(B) 71

(C) 53

(D) 76

$$\Rightarrow 7 \times 6 + (8 + 4) = 42 + 12 = 54$$

$$4 \times 8 + (12 + 7) = 32 + 19 = 51$$

$$5 \times 9 + (14 + 9) = 45 + 23 = \boxed{68}$$



(36.)

|   |    |   |    |    |    |    |    |
|---|----|---|----|----|----|----|----|
|   | 18 |   | 11 |    | 6  |    | 12 |
| 9 | 38 | 6 | 19 | 32 | 9  | 26 | 44 |
|   | 17 |   | 11 |    | 15 |    | 8  |

(A) 9 (✓) (B) 40 (C) 7 (D) 36

$\Rightarrow (18+17)+(9-6) = 38$

(SBI-PO-2015)

$(11+11)+(19-9) = 32$

$(15+6)+(26-3) = 44$

$(12+8)+(20-9) = \boxed{9}$

(37.) 7 8 5  $\Rightarrow$  In column wise (CGL-2014)

6 9 ?

$7 \times 6 \times 2 = 84$

2 3 6

$8 \times 9 \times 3 = 216$

84 216 900

$5 \times x \times 6 = 900$

(A) 90

(B) 70

$x = \boxed{30}$

(C) 65

(D) 30 (✓)

(38) 96 ? 168

$\Rightarrow$  2<sup>nd</sup> row is  $\frac{1}{3}$ <sup>rd</sup> of 1<sup>st</sup> row

32 48 56

3<sup>rd</sup> row is  $\frac{1}{6}$ <sup>th</sup> of 1<sup>st</sup> row.

16 24 28

(A) 52

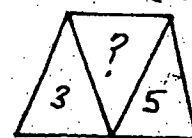
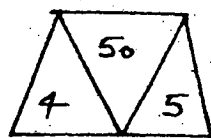
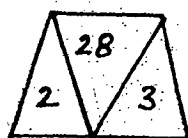
(B) 144 (✓)

(CGL-2011)

(C) 64

(D) 38

(39.)



(CPO-2010)

(A) 34

(B) 40 (✓)

(C) 49

(D) 53

$\Rightarrow 2, 3 \rightarrow 23 + 5 = 28$

$4, 5 \rightarrow 45 + 5 = 50$

$3, 5 \rightarrow 35 + 5 = \boxed{40}$

## \* REPEATED-SERIES \*

(1)  $a\_c\_ab\_\_bca\_ca\_c\_.$

(a)  $cabbc$  (b)  $abcca$  (c)  $abcab$  (d)  $bcaabb$  (✓)

$\Rightarrow \overline{abcab} \overline{c} \overline{abca} \overline{b} \overline{c} \overline{ab} \overline{c}.$

(2)  $\_ \_ aba \_ \_ ba \_ ab$

(CPO-2016)

$\Rightarrow \overline{a} \overline{a} \overline{aba} \overline{a} \overline{a} \overline{ba} \overline{aab}$

OR

$\overline{a} \overline{b} \overline{aba} \overline{b} \overline{aba} \overline{b} \overline{ab}$

$aaaaa/abbaab$

(3)  $ab \_ \_ baa \_ \_ ab \_$

$\Rightarrow \overline{ab} \overline{a} \overline{aba} \overline{a} \overline{b} \overline{a} \overline{ab} \overline{a}$

$aaabaa$

(4)  $bca \_ \_ aab \_ \_ a \_ caa$

$\Rightarrow \overline{bca} \overline{a} \overline{bca} \overline{ab} \overline{ca} \overline{a} \overline{bca} \overline{a}.$

$accab$

(5)  $\_ aa \_ ba \_ bb \_ ab \_ a \_ b$

$\Rightarrow \overline{baa} \overline{b} \overline{ba} \overline{abb} \overline{aab} \overline{baa} \overline{b}$

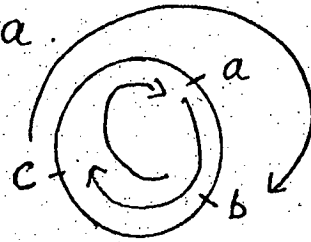
(CGL-2015)

$bbaaaba$

(6) bc ca aba c ca (FCI-2011)

$\Rightarrow$  a bc b ca c ab a b c b ca

abcbb



(7) a aaaa a ba ab

$\Rightarrow$  a a b aaaa a a a ba a a a

abaaaa

(8) aa aa ba aba cb (CGL-2015)

$\Rightarrow$  aa c b aa b ba a aba a cb

$\Rightarrow$ 

|   |   |   |   |
|---|---|---|---|
| a | a | c | b |
| a | a | b | b |
| a | a | a | b |
| a | a | c | b |

cbbaa

(9) bca ac bb aaac b (LDC-2013)

$\Rightarrow$  bca a ac b bb c aaac b

abc b

(10) a aa aaa a aa aa aab

$\Rightarrow$  a b aa b aaa b a aaa b aa a aab

bbbaaba

(11.)  $cc - cc bc - accb cc - c - b$  (ASI - CISE - 2012)

(1) acac (✓) (2) abac

(3) abab (4) aa bc

$\Rightarrow \overline{cc} \overline{a} \overline{cc} \overline{bc} \overline{c} \overline{acc} \overline{bcc} \overline{a} \overline{c} \overline{b}$

(12.)  $\_ nmmn \_ mmnn \_ mnnm \_$  (CPO-SI - 2011)

(1) nnmm (✓) (2) nmnm

(3) mnnm (4) nmnn

$\Rightarrow \overline{n} \overline{nm} \overline{mn} \overline{n} \overline{mm} \overline{nn} \overline{m} \overline{m} \overline{nn} \overline{m}$

[13.]  $\_ ST \_ OL \_ GY$

(UP PCS-14)

(1) ORA (2) RAO (3) ARO (✓) (4) AOR

$\Rightarrow \underline{A} \underline{S} \underline{T} \underline{R} \underline{O} \underline{L} \underline{O} \underline{G} \underline{Y}$

[14.]  $ac \_ cab \_ bac a \_ aba \_ aca \_$  (CGL-2011)

(1) acbcc

(2) aacbc (✓)

(3) babbb

(4) bcbb a

$\Rightarrow \overline{ac} \overline{a} \overline{c} / \overline{ab} \overline{a} \overline{b} / \overline{aca} \overline{c} / \overline{aba} \overline{b} / \overline{aca} \overline{c}$

[15]  $aed \_ fiji \_ Koln \_$

(1) cmh (2) chm (✓) (3) cgm (4) cjl

$\Rightarrow \overline{a} \overline{e} \overline{b} \overline{d} \overline{c} / \overline{f} \overline{i} \overline{j} \overline{i} \overline{h} / \overline{K} \overline{o} \overline{l} \overline{n} \overline{m}$

[16.]  $h \_ eg \_ fegh \_ eg hfe \_$

(CGL-2013)

(1) gffh

(2) hkgg

(3) ffgh

(4) fhfg (✓)

$\Rightarrow \overline{h} \overline{f} \overline{e} \overline{g} \overline{h} \overline{f} \overline{e} \overline{g} \overline{h} \overline{f} \overline{e} \overline{g} \overline{h} \overline{f} \overline{e} \overline{g}$

(17.) — — a b a — — b a — a b (Bank PO-2013)

(1) a b b b a (2) a b b a b (✓)

(3) b a a b b (4) b b a b a

$\Rightarrow$  a b a b a b a b a b

[18.] x x x — s x — z — t x y — — y z (LDC-2013)

(1) y y x x s

(2) y y s x x

(3) y y x s x (✓)

(4) y y x s s

$\Rightarrow$  x x x s x z t x y y z

[19.] a — c a — c a — c a a — (CGL-2015)

(1) c a c a

(2) c a c c (✓)

(3) c a a c

(4) c c c a

$\Rightarrow$  a c c a a c a a c

[20.] E — G F F — F G F — E (CPO-2016)

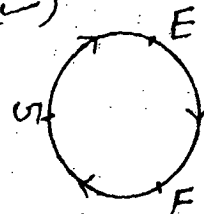
(1) E G F G

(2) G F E F

(3) F E G G

(4) F G E G (✓)

$\Rightarrow$  E F G F G E F G F G E



[21.] S A T I — H S A — I S — — A T — S H

(1) S T H S I (✓)

(2) S T A S I

(3) A T H S I

(4) S T H S A

$\Rightarrow$  S A T I S H S A I S H A T S H

# \* ANALOGY \*

(1.)  $CX : TH :: PG : ?$

$$\Rightarrow \begin{array}{ccc} \begin{array}{cc} 3 & 24 \\ C & X \end{array} & : & \begin{array}{cc} 20 & 8 \\ T & H \end{array} \\ \text{+17} & & \text{-16} \end{array} :: \begin{array}{cc} 16 & 7 \\ P & G \end{array} : \boxed{GQ}$$

$\Rightarrow 16+17=33 \Rightarrow (26+7)$   
 $7+26-16=17$   
 $\downarrow$   
 $Q$

OR

$$\begin{array}{ccc} \begin{array}{cc} & +5 \\ C & X \end{array} & : & \begin{array}{cc} & +5 \\ T & H \end{array} \\ \text{-4} & & \text{-4} \end{array} :: \begin{array}{cc} & +5 \\ P & G \end{array} : \boxed{CU}$$

(2.)  $6 : 210 :: 8 : ?$

$$\Rightarrow \begin{array}{ccc} 6 & : & 210 \\ \text{+204} & & \end{array} :: \begin{array}{ccc} 8 & : & \boxed{212} \\ \text{+204} & & \end{array}$$

OR

$$\begin{array}{ccc} 6 & : & 210 \\ \downarrow & & \downarrow \\ 6^3 - 6 & = & 210 \end{array} :: \begin{array}{ccc} 8 & : & \boxed{504} \\ \downarrow & & \downarrow \\ 8^3 - 8 & = & 504 \end{array}$$

## (\*) Word Analogy TYPE-1 :

(A.) Worker : Working Place

- (i) Doctor : Hospital
- (ii) Engineer : Site
- (iii) Pilot : cockpit
- (iv) Actor : Stage
- (v) Artist : Theatre

- (vi) Gambler : Casino
- (vii) chef : Kitchen
- (viii) Umpire : Pitch.

**[B] WORKER : TOOL**

- (i) Doctor : Stethoscope
- (ii) Teacher : Chalk
- (iii) Tailor : Needle
- (iv) Barber : Scissors
- (v) Labour : Spade
- (vi) Gardiner : Harrow
- (vii) Chef : Knife
- (viii) Butcher : Chopper

**[C] PHOBIA → Fear of**

- (i) Hydrophobia — Water
- (ii) Aerophobia — Air
- (iii) Acrophobia — Height
- (iv) Photophobia — Light
- (v) Chromophobia — Color
- (vi) Cynophobia — Dogs
- (vii) Phasmophobia — Ghosts
- (viii) Gamophobia — Marriage
- (ix) Phobophobia — Fear

**[D] Instrument : Measurement**

- (i) Thermometer : Temperature
- (ii) Anemometer : Wind
- (iii) Hygrometer : Humidity
- (iv) Seismograph : Earthquake
- (v) Barometer : Pressure
- (vi) Tensiometer : Strains
- (vii) Odometer : Speed
- (ix) Sphygmomanometer : B.P.

**[E] Individual — Group**

- (i) Man — Crowd
- (ii) Cattle — Herd
- (iii) Sheep — Flock
- (iv) Fish — Shoal
- (v) Bees — Swarm
- (vi) Flower — Bouquet
- (vii) Singer — Chorus/choir
- (viii) Nomads — Horde

**[F] Individual → class**

- (i) Man → Mammal
- (ii) Rat → Rodent
- (iii) Ostrich → Bird
- (iv) Butterfly → Insect
- (v) Snake → Reptile
- (vi) Whale → Mammal.



| STUDY → TOPICS        | STUDY → TOPICS            |
|-----------------------|---------------------------|
| Virology — Virus      | Astrology — Future        |
| Histology — Tissues   | Craniology — Skull        |
| Oology — Eggs         | Concology — Shells        |
| Pedology — Soil       | Ichthyology — Fish        |
| Pathology — Disease   | Paleography — Writing     |
| Ornithology — Birds   | Nephrology — Kidney       |
| Nidology — Nest       | Anthropology — Mankind    |
| Mycology — Fungi      | Cosmology — Universe      |
| Orography — Mountains | Neurology — Brain/Nervous |
| Semantics — Language  | Demography — Population   |
| Cardiology — Heart    | Botany — plants           |
| Oncology — Cancer     | Serpentology — Snakes     |
| Entomology — Insects  | Seismology — Earth quake  |
| Haematology — Blood   |                           |

| Word — Synonym     | Word — Antonym      |
|--------------------|---------------------|
| Dearth — Scarcity  | Best — Worst        |
| Sedate — Calm      | Fresh — Stale       |
| Abduct — Kidnap    | Gradual — Abrupt    |
| Solicit — Request  | Condense — Expand   |
| Haughty — Proud    | Robust — Weak       |
| Fallacy — Illusion | Ignore — Notice     |
| Mend — Repair      | Lethargy — Alert    |
| Ban — Prohibition  | Kindle — Extinguish |
| Blend — Mix        |                     |



## TYPE-2 :

(1) Metals : Flaws :: Book : **ERRATA**

(2) Breeze : cyclone :: Drizzle : **Downpour**

(3)  $\begin{matrix} +8 & +8 \\ A & G & : & I & O \\ 1 & 7 & & 9 & 15 \end{matrix} :: \begin{matrix} E & K \\ 5 & 11 \end{matrix} : \boxed{MS}$

(4)  $\begin{matrix} 2 & 4 & 6 & & 8 & 9 & 10 \\ B & D & F & : & H & I & J \\ +6 & & +5 & & +4 \end{matrix} :: \begin{matrix} 13 & 15 & 17 \\ M & O & Q \\ +6 & +5 & +4 \end{matrix} : \boxed{STU}$

(5)  $\begin{matrix} 5 & 7 & 9 & 11 \\ E & G & I & K \\ +1 & +2 & +3 & +4 \end{matrix} : \begin{matrix} 6 & 9 & 12 & 15 \\ F & I & L & O \\ +1 & +2 & +3 & +4 \end{matrix} :: \begin{matrix} 6 & 8 & 10 & 12 \\ F & H & J & L \\ +1 & +2 & +3 & +4 \end{matrix} : \begin{matrix} 7 & 10 & 13 & 16 \\ G & J & M & P \\ +1 & +2 & +3 & +4 \end{matrix}$

(6)  $\begin{matrix} 1 & 2 & 3 & 4 & & 15 & 16 & 17 & 18 \\ A & B & C & D & : & O & P & Q & R \\ +14 & & +14 & & +4 \end{matrix} :: \begin{matrix} 20 & 19 & 23 & 11 \\ T & S & W & K \\ +14 \end{matrix} : H G K Y$   
 $20 + 14 = 34 = 26 + 8 \rightarrow H$  (FCE-2012)

(7)  $\begin{matrix} 4 & 8 & 12 & 16 \\ D & H & L & P \\ \text{Half} \end{matrix} : \begin{matrix} 2 & 4 & 6 & 8 \\ B & D & F & H \\ \text{Half} \end{matrix} :: \begin{matrix} 6 & 12 & 18 & 24 \\ F & L & R & X \\ ? \end{matrix}$

(a) FHLK

(b) CFIL (✓)

(c) KHPL

(d) DHMK

(8)  $\begin{matrix} 3 & 4 & 5 & 6 \\ C & D & E & F \\ +2 & +3 & +5 & +7 \end{matrix} : \begin{matrix} 5 & 7 & 10 & 13 \\ E & G & J & M \\ +2 & +3 & +5 & +7 \end{matrix} :: \begin{matrix} 16 & 17 & 18 & 19 \\ P & Q & R & S \\ ? \end{matrix}$  (CPO-2011)

(a) RTUZ (✓)

(b) SUMK

(c) ADIL

(d) ALNP

(9)  $48 : 122 :: 168 : ?$

(CPO-2011)

(a) 215

(b) 225

(c) 290 (✓)

(d) 292

$$\Rightarrow \begin{cases} 7^2 - 1 = 48 \\ 11^2 + 1 = 122 \\ 13^2 - 1 = 168 \\ 17^2 + 1 = 290 \end{cases}$$

Co-prime Number Series

(10)  $81 : 32 :: 64 : ?$

(CGL-2015)

(a) 25 (✓)

(b) 16

(c) 35

(d) 49

$$\Rightarrow 81 : 32 :: 64 : \boxed{25}$$

$\downarrow \quad \downarrow \quad \downarrow \quad \uparrow$   
 $3^4 \quad 2^5 \quad 4^3 \quad 5^2$

(11)  $520 : 738 :: ? : 350$

(CGL-2012)

(a) 111

(b) 222 (✓)

(c) 291

(d) 325

$$\Rightarrow 520 : 738 :: \boxed{222} : 350$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$   
 $8^3 + 8 \quad 9^3 + 9 \quad 6^3 + 6 \quad 7^3 + 7$

(12)  $72 : 18 :: 56 : ?$

(CPO-2012)

$$\Rightarrow 72 : 18 :: 56 : \underline{\hspace{2cm}}$$

$$7 + 2 = 9 \times 2 = 18$$

$$5 + 6 = 11 \times 2 = \boxed{22}$$

$$72 : 18 :: 56 : \underline{\hspace{2cm}}$$

$$72 \div 4 = 18$$

$$56 \div 4 = \boxed{14}$$

$(41, 14, 81)$

(ii)  $13 : 19 :: 21 : \underline{14}$

$$13 = 1 + 3 = 4 \times 5 - 1 = 19$$

$$21 = 2 + 1 = 3 \times 5 - 1 = \boxed{14}$$

(CGL-2013)

$$21 = 2 + 1 = 3 \quad 8 + 1$$

$$\frac{3}{6} = \frac{1}{2}$$

(CGL-2015)

(d) 115 (L)

$$23 \times (2+3) = 115$$

(CGL-2014)

(d) 7

$$8 \div 2 = 4 - 1 = \boxed{3}$$

(6, 5, 8) (CGL-2001, 13, 11)

(ii)  $1+7=\frac{8}{2}=4$

(iii)  $17 + 3 = 20 \div 5$   
 $= 4$

$$3 + 7 = \frac{10}{2} = \boxed{5}$$

$$37 + 3 = 40 \div 5$$

$$= 8$$

(17) DEF : EFD :: FGH : ?

(CGL-2016)

- (A) FHG (B) HGF (C) HFG (D) GHF (✓)

⇒ DEF → EFD

Similarly, FGH → GHF

(18) Governor : State :: President : ?

(CGL-2015)

- (A) Organisation (B) constituency  
(C) District (D) country (✓)

(19) King : Throne :: Rider : ?

(CGL-2015)

- (A) Saddle (✓) (B) Chair  
(C) Horse (D) Seat

⇒ A king sits in a throne similarly, a rider sits on a saddle on the back of a horse.

(20) Macabre : Lovely :: Baneful : ?

(CPO-2014)

- (A) Unharmful (✓) (B) Churlish  
(C) Filter (D) Ugly.

⇒ Antonym of macabre is lovely and antonym of baneful is ~~ex~~ harmless.

(21) Hive : bee :: Eyrie : ?

(CPO-2014)

- (A) Pigeon (B) Parrot  
(C) Sparrow (D) Eagle (✓)

⇒ Bee lives in hive.

Eagle lives in Eyrie.

- (22.) (A) Sharpener (B) calculator (✓) (FCI-2014)  
(C) Eraser (D) Pencil.

⇒ Except calculator, all others are stationary item.

- (23.) (A) Nagas (B) Hindus (✓) (C) Bhils (D) Todas (LDC-2012)

⇒ Except hindu, all others are tribes.

- (24.) (A) Troop (B) Group (C) Mob (✓) (D) class

⇒ Except Mob, all others are organized crowd.

- (25.) (i) Guitar (✓) (ii) Piano (CGL-2013)  
(iii) Harmonium (iv) Banto

⇒ Guitar is only string instrument.

- (26.) (A) Triangle (✓) (B) Pentagon (CGL-2011)  
(C) Hexagon (D) Quadrilateral.

⇒ Except triangle, all others have diagonal.

- (27.) (A) 50 (B) 63 (C) 55 (D) 49 (✓)

⇒ 49 → it is a perfect square.

- (28.) (A) Sam Veda (B) Atharv Veda (CPO-2011)  
(C) Rig Veda (D) Ayur Veda (✓)

⇒ Except Ayur Veda, all others are Vedas.

- (29.) (A) Fog (B) cloud (C) Rain (D) Mist

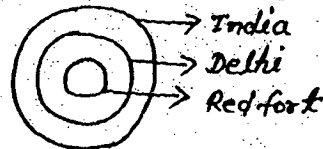
⇒ Except cloud, all others are different form of precipitation.

## \* VENN-DIAGRAM \*

Direction : The following Venn diagram is best represents the relation between given classes ?

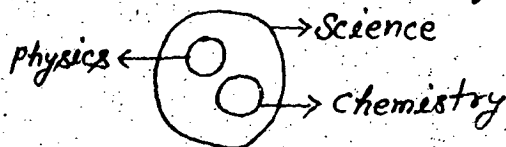
TYPE-1 :

Questions - (1.) India, Delhi, Red fort.



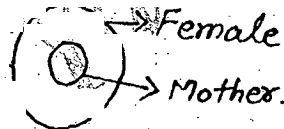
⇒ Red fort is in Delhi and Delhi is in India.

(2.) Science, Physics, Chemistry.



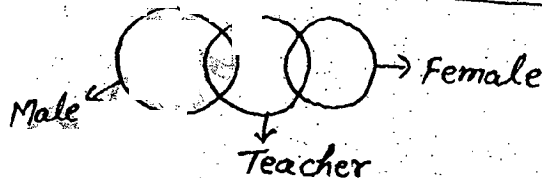
⇒ Physics and chemistry are different subjects but they comes under Science.

(3.) Female, Mother.



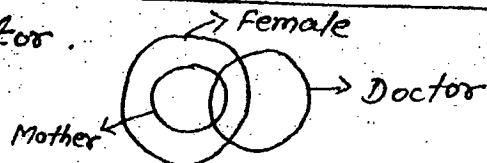
⇒ All female is not mother but all mother is female.

(4.) Male, Teacher, Female (Female).



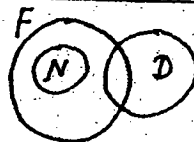
⇒ Female and Male can be teacher.

(5.) Female, Mother, Doctor.



⇒ All mothers are female and some doctor are Female or mother.

(6.) Female, Nurse, Doctor.



⇒ All nurse are female and some female can be doctor.

- (15.) Consider the figure. The number obtained from subtracting the sum of numbers in the triangle only from the number present in all the figures is.

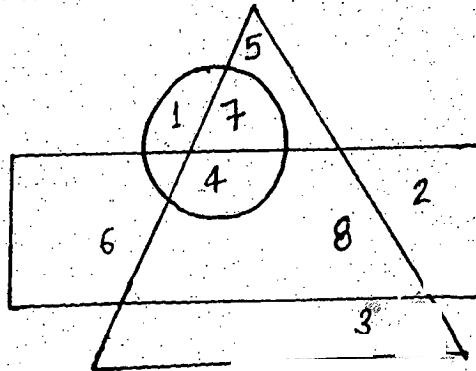
(UPPCS-2015)

(A)  $-4$  (✓)

(B)  $-3$

(C)  $4$

(D)  $11$



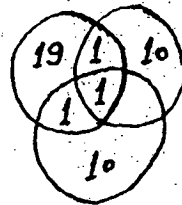
- (16.) In a group of person, 11 persons speak Kannada, 20 persons speak Tamil and 11 persons speak Telugu. In that group, if two persons speak two languages and one person speak all the languages, then how many persons are there in the group? (CGL-2014)

(A)  $40$

(B)  $41$

(C)  $42$  (✓)

(D)  $43$



- (17.) In the diagram, the rectangle represent males, circle represent actors and the triangle singers. Which numbered area represents male singers?

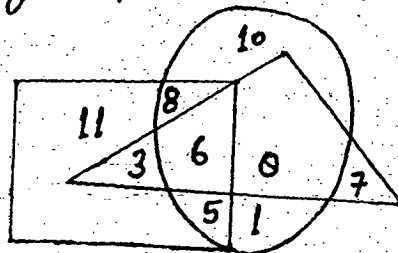
(CGL-2013)

(A)  $8$

(B)  $6$

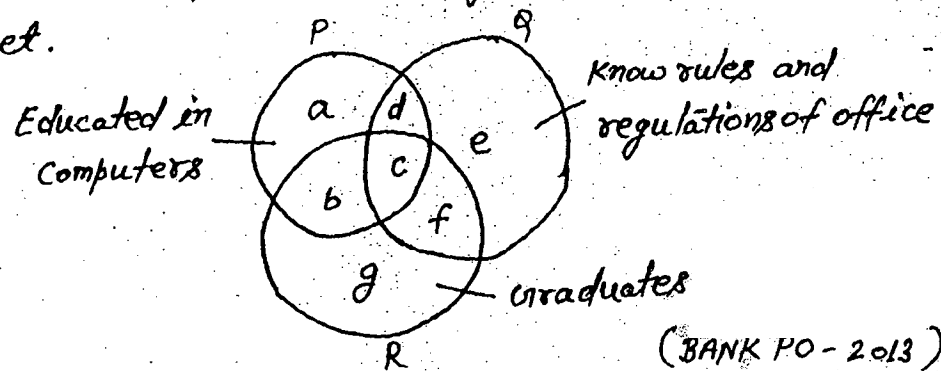
(C)  $5$

(D)  $3$  (✓)





- (18.) In the following diagram, there are 3 intersecting circles showing 3 different classes, one showing those candidates who have knowledge in computers, second showing those who know rules and regulations of office and third showing graduates. Each part of the diagram is represented by an alphabet.



- (i) Which alphabet represents those candidates who have knowledge in computers, know rules and regulations of office but not graduates?

$\Rightarrow [d]$

- (ii) Which alphabet represents those candidates who have knowledge in computers but are neither graduates nor they know rules and regulations?

$\Rightarrow [a]$

- (iii) Which alphabet represents those candidates who know computers and are graduates but do not know about office rules?

$\Rightarrow [b]$

- (iv) Which alphabet represents those candidates who have knowledge in computers, know rules and regulations of office and are also graduates?

$\Rightarrow [c]$

- (v) Which alphabet represents those candidates who are graduates, know rules and regulations of office but don't know computers?

$\Rightarrow [f]$



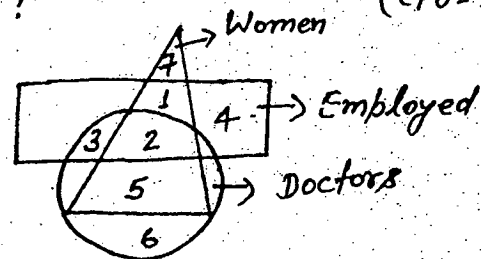
(19) In the following figure, which number represents the Women Doctors who are not Employed? (CPO-2011)

(A) 5 (✓)

(B) 3

(C) 1

(D) 6



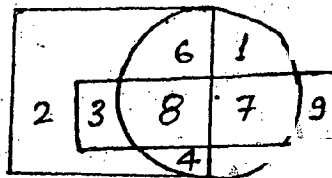
(20) Which are the numbers that make their presence felt in only one of the geometric figures? (CPO-2010)

(A) 4, 6, 7

(B) 1, 2, 9 (✓)

(C) 3, 7, 9

(D) 2, 3, 8



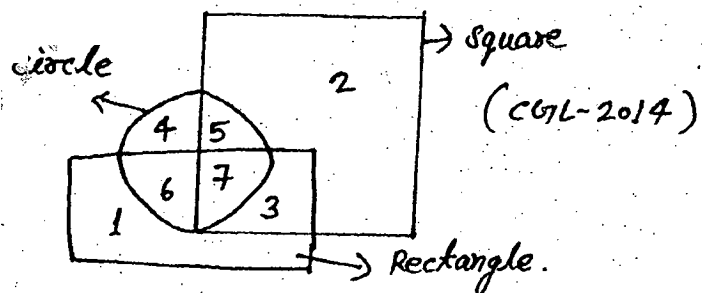
(21) In the following diagram, Square represents Doctors, Circle represents players and Rectangle represents Artists. Which number represents those Doctors who are players as well as Artists?

(A) 7 (✓)

(B) 2

(C) 3

(D) 6



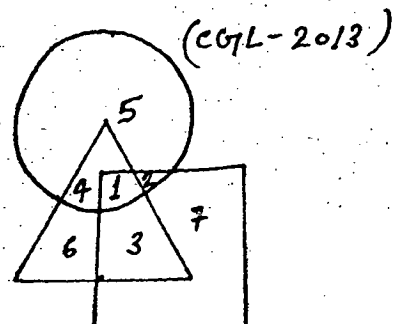
(22) In the following figure  $\bigcirc$  represents hardworking,  $\triangle$  represents Sincere and  $\square$  represents intelligent. Find out the hardworking who are intelligent but not sincere.

(A) 1

(B) 2 (✓)

(C) 3

(D) 4

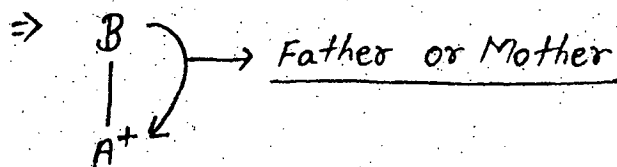


## \* "BLOOD-RELATION" \*

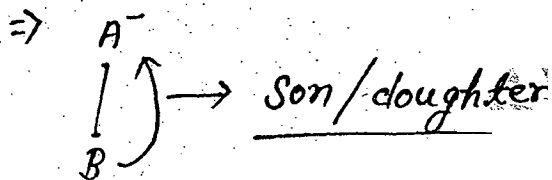
Note:- Important Symbol →

- (i) '+' → MALE
- (ii) '-' → Female
- (iii) 'I' → Next or old generation.
- (iv) '—' → Same generation.
- (v) '== ' → This symbol is used for husband and wife.

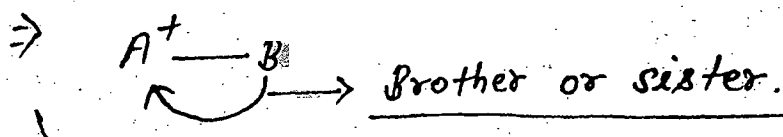
(1) A is the son of B. Find out the relation of B to A?



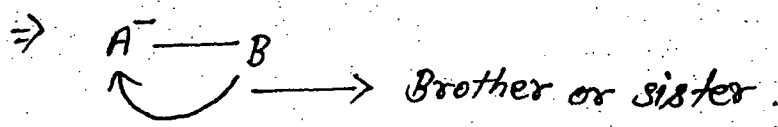
(2) A is the mother of B. Find out the relation of B to A?



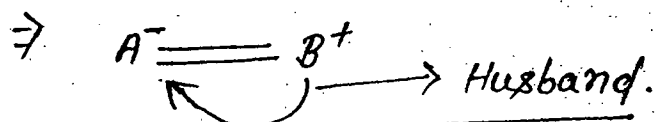
(3) A is the brother of B. Find out the relation of B to A?



(4) A is sister of B. Find out the relation of B to A?



(5) A is wife of B. Find out the relation of B to A?



⊗ TYPE-1 :

Question [1]  $A+B$  - Means -  $A$  is the Son of  $B$ .

$A-B$  - means -  $A$  is wife of  $B$ .

$A \times B$  - means -  $A$  is the brother of  $B$ .

$A \div B$  - means -  $A$  is the sister of  $B$ .

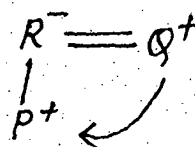
$A=B$  - means -  $A$  is the mother of  $B$

Given.

\* How is  $P$  related to  $Q$  ?

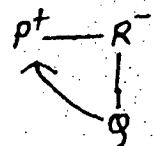
(i)  $P+R-Q$

$\Rightarrow$  Son



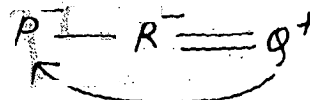
(ii)  $P \times R = Q$

$\Rightarrow$  Maternal Uncle.



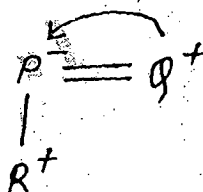
(iii)  $P \div R - Q$

$\Rightarrow$  Sister-in-law.



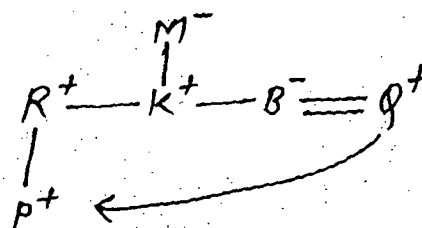
(iv)  $P = R + Q$

$\Rightarrow$  Wife.



(v)  $P+R \times K+M=B-Q$

$\Rightarrow$  Nephew.



[2] Which of the following shows that ' $P$ ' is the grand-mother of  $Q$  ?

(a)  $P \div R \times Q$

(b)  $P - R \times Q$

(c)  $P \times R + Q$

(d)  $P = R = Q$  (✓)

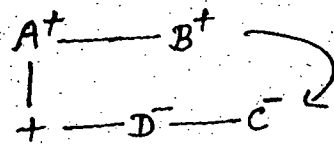
(e) None of these.



## TYPE-2 :

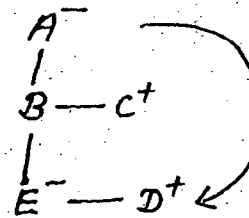
- (1) A and B are brothers. C and D sister. A's son is D's brother. So how is B related to C ?

⇒ Uncle.



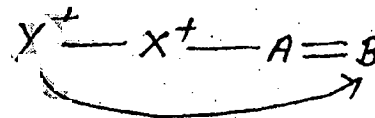
- (2) A is the mother of B. C is the son of A. D is the brother of E. E is daughter of B. So how is A related to D ?

⇒ Grand-Mother.



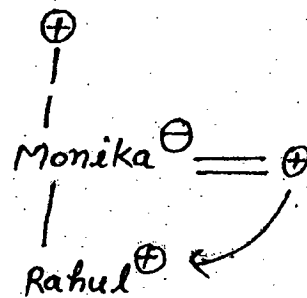
- (3) A and B is a married couple. X and Y are brother's. X is brother of A. So how is Y related to B ?

⇒ Brother-in-law.



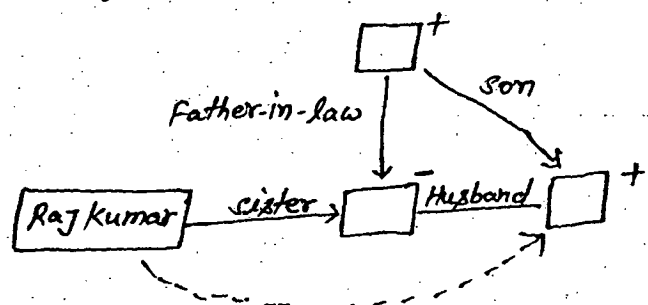
- (4) Rahul's mother is the only daughter of Monika's father. So how is Monika's husband related to Rahul ?

⇒ Father



- (5) What is the relation between the son of Rajkumar's sister's father in law and Rajkumar.

- (i) Father  
(ii) Son  
(iii) Brother-in-law (✓)  
(iv) Brother



### TYPE-3 :

(1) There are family of six member 'a', 'b', 'c', 'd', 'e' and 'f'.

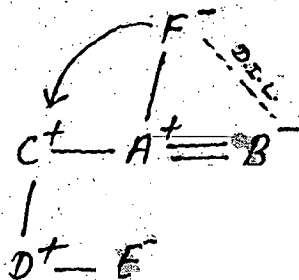
'a' is the husband of 'b'. 'd' is the son of 'c'. Who is the brother 'a'. 'e' is the sister of 'd'. 'b' is the daughter-in-law of 'f' whose husband has died?

(i) How is f related to c?

(ii) How many male member of this family?

⇒ (i) Mother.

(ii) Three male.



(2) There is the family of six member A, B, C, D, E and F.

They are doctor, lawyer, teacher, Engineer, Inspector, Salesman. There are two married couple in the family. D is a salesman is married to the lady teacher. Doctor is married to a Lawyer. F is inspector is son of B and brother of E. C is lawyer is daughter in-law of A. E is unmarried engineer. A is grandmother of F.

(i) What is profession of A?

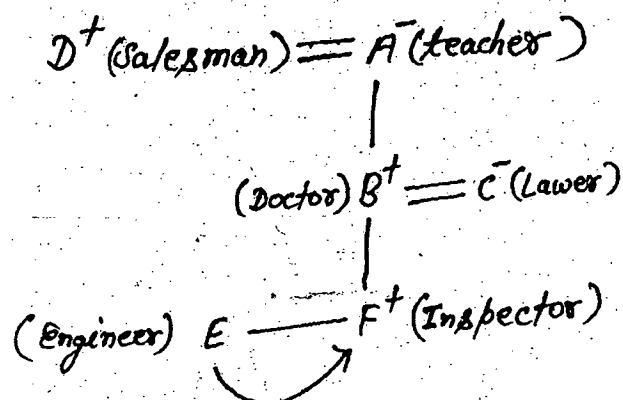
(ii) What is profession of B?

(iii) Who is E to F?

⇒ (i) Teacher.

(ii) Doctor.

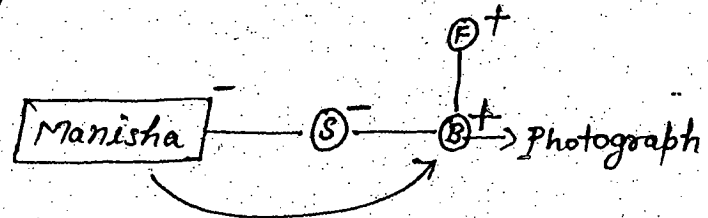
(iii) Brother/sister.



#### TYPE-4:

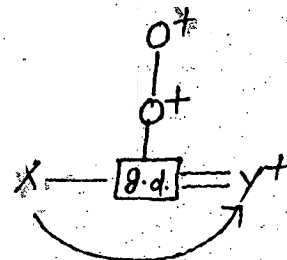
- (1) Pointing to a photograph Manisha said, "He is the only son of the father of my sister's brother. So how is photograph relating to Manish?"

⇒ Brother.



- (2) X introduces to Y. He is the husband of grand daughter of the father of my father. How is X related to Y?

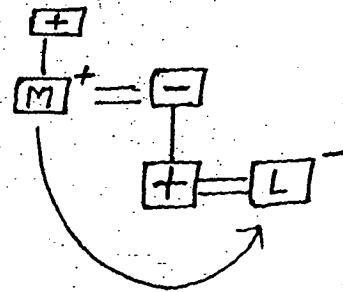
⇒ Brother-in-law/sister-in-law.



Note:- If brother-in-law or sister-in-law is not in option than wife is its Answer.

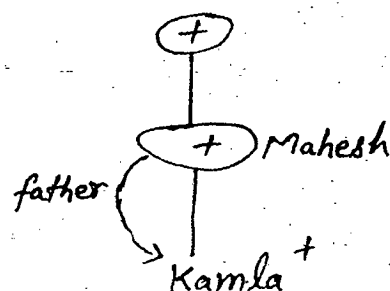
- (3) Pointing to a lady a man said, "Her husband's mother is the wife of my father's only son. So who is man to lady?"

⇒ Father-in-law.



- (4) Introducing Kamla, Mahesh said, "Her father is the only son of my father." How was Mahesh related to Kamla?

⇒ Father



(UPPCS-2015)

(5) Given that,

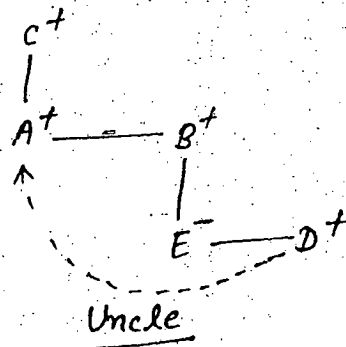
(i) A is the brother of B. (ii) C is the father of A.

(iii) D is the brother of E. (iv) E is the daughter of B.

Then, the uncle of D is

(CSAT-2012)

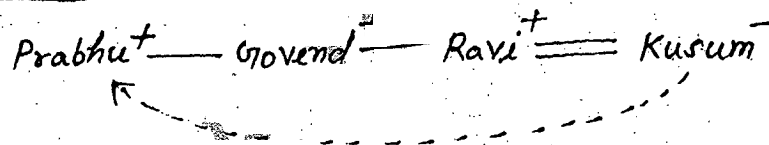
⇒ A



(6) Kusuma is the wife of Ravi. Govind and Prabhu are brothers. Govind is the brother of Ravi. Prabhu is Kusum's.

(FCT-2012)

⇒ Brother-in-law



(7) Mr. and Mrs. Gopal has 3 daughters and each daughter has one brother. How many persons are there in the family?

(CGL-2011)

⇒ Gopal and wife → 2

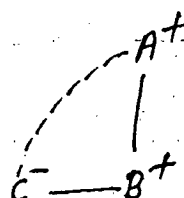
3 daughters → 3

1 brother → 1

∴ Total member = 2+3+1 = 6

(8) A is the father of B and C. B is the son of A. But C is not the son of A. How is C related to A? (CPO-2010)

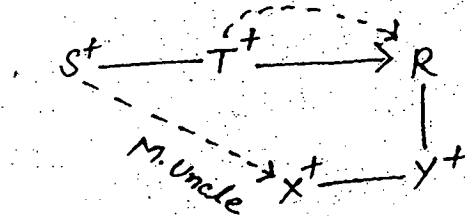
⇒ Daughter





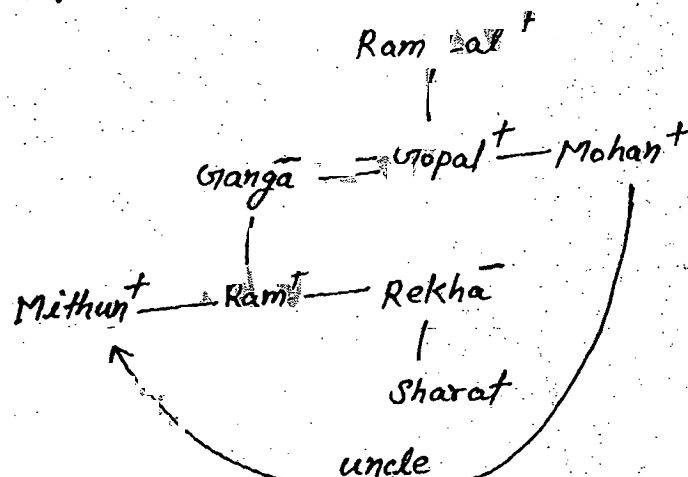
(9) X and Y are brothers. R is the father of Y. S is the brother of T and maternal uncle of X. What is T to R? (CGL-2010)

⇒ Brother-in-law



(10) Among her children, Ganga's favourites are Ram and Rekha. Rekha is the mother of Sharat, who is loved most by his uncle Mithun. The head of the family is Ram Lal, who is succeeded by his son Gopal and Mohan. Gopal and Ganga have been married for 35 years and have 3 children. What is the relation between Mithun and Mohan? (CGL-2014)

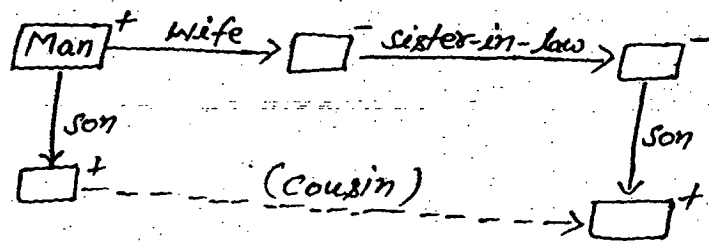
⇒ Uncle



(Mohan is the uncle of Mithun.)

(11) A man showed a boy next to him and said - "He is the son of my wife's sister-in-law, but I am the only child of my parents." How is my son related to him? (CPO-2009)

⇒ Cousin



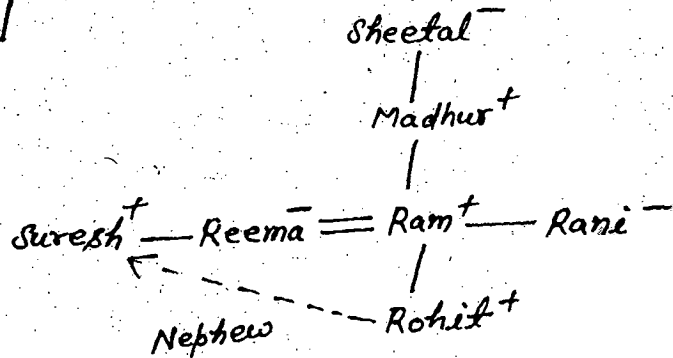
[Son of Man and that boy are cousin.]



- (12.) Suresh's sister is the wife of Ram. Ram is Rani's brother. Ram's father is Madhur. Sheetal is Ram's grandmother. Reema is Sheetal's daughter-in-law. Rohit is Rani's brother's son. Who is Rohit to Suresh?

⇒ Nephew

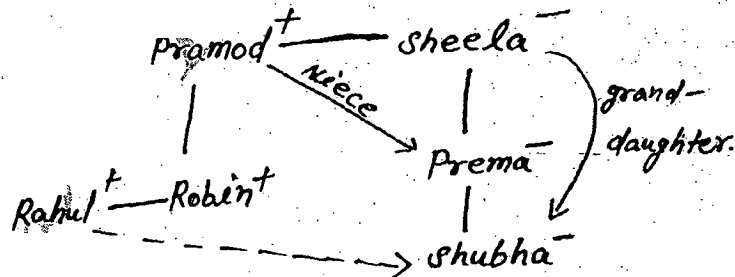
(FCI-2012)



- (13.) Rahul and Robin are brothers. Pramod is Robin's father. Sheela is Pramod's sister. Prema is Pramod's niece. Shubha is Sheela's grand daughter. How is Rahul related to Shubha?

(CPO-2015)

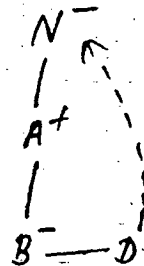
⇒ Maternal Uncle



- (14.) If 'P\$Q' means 'P is father of Q', 'P#Q' means 'P is mother of Q', 'P\*Q' means 'P is sister of Q', then how is D related to N in N#A\$B\*D?

(IBPS-2015)

- (i) Nephew
- (ii) Grandson
- (iii) Grand daughter
- (iv) Data Inadequate. (✓)



Here gender of D is not given so, D is either grand daughter or grand son of N.

## \* → "Logical Arrangement of Words" ← \*

Direction: In each of the following questions, which one of the given responses would be a meaningful order of the following?

(1.) Tree, Seed, Wood, Plant, Furniture.

⇒ (a) 2, 4, 1, 3, 5 (✓)

(b) 4, 1, 5, 2, 3

(c) 5, 1, 3, 2, 4

(d) \* 3, 1, 4, 2, 5

Note ⇒ (i) Ascending Order (✓) or Descending Order (✓)

(ii) Ascending order (✓) and descending order. (X)

Level-1 :

(2.) Family, Community, Member, Locality, Country.

⇒ (3 1 2 4 5)

(3) Consultation, Illness, Doctor, Treatment, Recovery.

⇒ (2 3 1 4 5)

(4) Gold, Iron, Sand, Platinum, Diamond.

⇒ (3 2 1 5 4)

(5) Banglow, Flat, Cottage, Palace, Mansion.

⇒ (3 2 1 5 4)

(6) Grand father, Brother, father, Son, Grand Son.

⇒ (i) 4 5 2 1 3

(ii) 5 1 3 2 4

(iii) 5 4 2 3 1 (✓)

(iv) 5 4 1 3 2

(7.) Mega, Kilo, Tera, Giga.

⇒ 2, 1, 4, 3

Level-2 :

(8) Arrest, Punishment, Crime, Prison, Judgment.

⇒ (31524)

---

(9) Key, Door, Lock, Room, Switch on.

⇒ (13245)

---

(10) Travell, Destination, Payment, Seat no., Reservation, Availability of seat.

⇒ (263541)

---

(11) Never, Some times, Generally, Seldom, Always.

⇒ (14235)

---

(12) Milky way, Sun, Moon, Earth, Stars.

⇒ (34251)

---

(13) Work, Interview, selection, appointment, Advertisement, Application.

⇒ (562341)

---

Level-3 :

(14) (1) Mohd. Hedayatullah.

(2) Giani Zail Singh.

(3) Neelam Sanjeeva Reddy.

(4) Fakhruddin Ali Ahmed.

(5) Pranab Mukherjee.

(6) Zakir Hussain.

⇒ (614325)

---

(15) (1) Major.

(2) Captain.

(3) LT. Colonel.

(4) Colonel.

(5) Lieutenant.

⇒ (52134)

---

Directions: In each of the following questions, which one of the given responses would be a meaningful order of the following?

(1.) Arrange the words in ascending order of their magnitude.

1. Mega    2. Kilo    3. Tera    4. Giga.    (CGL-2013)

(A) 1, 3, 2, 4    (B) 2, 1, 4, 3 (✓)    (C) 1, 2, 3, 4    (D) 2, 4, 3, 1

(2.) 1. Crop    2. Root    3. Stem    4. Seed    5. Flower

(A) 2, 3, 4, 1, 5    (B) 2, 4, 5, 1, 3    (SAT-2015)

(C) 4, 2, 3, 5, 1 (✓)    (D) 2, 3, 5, 1, 4

(3.) 1. Epilogue    2. Chapter    3. Index

4. Prologue    5. Cover.

(UPSI-2011)

(A) 3, 4, 2, 1, 5    (B) 1, 3, 4, 2, 5    (C) 1, 5, 2, 4, 3    (D) 5, 3, 4, 2, 1 (✓)

(4.) 1. Country    2. Furniture    3. Forest

4. Wood    5. Tree.

(FCI-2009)

(A) 1, 3, 5, 4, 2 (✓)    (B) 1, 4, 3, 2, 5    (C) 2, 4, 3, 1, 5    (D) 5, 2, 3, 1, 4

(5.) 1. Earth    2. Jupiter    3. Venus

4. Mars

5. Mercury.

(LDC-2013)

(A) 5, 3, 1, 2, 4    (B) 5, 3, 4, 1, 2 (✓)    (C) 5, 3, 1, 4, 2    (D) 5, 3, 2, 4, 1

(6.) 1. Adulthood    2. Infancy    3. Childhood

4. Adolescence.

(CGL-2012)

(A) 1, 3, 4, 2    (B) 2, 3, 4, 1 (✓)    (C) 2, 4, 3, 1    (D) 1, 2, 3, 4

⇒ Stages of human life.

(7.) 1. Indigo    2. Red    3. Violet    4. Blue

5. Green

6. Yellow

7. Orange

(CGL-2014)

(A) 3, 1, 4, 5, 2, 6, 7

(B) 3, 1, 4, 5, 6, 2, 7

(C) 1, 5, 6, 7, 3, 4, 2

(D) 3, 1, 4, 5, 6, 7, 2 (✓)

⇒ Sequence of colour of Rainbow

(8.) 1. Plant 2. Seed 3. Fruit 4. Seedling. (CGL-2014)  
 (A) 4,2,3,1 (B) 2,3,1,4 (C) 4,1,3,2 (✓) (D) 3,2,4,1

(9.) 1. Fruit 2. Pollen 3. Corolla 4. Stamen. (Cib. Sec-2013)  
 (A) 2,4,3,1 (✓) (B) 2,3,4,1 (C) 3,1,4,2 (D) 2,1,3,4

(10.) 1. Pupa 2. Larva 3. Moth 4. Egg. (CGL-2013)  
 (A) 4,3,2,1 (B) 4,3,1,2 (C) 4,2,1,3 (✓) (D) 4,1,2,3

(11.) 1. South America 2. Africa 3. Europe. (CGL-2013)  
 4. Australia 5. North America.

(A) 2,5,4,3,1 (B) 1,5,4,3,2 (C) 1,2,4,5,3 (D) 1,4,2,5,3 (✓)

⇒ according to the size of continents from larger to smaller.

(12.) 1. Pages 2. Book rack 3. Library (CGL-2013)  
 4. Books 5. catalogue.

(A) 2,5,4,3,1 (B) 1,5,4,3,2 (C) 1,4,2,5,3 (✓) (D) 1,2,4,5,3

(13.) 1. Sir Radhakrishnan 2. Rajendra Prasad 3. Giani Zail Singh  
 4. V.V. Giri 5. A.P.J. Abdul Kalam. (FCI-2012)

(A) 2,1,4,5,3 (B) 2,1,4,3,5 (C) 2,1,3,4,5 (✓) (D) 2,1,5,4,3

⇒ Chronology order of their services.

2. Rajendra Prasad (1950-1962) → 1. S. Radhakrishnan (1962-1967) →  
 → 4. V.V. Giri (1969-1974) → 3. Giani Zail Singh (1982-1987)  
 → 5. APJ. Abdul Kalam. (2002-2007)

(14.) 1. District 2. Continent 3. Country (CPO-2014)  
 4. State 5. Village.

(A) 3,4,1,5,2 (B) 5,1,4,3,2 (✓) (C) 5,1,4,2,3 (D) 3,2,4,1,5

⇒ From smaller to bigger (On the basis of physical area)

## \* CODING - DECODING \*

TYPE-1 :

(1) In a coded language if MYSTIFY is coded as NZTUGZ then how NEMESIS would be coded in the similar language?

⇒ MYSTIFY → <sup>+1 +1 +1 +1 +1 +1</sup> NZTUGZ

Similarly,

NEMESIS → OFNFTJT

(2) In a coded language if MYSTIFY is coded as NZTUGZ then which word would be coded as NEMESIS?

⇒ MYSTIFY → NZTUGZ | NEMESIS → <sup>-1 -1 -1 -1 -1 -1</sup> MDLDRHR

So,

MDLDRHR → NEMESIS

(3) IF SIKKIM is written as THLJJL, how is TRAINING written?

⇒ SIKKIM → <sup>+1 +1 +1 +1 +1</sup> THLJJL

Similarly,

TRAINING → UQBHOHOF

(4) If DELHI is written as CCIDD, how is BOMBAY written?

⇒ DELHI → <sup>-1 -2 -3 -4 -5</sup> CCIDD

Similarly,

BOMBAY → AMJXVS

(5) IF PAPER is written as SCTGW, how is MOTHER written?

⇒ PAPER → <sup>+3 +2 +4 +2 +5</sup> SCTGW

Similarly,

MOTHER → PQXJJT

(6) IF FASHION is written as FOIHSAN, how is PROBLEM written?

⇒ FASHION → FOIHSAN

Similarly,

PROBLEM → PELBORM

(7) If SUBSTITUTION is written as ITSBUSNOITUT, how is DISTRIBUTION written?

⇒ SUBSTITUTION → ITSBUSNOITUT

Similarly,

DISTRIBUTION → IRTSIDNOITUB

(8) If FASHION is written as INAOFSH, how is PROBLEM written?

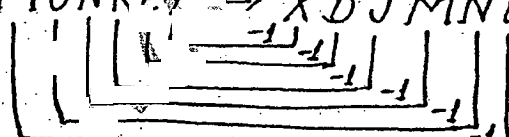
⇒ <sup>1 2 3 4 5 6 7</sup> FASHION → <sup>5 7 2 6 1 3 4</sup> INAOFSH

Similarly,

<sup>1 2 3 4 5 6</sup> PROBLEM → LMREPOB

(9) IF MONKEY is written as XDJMNL, how is TIGER written?

⇒ MONKEY → XDJMNL



Similarly —

TIGER → QDFHS

(10) IF NATURE is written as MASUQE, how is FAMINE written?

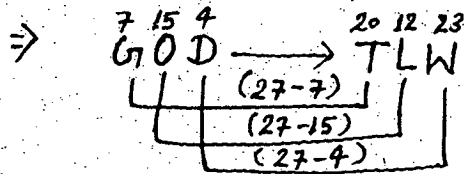
⇒ NATURE → <sup>1 2 3 4 5 6</sup> MASUQE

Similarly,

FAMINE → EALIME



(11.) If GOD is written as TLW, how is TEMPLE written?



Similarly,

TEMPLE → GVN KOV

Note ⇒

$\frac{27}{-SP} \leftarrow$  (Starting Position)

$\frac{LP}{-LP} \leftarrow$  (Last Position)

(12.) IF NATURE is written as MASUQE, how is POND written?

⇒ NATURE → <sup>1</sup>M<sup>0</sup>A<sup>1</sup>S<sup>0</sup>U<sup>1</sup>Q<sup>0</sup>E

Similarly,

POND → O O M C

Vowel → No change, Consonant → (-1).

TYPE-2:

(13.) In a coded language if 'PAINT' is coded as 53241 and likewise 'EXCEL' is 78679 then how 'ACCEPT' would be coded in similar language?

⇒ PAINT → 53241

EXCEL → 78679

∴ ACCEPT → 366751

(14.) D = 4, COVER = 63 then find out BASIS = ?

⇒ D = 4

COVER = (3+15+22+5+18) = 63

So, BASIS = 2+1+19+9+19 = 50

(15.) PALAM = 34 , SANTACRUZ = ?

$$\Rightarrow \text{PALAM} = 34 \quad (16+1+12+1+13=43)$$

[Reverse the sum of place value]

So,  $\text{SANTACRUZ} = 19+1+14+20+1+3+18+21+26 = 123$

Its reverse =  $\boxed{321}$

---

(16.) Z = 52 , ACT = 32 , TAB = ?

$$\Rightarrow Z = 26 = 26-1 = 25 \quad \text{Its reverse} = 52$$

$$\text{ACT} = 1+3+20 = 24-1 = 23 \quad \text{its reverse} = 32$$

So,  $\text{TAB} = 20+1+2 = 23-1 = 22$  Its reverse = 22

$$\boxed{\text{TAB} = 22}$$

---

(17.) GO = 32 , SHE = 49 , SOME = ?

$$\Rightarrow \overset{20}{G} \overset{12}{O} \longrightarrow 20+12 = 32$$

[Value of reverse position in alphabets]

$$\text{SHE} \longrightarrow 0+19+22 = 49$$

So,  $\text{SOME} \longrightarrow 8+12+14+22 = \boxed{56}$

---

(18.) AT = 20 , CAT = 60 , BAT = ?

$$\Rightarrow \text{AT} = 1 \times 20 = 20$$

$$\text{CAT} = 3 \times 1 \times 20 = 60$$

So,  $\text{BAT} = 2 \times 1 \times 20 = \boxed{40}$

(19.) In a coded language if MOBILITY is coded as 46293927 then how EXAMINATIONS would be coded in similar language?

$$\begin{array}{l|l}
 \Rightarrow M = 13 = 1+3 = 4 & E = 5 \\
 O = 15 = 1+5 = 6 & X = 24 = 2+4 = 6 \\
 B = 2 & A = 1 \\
 I = 9 & M = 13 = 1+3 = 4 \\
 L = 12 = 1+2 = 3 & I = 9 \\
 I = 9 & N = 14 = 1+4 = 5 \\
 T = 20 = 2+0 = 2 & A = 1 \\
 Y = 25 = 2+5 = 7 & T = 20 = 2+0 = 2 \\
 & I = 9 \\
 & O = 15 = 1+5 = 6 \\
 & N = 14 = 1+4 = 5 \\
 & S = 19 = 1+9 = 10 = 1+0 = 1
 \end{array}$$

So,

EXAMINATIONS = 561495129651

(20.) IF MACHINE is written as 197914152011, how is DANGER written?

$$\begin{array}{ccccccc}
 13 & 1 & 3 & 8 & 9 & 14 & 5 \\
 \text{MACHINE} & \rightarrow & 19 & 7 & 9 & 14 & 15 & 20 & 11 \\
 \begin{array}{c} \text{Diagram showing the addition of 6 to each letter's value to get the next letter's value in the sequence.} \\ \text{M (13) + 6 = 19 (P)} \\ \text{A (1) + 6 = 7 (G)} \\ \text{C (3) + 6 = 9 (I)} \\ \text{H (8) + 6 = 14 (N)} \\ \text{I (9) + 6 = 15 (O)} \\ \text{N (14) + 6 = 20 (T)} \\ \text{E (5) + 6 = 11 (L)} \end{array}
 \end{array}$$

Similarly,

DANGER  $\rightarrow$  10720131124

(21) If ASHA is coded as 79, how is VIDYABHUSHAN coded?

$$\Rightarrow \begin{matrix} 26 & 8 & 19 & 26 \\ A & S & H & A \\ 1 & 19 & 8 & 1 \end{matrix} = 79$$

[Sum of the (27-Place Value)]

$$\therefore \text{VIDYABHUSHAN} = 5+18+23+2+26+25+19+6+8+19 \\ +26+13 = \boxed{190}$$

TYPE-3 :

(22.) IF

Tee See Pee  $\rightarrow$  Means  $\rightarrow$  Drink Fruit Juice  
 See Kee Tee  $\rightarrow$  Means  $\rightarrow$  (Juice) is Sweet  
 Tee Ree Mee  $\rightarrow$  Means  $\rightarrow$  He is Intelligent.

(i) What is the code for Sweet?

$\Rightarrow$  Kee

(ii) What is the code for He?

|            |                          |
|------------|--------------------------|
| (1) See(x) | (2) See(x)               |
| Ree(x)     | Ree(x)                   |
| Mee(x)     | Mee(x)                   |
| Tee(✓)     | Tee(x)                   |
|            | can not be determine (✓) |

(3) See(x)  
 Ree(x)  
 Mee(x)  
 Tee(x)  
 can not be determine.(x)  
 Data insufficient. (✓)

(23)(i) Pic Vic Nic  $\rightarrow$  means  $\rightarrow$  Winter is cold.

(ii) To Nice Re  $\rightarrow$  means  $\rightarrow$  Summer is hot.

(iii) Re Pic boo  $\rightarrow$  means  $\rightarrow$  Winter and Summer.

(iv) Vic Tho Paa  $\rightarrow$  means  $\rightarrow$  Night are cold.

(1) What is the code for Summer?

$\Rightarrow$  [Re]

(2) Which is superfluous statements?

$\Rightarrow$  [i, iv]

---

(24.) (i) SOUND  $\rightarrow$  a b i (v) (a) SON = ?

(ii) ADDRESS  $\rightarrow$  c j m v (iii) (b) DOOR = ?

(iii) CRUX  $\rightarrow$  i k m o p (i or v)

(iv) CRONY  $\rightarrow$  j k g o v (vi)

(v) NET  $\rightarrow$  i j k v (i or iv)

(vi) CROWDY  $\rightarrow$  b k o o p v (ii)

$\Rightarrow$  (a) SON = [PK i]

(b) DOOR = [OKK v]

---

TYPE-4:

(25) IF,

Cook is called Tailor.

Tailor is called Manager.

Manager is called Teacher.

Teacher is called President.

President is called clerk.

clerk is called Principal.

Question- Who will teach in a class?

(i) President (✓)

(ii) Manager

(iii) Teacher

(iv) Principal.

(26) IF,

Gold is called Diamond.

Diamond is called Silver.

Silver is called Ruby.

Ruby is called Emerald.

Emerald is called Sapphire.

Sapphire is called Coral.

Then,

(i) Which is the green coloured Jewel?

⇒ Sapphire.

(ii) Which is cheapest Jewel?

⇒ Ruby.

TYPE-5:

Question.-(27)

| I |   |   |   |   |   | II |   |   |   |   |   |
|---|---|---|---|---|---|----|---|---|---|---|---|
|   | 0 | 1 | 2 | 3 | 4 |    | 5 | 6 | 7 | 8 | 9 |
| 0 | I | A | C | B | E | 5  | T | S | H | M | R |
| 1 | E | I | A | B |   | 6  | M | R | T | S | H |
| 2 | A | C | I | E | B | 7  | R | T | M | H | S |
| 3 | E | B | A | I | C | 8  | H | M | S | R | T |
| 4 | B | E | C | A | I | 9  | S | T | R | H | M |

(i) BARE

⇒ (a) 31, 01, 59, 33

(b) 03, 20, 65, 23

(c) 40, 13, 66, 13

(d) 03, 20, 75, 41 (✓)

(ii) HEAT

⇒ (a) 98, 29, 14, 96

(b) 85, 30, 32, 79

(c) 78, 41, 01, 56 (✓)

(d) 65, 13, 16, 69.

TYPE-6:

$$\begin{array}{ll} (28) & 25 \div 5 = 5 \quad \Rightarrow \quad 25 \times 5 = 12\textcircled{5} = 5 \\ & 8 \div 4 = 2 \quad \quad \quad 8 \times 4 = 3\textcircled{2} = 2 \\ & 9 \div 3 = \boxed{7} \quad \quad \quad 9 \times 3 = 2\textcircled{7} = 7 \\ & \text{options} - (1, 4, 6, 7) \end{array}$$

---

$$\begin{array}{ll} (29) & 8 \div 3 = 4 \quad \Rightarrow \quad 8 \times 3 = 2\textcircled{4} = 4 \\ & 7 \div 6 = 2 \quad \quad \quad 7 \times 6 = 42 = 2 \\ & 9 \div 5 = \boxed{5} \quad \quad \quad 9 \times 5 = 45 = 5 \end{array}$$

---

$$\begin{array}{ll} (30) & 2 - 2 = 2 \quad \Rightarrow \quad 2^2 - 2 = 2 \\ & 4 - 4 = 12 \quad \quad \quad 4^2 - 4 = 12 \\ & 6 - 6 = 30 \quad \quad \quad 6^2 - 6 = 30 \\ & 8 - 8 = \boxed{56} \quad \quad \quad 8^2 - 8 = 56 \end{array}$$

---

$$\begin{array}{ll} (31) & 16 - 2 = 2 \quad \Rightarrow \quad \sqrt{16} - 2 = 4 - 2 = 2 \\ & 9 - 3 = 0 \quad \quad \quad \sqrt{9} - 3 = 3 - 3 = 0 \\ & 81 - 1 = 8 \quad \quad \quad \sqrt{81} - 1 = 9 - 1 = 8 \\ & 64 - 4 = \boxed{4} \quad \quad \quad \sqrt{64} - 4 = 8 - 4 = 4 \end{array}$$

---

$$\begin{array}{ll} (32) & 54 + 43 = 1 \quad \Rightarrow \quad (5-4) + (4-3) = 1+1 = 2 \\ & 60 + 51 = 10 \quad \quad \quad (6-0) + (5-1) = 6+4 = 10 \\ & 62 + 72 = \boxed{9} \quad \quad \quad (6-2) + (7-2) = 4+5 = 9 \end{array}$$

---

$$\begin{array}{ll} (33) & 20 + 15 = 24 \quad \Rightarrow \quad (2+0) + (1+5) = 2+6 = 8 \times 3 = 24 \\ & 64 + 13 = 42 \quad \quad \quad (6+4) + (1+3) = 10+4 = 14 \times 3 = 42 \\ & 11 + 28 = \boxed{36} \quad \quad \quad (1+1) + (2+8) = 2+10 = 12 \times 3 = 36 \end{array}$$

---

$$\begin{array}{ll} (34) & 33 + 45 = 30 \quad \Rightarrow \quad (3+3) + (4+5) = 6+9 = 15 = 1+5 = 6 \times 5 = 30 \\ & 90 + 26 = 40 \quad \quad \quad (9+0) + (2+6) = 9+8 = 17 = 1+7 = 8 \times 5 = 40 \\ & 30 + 45 = \boxed{15} \quad \quad \quad (3+0) + (4+5) = 3+9 = 12 = 1+2 = 3 \times 5 = 15 \end{array}$$



$$\begin{array}{ll}
 (35.) \quad 73 + 46 = 42 & \Rightarrow (7-3) + (4+6) = 4 + 10 = 14 \times 3 = 42 \quad (\text{CGL-2014}) \\
 95 + 87 = 57 & (9-5) + (8+7) = 4 + 15 = 19 \times 3 = 57 \\
 62 + 80 = \boxed{36} & (6-2) + (8+0) = 4 + 8 = 12 \times 3 = 36
 \end{array}$$

$$\begin{array}{ll}
 (36.) \quad 98 - 64 = 14 & \Rightarrow (9+8) - (6+4) = 17 - 10 = 7 \times 2 = 14 \quad (\text{CPO-2008}) \\
 06 - 23 = 27 & (8+6) - (2+3) = 14 - 5 = 9 \times 3 = 27 \\
 40 - 11 = \boxed{8} & (4+0) - (1+1) = 4 - 2 = 2 \times 4 = 8
 \end{array}$$

$$\begin{array}{ll}
 (37.) \quad 94 + 31 + 22 = 43 & \Rightarrow 9 \times 4 + 3 \times 1 + 2 \times 2 = 36 + 3 + 4 = 43 \quad (\text{FCI-2011}) \\
 56 + 85 + 28 = 86 & 5 \times 6 + 8 \times 5 + 2 \times 8 = 30 + 40 + 16 = 86 \\
 27 + 45 + 25 = \boxed{44} & 2 \times 7 + 4 \times 5 + 2 \times 5 = 14 + 20 + 10 = 44
 \end{array}$$

$$\begin{array}{ll}
 (38.) \quad 16 \times 13 = 208 & \Rightarrow 16 \times 13 = 208 = 280 \quad (\text{CGL-2014}) \\
 14 \times 13 = 128 & 14 \times 13 = 182 = 128 \\
 15 \times 11 = \boxed{156} & 15 \times 11 = 165 = 156
 \end{array}$$

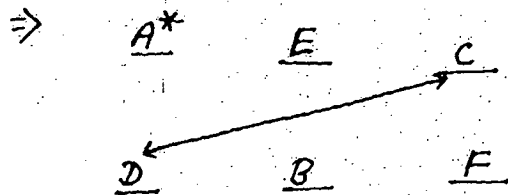
$$\begin{array}{ll}
 (39.) \quad 54 \times 45 = 40 & \Rightarrow (5 \times 4) + (4 \times 5) = 20 + 20 = 40 \quad (\text{CPO-2012}) \\
 73 \times 21 = 23 & (7 \times 3) + (2 \times 1) = 21 + 2 = 23 \\
 84 \times 52 = 42 & (8 \times 4) + (5 \times 2) = 32 + 10 = 42 \\
 62 \times 43 = \boxed{24} & (6 \times 2) + (4 \times 3) = 12 + 12 = 24
 \end{array}$$

$$\begin{array}{ll}
 (40.) \quad 5472 = 18 & \Rightarrow 5 + 4 + 7 + 2 = 18 \quad (\text{CGL-2015}) \\
 6342 = 15 & 6 + 3 + 4 + 2 = 15 \\
 7584 = 24 & 7 + 5 + 8 + 4 = 24 \\
 9236 = \boxed{20} & 9 + 2 + 3 + 6 = 20
 \end{array}$$

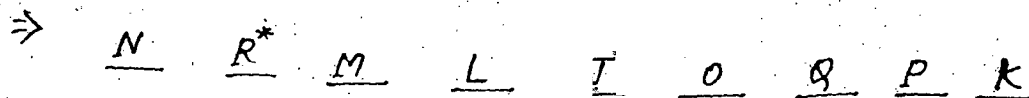
$$\begin{array}{ll}
 (41.) \quad 38 + 15 = 66 & \Rightarrow (3+8) \times (1+5) = 11 \times 6 = 66 \quad (\text{CGL-2011}) \\
 29 + 36 = 99 & (2+9) \times (3+6) = 11 \times 9 = 99 \\
 82 + 44 = \boxed{88} & (8+2) \times (4+4) = 10 \times 8 = 88
 \end{array}$$

\* SITTING ARRANGEMENT \*

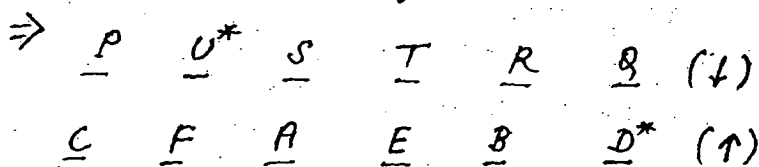
- (1) A, B, C, D, E and F are sitting in two rows. E is not at end of any row. D is second to the left of F. C is the neighbour of E sitting diagonally opposite to D. B is neighbour of F.



- (2) Nine friends J, K, L, M, N, O, P, Q and R are watching a match in a stadium sitting in one row. L is at the right of M and third to the right of N. K is at end to the row. Q sits adjacent between O and P. O sits to left of K. J is right next to the left of O.

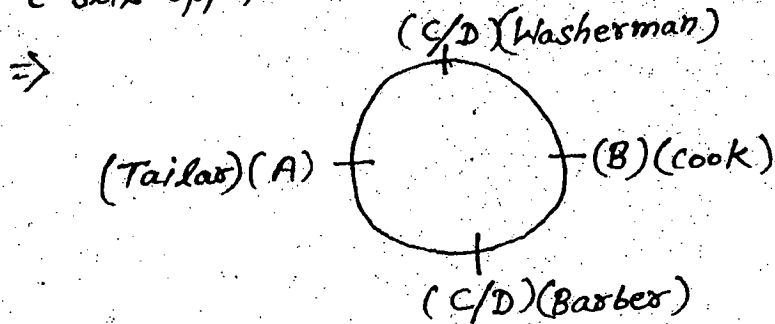


- (3) 12 Person is sitting in two row in six each. In row one P, Q, R, S, T, U sitting and all are facing south. In other row A, B, C, D, E, F are sitting and all are facing North. All the member are sitting such a way that member of one row are facing member of other row. S sits to third to right of Q. Either S or Q sitting in end of row. The one who faces Q sits second to right of E. Two person between B & F. Neither B nor F sitting in the end of row. The neighbour of B faces the person who sits third to the left of P. R & T are neighbour to each other. C sits second to the left of A. T does not face neighbour of D.



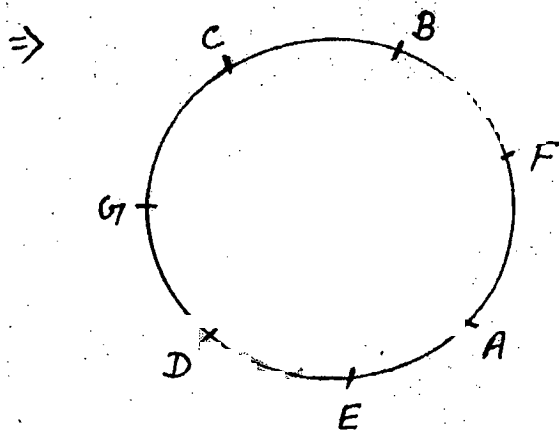
Note ⇒ Clock wise → Left  
Anti clock wise → Right

- (4.) Four persons A, B, C and D are sitting around a table and discussing their trades. A sits opposite to cook. B sits right to the barber, washerman is on the left of Tailor. C sits opposite to D. What is the trade of C?



Data Inadequate

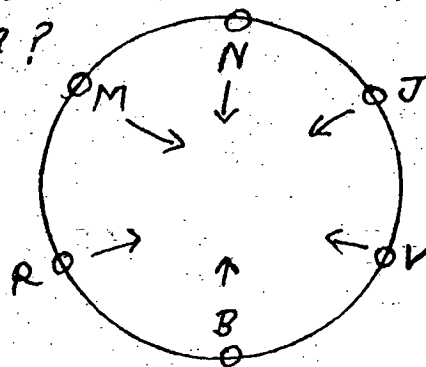
- (5.) 7 persons A, B, C, D, E, F, G are sitting in circle. E is the neighbour of A and D. G is not between F and C. F is immediate to right of A



- (6.) Six girls are standing in circle facing the centre. S is to the left of V. R is between B and M. T is between V and N. Who is to left of M?

(CGL-2016)

⇒ N



(7.) Seven persons A, B, C, D, E, F and G are standing in a straight line. D is to the right of G. C is between A and B. E is between F and D. There are three persons between G and B. Who is on the extreme left? (CGL-2015)

- (i) A (ii) B (iii) D (iv) G (✓)

⇒ G    D    E    F    B    C    A

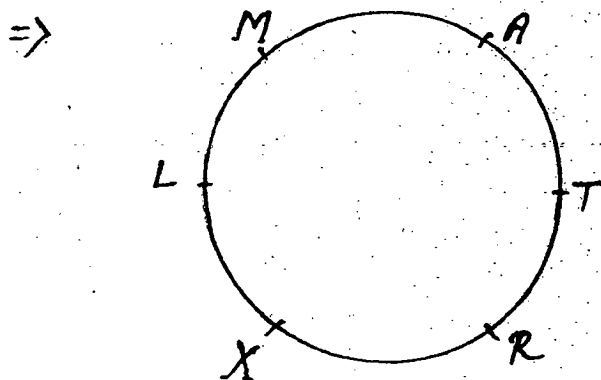
(8) On one side of a street are even numbers and on the other side are odd numbers. No. 1 is exactly in front of No. 2. My house is No. 9. From my house, a man comes up from No. 2 and knocks at the door, five doors beyond the house in front of me. What is the No. of that house? (CGL-2014)

- (A) 18 (B) 20 (✓) (C) 22 (D) 26

⇒ 1    3    5    7    9  
2    4    6    8    10    12    14    16    18    20 (✓)

(9.) Six persons R, L, M, A, T standing in a circle facing the centre of the circle. L is between X and M, A is between M and T. X is between R and L. A is standing to the left of M and right of T. Who is standing next to R? (CPO-2015)

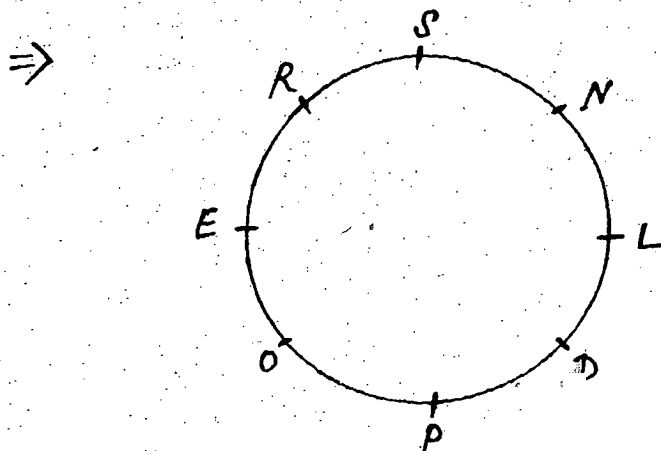
- (i) X (✓) (ii) L (iii) A (iv) M



X is standing next to R

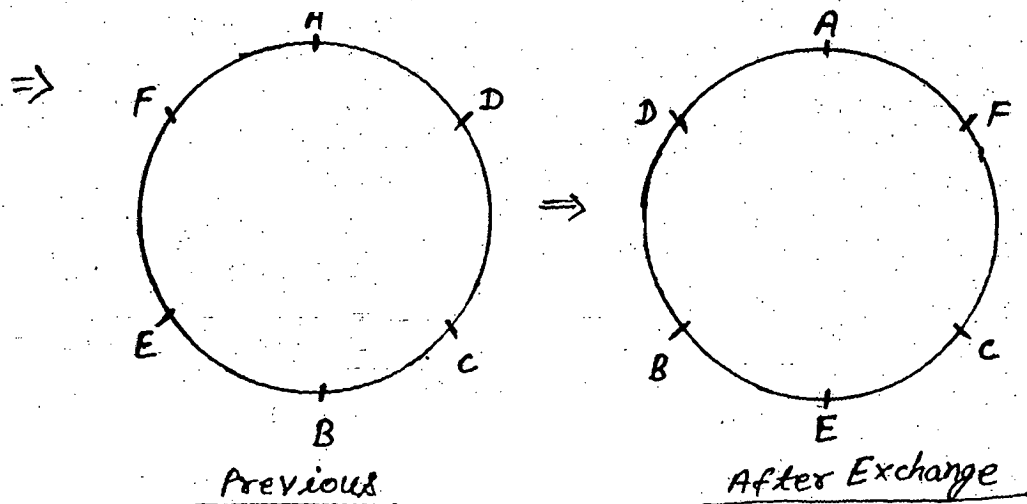
(10.) A group of eight friends S, P, L, E, N, D, O, R are seated at equidistance in a circle. S and P are seated in the North-South direction and L and E are in East-West direction. N is between S and L and is diagonally opposite to O. If R is next to E then where is D seated? (LDC-2012)

- (i) Between O and T      (ii) Between E and T  
 (iii) Between P and L (✓)      (iv) Between E and S



(11.) Six persons are sitting in a circle. A is facing B, B is to the right of E and left of C. C is to the left of D. F is to the right of A. Now D exchanges his seat with F and E with B. Who will be sitting to the left of D? (LDC-2016)

- (i) D      (ii) E      (iii) A (✓)      (iv) B



(12.) Six girls are standing in such a way that they form a circle, facing the centre. Subbu is to the left of Pappu, Revathi is between Subbu and Nisha. Aruna is between Pappu and Keertana. Who is the left of Pappu? (FCT-2012)

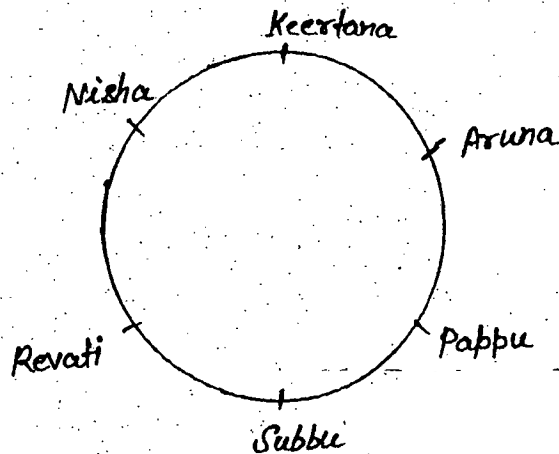
(A) Subbu (✓)

(B) Keertana

(C) Nisha

(D) Aruna

⇒



Subbu is sitting left of Pappu

(13.) There are five buses M, N, O, P, Q in a row on a road. Bus M is standing at the front and Q is standing at the back end. Bus N stands between M and O. Bus P stands between O and Q which bus is in the middle of the five? (CPO-2012)

(A) M

(B) P

(C) N

(D) O (✓)

⇒ Q   P   O   N   M

(14.) A, B, C, D and E are five schools facing towards North. A is in the middle of E and B. E is to the right of D. If C and D are at two ends. Which school is on the left of C?

(i) E

(ii) A

(iii) D

(iv) B (✓) (CPO-2012)

⇒ D   E   A   B   C

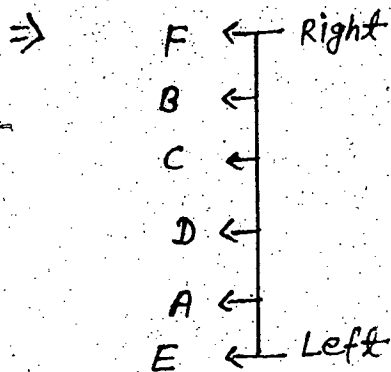
B is on the left side of C



(15.) 6 boys A, B, C, D, E, F are sitting in a row facing West. D is between A and C. B is just right of C but left of F. E is not at the right end. Who is at the right end?

(Bank PO- 2014)

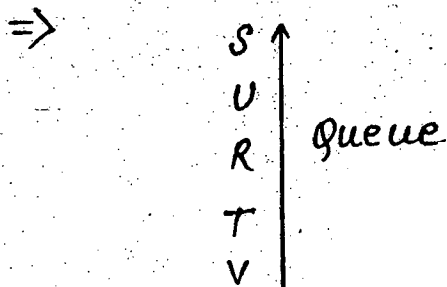
- (i) C                      (ii) F (✓)                      (iii) D                      (iv) B



F is at the right end

(16.) Five persons (R, S, T, U and V) are in a queue facing a reservation counter. Immediately behind S is U. T is standing between R and V. In between R and U, No one is there. Then, where is S standing in the queue?

(CPO- 2013)



S is standing at the 1<sup>st</sup>

(17.) Of the six members of a panel sitting in a row X is to the left of Q but on the right of P. Y is on the right of Q but is on the left of Z, Z is to the left of R. Find the members who are at the extreme?

(CGL- 2012)

- (A) QZ                      (B) XZ  
(C) PR (✓)                      (D) QY

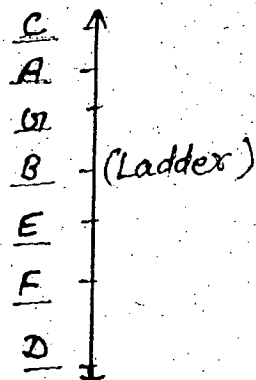




(18.) There are seven persons up on a ladder. A, B, C, D, E, F and G (not in that order). A is further up than E but is lower than C. B is in the middle. G is between A and B. E is between B and F. If F is between E and D. The person on the bottom step of the ladder will be ?

- (i) B (ii) F (iii) D (✓) (iv) E

⇒



(CGL-2011)

D is at the bottom of the ladder

(19.) In a circular arrangement of 3 boys ( $B_1, B_2$  and  $B_3$ ) and 3 girls ( $G_1, G_2$  and  $G_3$ ) sitting for a dinner successively, what will be the position of  $G_3$  and  $B_3$ , if no two girls sit together ?

(CSAT-2014)

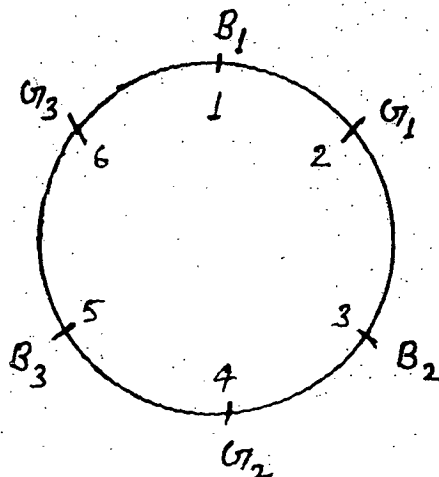
(A) 4<sup>th</sup> and 5<sup>th</sup>

(B) 5<sup>th</sup> and 6<sup>th</sup> (✓)

(C) 3<sup>rd</sup> and 4<sup>th</sup>

(D) 3<sup>rd</sup> and 6<sup>th</sup>

⇒



(20.) Six Persons A, B, C, D, E, F sit in 2 rows, 3 in each. If E is not at any end, D is second to left of F, C is neighbour of E and is sitting diagonally opposite to D and B is neighbour of F, who will be opposite to B?

(i) A

(ii) E (✓)

(FCI-2011)

(iii) C

(iv) D

⇒ 

|          |          |          |
|----------|----------|----------|
| <u>A</u> | <u>E</u> | <u>C</u> |
| <u>D</u> | <u>B</u> | <u>F</u> |

E is in opposite of B

(21.) A group of friends are sitting in an arrangement on each at the corner of an octagon. All are facing the centre. Mahima is sitting diagonally opposite Rama, who is on Sushma's right. Ravi is next to Sushma and opposite Girdhar, who is on Chandra's left. Savitri is not on Mahima's right but opposite Shalini. Who is on Shalini's right?

(CGL-2014)

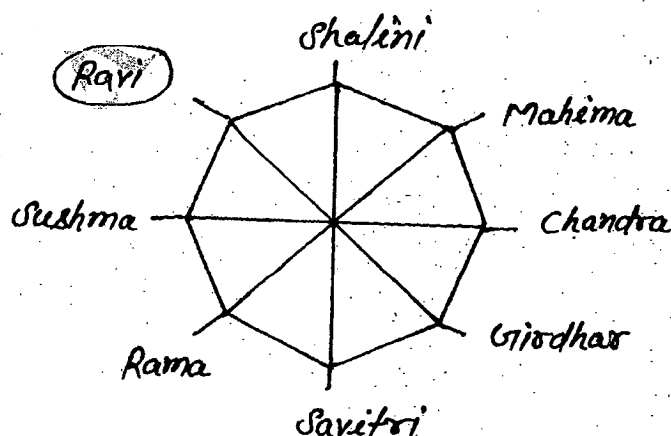
(A) Ravi (✓)

(B) Mahima

(C) Girdhar

(D) Rama

⇒



Ravi is sitting is right of Shalini.

## \* RANKING ARRANGEMENT \*

(1.) In the row of 39 boys. Rahul is 19<sup>th</sup> from the bottom, so what is his rank from the top?

$$\Rightarrow \text{Total boys} = \text{Top rank} + \text{Bottom rank} - 1$$

$$39 = \text{Top} + 19 - 1$$

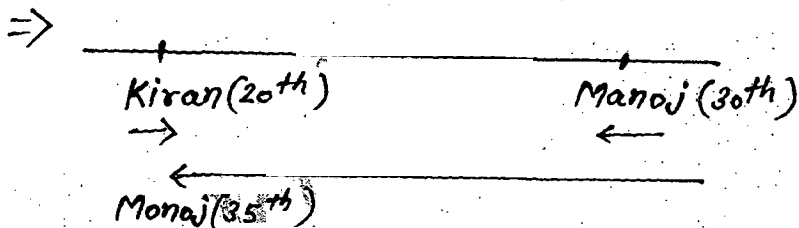
$$\boxed{\text{Top} = 21^{\text{st}}}$$

Note  $\Rightarrow$   $\overset{x}{\text{Position from one End}} + \overset{y}{\text{Position from other End}} - 1$   
 $= \text{Total}$

$$\Rightarrow \boxed{(x + y) - 1 = \text{Total/Number}}$$

(2.) In a row of men, Manoj is 30<sup>th</sup> from the right and Kiran is 20 from the left. When they interchange their position. Manoj becomes 35 from the right. What is the total number of men in the row?

(A) 45      (B) 44      (C) 54 (✓)      (D) 34



$$\text{Total No. of Men} = 35 + 20 - 1 = \boxed{54}$$

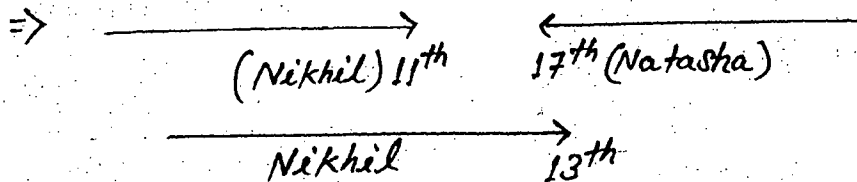
(3.) In a class Mohan is at the 15<sup>th</sup> position from top and 50<sup>th</sup> from bottom. How many total students are there in the class?

(A) 65      (B) 64 (✓)      (C) 66      (D) 67

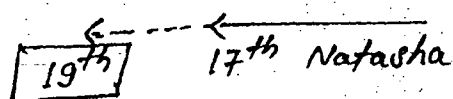
$$\Rightarrow \text{Total No. of students} = 50 + 15 - 1$$
$$= 65 - 1 = \boxed{64}$$

(4.) In a row of children. Nikhil is 11<sup>th</sup> from left and Natasha is 17<sup>th</sup> from right. When they exchange their places, Nikhil will be 13<sup>th</sup> from left. Which one of the following will be the new position of Natasha from right?

- (A) 21<sup>th</sup>      (B) 29<sup>th</sup>      (C) 11<sup>th</sup>      (D) 19<sup>th</sup> (✓)



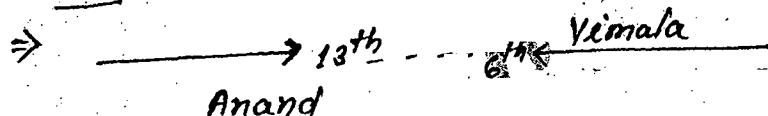
∴ Natasha also moves two steps.



Natasha become 19<sup>th</sup> from right

(5.) Anand ranks 13<sup>th</sup> in a class of 33 (thirty three). There are five students below Vimala rank wise. How many students are there between Anand and Vimala?

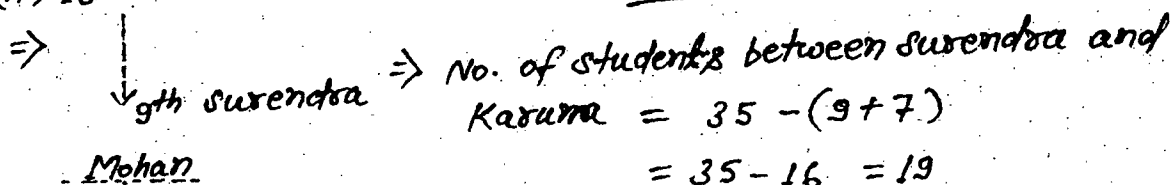
- (A) 14 (✓)      (B) 12      (C) 16      (D) 15



$$\begin{aligned} \text{Student between A \& V are} &= 33 - (13 + 6) \\ &= 33 - 19 = \boxed{14} \end{aligned}$$

(6.) In a class of 35, first year students, Karuna is placed 7<sup>th</sup> from the bottom where as Surendra is placed 9<sup>th</sup> from the top. Mohan is placed exactly between the two. What is Karuna's position from Mohan?

- (A) 13<sup>th</sup>      (B) 9<sup>th</sup>      (C) 10<sup>th</sup> (✓)      (D) 12<sup>th</sup>



$$\begin{aligned} \text{No. of students between Surendra and Karuna} &= 35 - (9 + 7) \\ &= 35 - 16 = 19 \end{aligned}$$

$$\text{Middle position} = 10^{\text{th}}$$

So Karuna is at 10<sup>th</sup> position from Mohan.

(7.) In a row Mohan is at the 12<sup>th</sup> position from top and Rakesh is at the 18<sup>th</sup> from bottom and there is five person between them. Then find out maximum and minimum number of person?

$$\Rightarrow \text{(i) Maximum} = 12 + 18 + 5 = \boxed{35}$$

$$\text{(ii) Minimum} = (12 - 5 - 2) + 18 = \boxed{23}$$

OR

$$5 \times 2 + 2 = 12 \quad \text{then } 35 - 12 = 23$$

Trick Note  $\Rightarrow$

$$\text{Maximum} = S_R + L_R + \underline{B/W} \rightarrow \text{between}$$

$$\text{Minimum} = S_R + L_R - \text{Between No.} - 2$$

(8.) There were 27 monkeys seating on road fencing. The monkey named 'Jittu' was on 11<sup>th</sup> position from right. Its position from left was -

(A) 16<sup>th</sup>      (B) 17<sup>th</sup> (✓)      (C) 18<sup>th</sup>      (D) 15<sup>th</sup>

$$\Rightarrow \text{The Position of Jittu from left is } (27 - 11 + 1) = \boxed{17^{\text{th}}}$$

(9.) Kartik is ranked 14<sup>th</sup> from the top and 27<sup>th</sup> from the bottom among those who passed in the the class test. If 6 student failed, what was the total number of students who appeared in the examination?

(A) 45      (B) 38      (C) 46 (✓)      (D) 50

$$\Rightarrow \text{Total students} = (14 + 27 - 1) + 6 = \boxed{46}$$

(10) In a row of boys, if A who is 10<sup>th</sup> from the left and B who is 9<sup>th</sup> from the right interchange their positions, A becomes 15<sup>th</sup> from left. How many boys are there in the row?

- (A) 23 (✓) (B) 27 (C) 28 (D) 31

⇒ According to question B is 9<sup>th</sup> from the right end and 15<sup>th</sup> from the left end.

Therefore, total number of boys in the row  
 $= 9 + 15 - 1 = \boxed{23}$

(11) In a row of 16 girls, when Hema was shifted by two places towards the left become 7<sup>th</sup> from the left end. What was her earlier position from the right end?

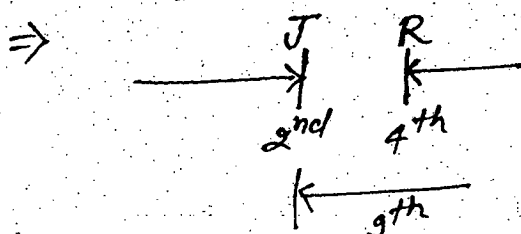
- (A) 7<sup>th</sup> (B) 8<sup>th</sup> (✓) (C) 9<sup>th</sup> (D) 10<sup>th</sup>

⇒ The original position of Hema from the left = 9<sup>th</sup>

Therefore, her position from right  
 $= 16 - 9 + 1 = 8^{\text{th}}$

(12) In a row of children, Ravi is fourth from right and Jitendra is second from left. When they interchange positions Ravi is ninth from right. What will be Jitendra position from left?

- (A) 5<sup>th</sup> (B) 6<sup>th</sup> (C) 7<sup>th</sup> (✓) (D) 8<sup>th</sup>



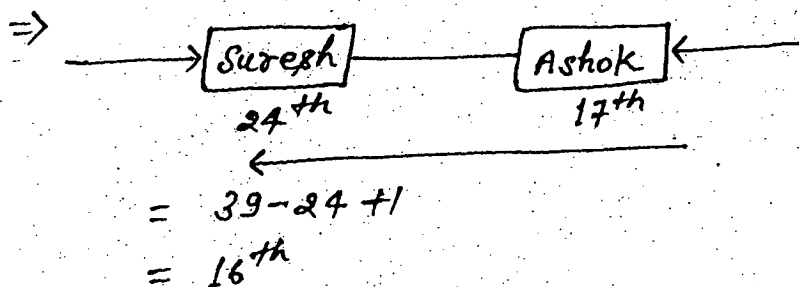
$$\text{Total} = 9 + 2 - 1 = 10$$

$$\text{Jitendra} = 10 - 4 + 1 = \boxed{7^{\text{th}}}$$



(13) Suresh is 7 ranks ahead of Ashok in the class of 39 students. If Ashok's rank is 17<sup>th</sup> from the last. What is Suresh's rank from the start?

- (A) 16<sup>th</sup> (✓) (B) 23<sup>th</sup> (C) 24<sup>th</sup> (D) 15<sup>th</sup>



(14) Sushmita ranks seventh from the top and 28<sup>th</sup> from bottom. How many students are there in the class?

- (A) 34 (✓) (B) 35 (C) 28 (D) 21

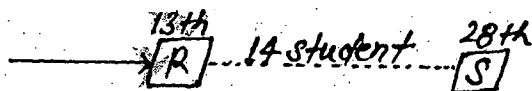
⇒ Total no. of the students in the class

$$= 7 + 28 - 1 = \boxed{34}$$

(15) Ramesh ranks 13<sup>th</sup> in a class of 33 students. There are 5 students below Suresh rank wise. How many students are there between Ramesh and Suresh?

- (A) 12 (B) 14 (✓) (C) 15 (D) 16

⇒ The rank of Suresh = 28<sup>th</sup>



(16) In a class Sushmita got 11<sup>th</sup> rank and she was 31<sup>st</sup> from the bottom of the list of girls passed. Three girls did not take the examination and one failed. What is the total strength of the class?

- (A) 32 (B) 42 (C) 45 (✓) (D) 46

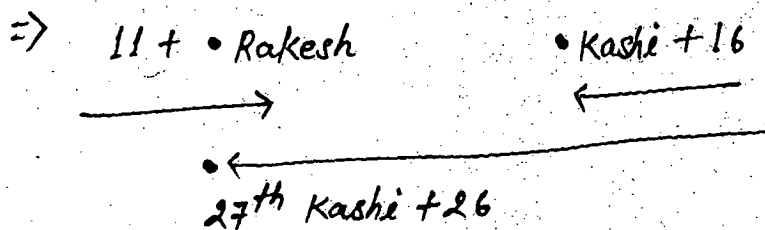
⇒ Total no. of the girls passed =  $11 + 31 - 1 = 41$

Now, total number of girls =  $41 + 3 + 1 = \boxed{45}$



(17.) In a row of students Rakesh is  $12^{\text{th}}$  from the left and Kashi is  $17^{\text{th}}$  from right. When they interchange their positions Kashi becomes  $27^{\text{th}}$  from the right. How many students are there between Kashi and Rakesh.

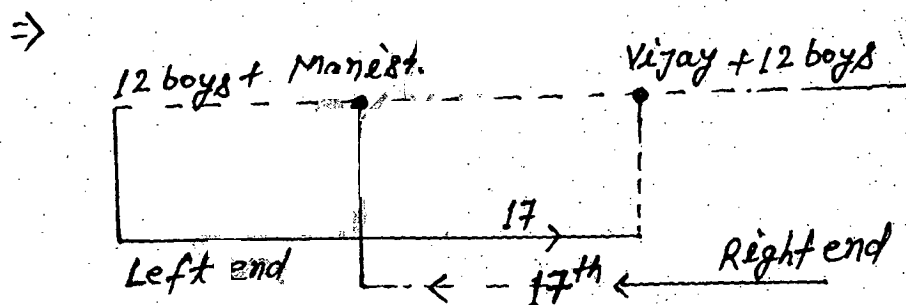
- (A) 9 (✓) (B) 12 (C) 7 (D) 10



No. of students between Rakesh and Kashi  
 $= 26 - 17 = \boxed{9}$

(18.) Vijay is  $17^{\text{th}}$  from the left and of a row of 29 boys and Manish is  $17^{\text{th}}$  from the right end in the same row. How many boys are there between them in the same row?

- (A) 5 (B) 6 (C) 3 (✓) (D) Data Inadequate



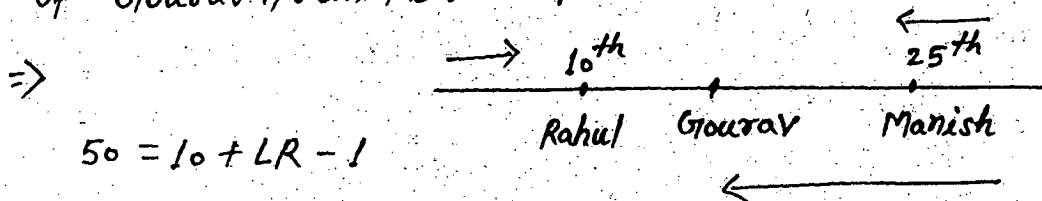
It is clear from the diagram, that there are 3 boys between Manish and Vijay.

(19.) In a sport competition, the position of one of the players is  $8^{\text{th}}$  from top and  $84^{\text{th}}$  from the bottom. What is the total no. of the competitors?

- (A) 93 (B) 91 (✓) (C) 89 (D) 88

$\Rightarrow$  Total No. of Competitors  $= 8 + 84 - 1 = \boxed{91}$

(20.) In a row Rahul is 10<sup>th</sup> from the front and Manish is 25<sup>th</sup> from the behind and Gourav is just middle in the two. If total 50 in the row than what position of Gourav from the last? (CGL-2011)



$$50 = 10 + LR - 1$$

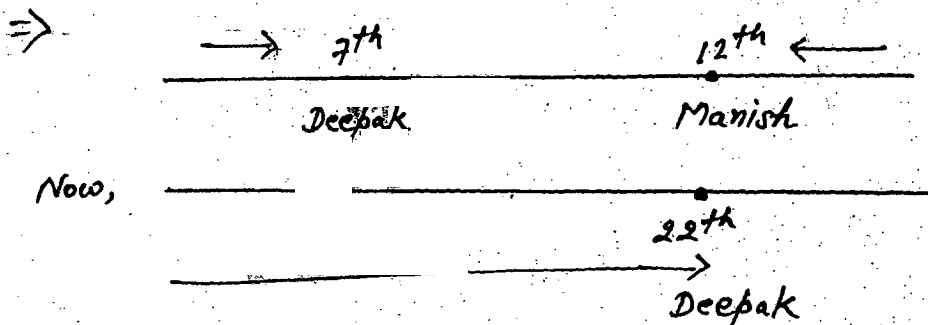
$$50 = 9 + LR (\text{last rank})$$

$$\text{Rahul } LR = 41$$

$$\text{Gourav} = \frac{41 + 25}{2} = \frac{66}{2} = \boxed{33}$$

(21.) In a row of boys Deepak is 7<sup>th</sup> from the left and Manish is 12<sup>th</sup> from right if they interchange their position themselves than Deepak become 22<sup>th</sup> from left so how many boys in row? (CPO-2009)

(A) 27      (B) 33      (C) 32      (D) can't be determine



$$\text{Total No. of boys} = 22 + 12 - 1 = \boxed{33}$$

(22.) In the row of 40 boys. Rahul is 11<sup>th</sup> from the bottom so what is his rank from top? (FCT-2011)

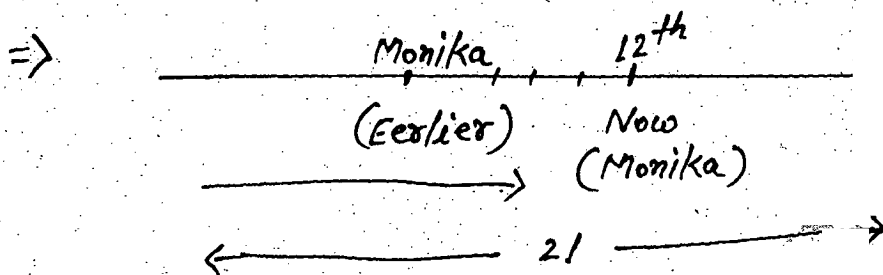
⇒  $40 = \text{Top} + 11 - 1$

$\text{Top} = 30^{\text{th}}$

(23.) In the row of 21 girls. When Monika is shifted to four place to right. She become 12<sup>th</sup> from left.

So what was her earlier rank from right? (UPSI-2011)

- (A) 14 (✓) (B) 13 (C) 10 (D) 6

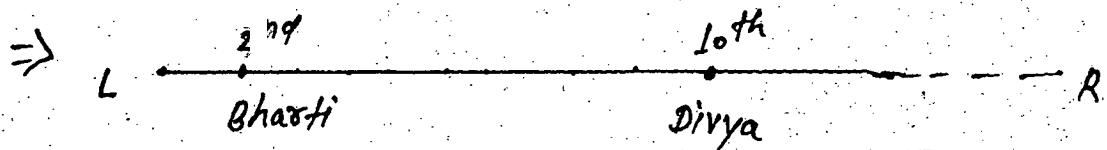


$$T_n = S_R + L_R - 1$$

$$21 = 8 + L_R - 1$$

$$L_R = \boxed{14^{th}}$$

(24.) In a class of 42 girls. Bharti is 8<sup>th</sup> Rank ahead of Divya whose ranks 10<sup>th</sup> in the same class. What is Bharti rank from last. (CSAT-2015)



$$T_n = L_R + S_R - 1$$

$$42 = L_R + 2 - 1$$

$$L_R = \boxed{41^{th}}$$

\*

## DICE

\*

Dice is a 3 dimensional object, which when seen from any direction shows its 3 surfaces to us. Dice are of two types. —

- (i) Standard dice      (ii) Ordinary dice

(i) Standard dice — In a standard dice, sum of its any two opposite sides is 7, and no two adjacent sides can give sum of 7. Such dice is known as standard dice. Like —



$$1 + 3 = 4$$

$$1 + 5 = 6$$

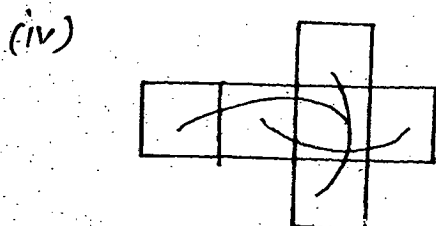
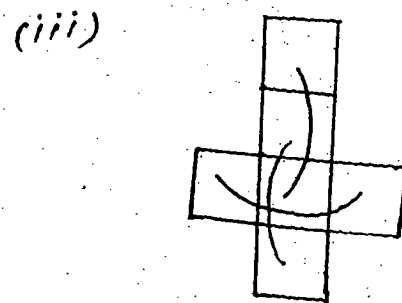
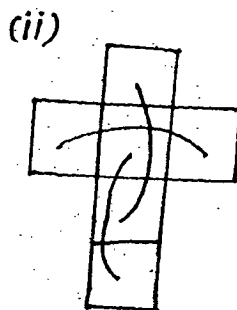
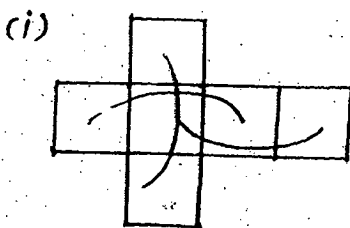
$$5 + 3 = 8$$

Note  $\Rightarrow$  In a standard dice 1 is always opposite to 6, 2 is opposite to 5 and 3 is opposite to 4.

(ii) Ordinary dice — In a ordinary dice, any two opposite surfaces can't give sum of 7, but its adjacent sides may sum to 7, such dice is known as ordinary dice. Like —



⊛ Explanatory figure of dice — There is four important Position of explanatory figure. —



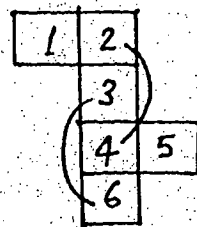
# Type-1 :

(1.)

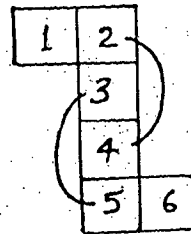
1-5

3-6

2-4



(2.)

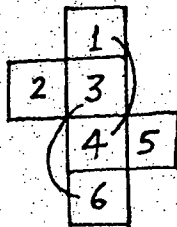


1-6

3-5

2-4

(3.)

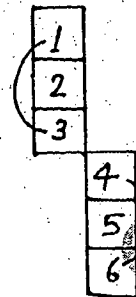


1-4

3-6

2-5

(4.)

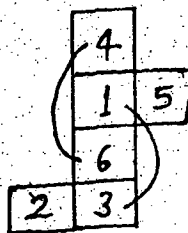


4-6

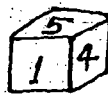
1-3

2-5

(5.)



(i)



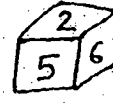
(✓)

(ii)



(x)

(iii)



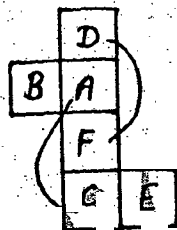
(x)

(iv)



(x)

(6.)

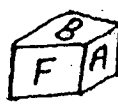


(i)



(x)

(ii)



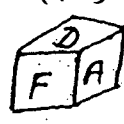
(✓)

(iii)



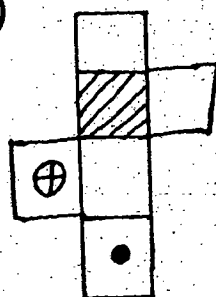
(x)

(iv)



(x)

(7.)



(i)



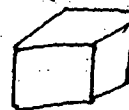
(✓)

(ii)



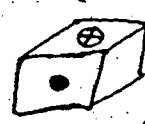
(x)

(iii)



(x)

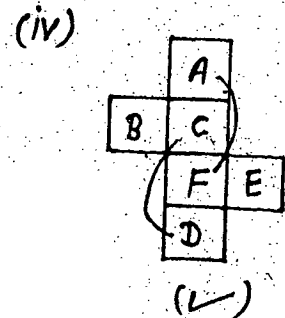
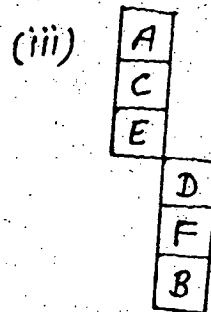
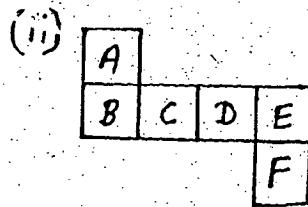
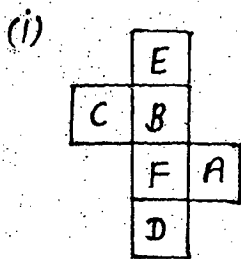
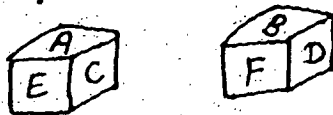
(iv)



(✓)

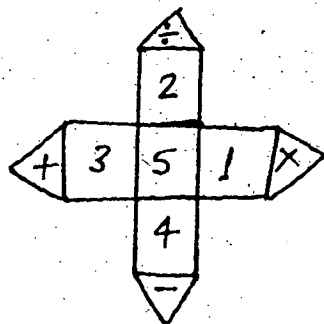
Both (i) & (ii)

(8.) Two positions of a dice are given, find out which is correct positions?



option (iv) is correct

(9.) Which will be on the opposite surface to 5?



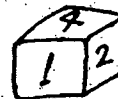
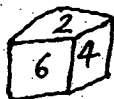
$\Rightarrow$  opposite surface to 3 = 1

2 = 4

$\therefore$  opposite surface to 5 =

### TYPE-2

(10.) Four positions of a dice are given, find out which digit will be opposite to 6?



(i) 3

(ii) 4

(iii) 5

(iv) 1 (✓)

$\Rightarrow$  6 = 1 ~~2~~ ~~3~~ ~~4~~ ~~5~~



(11.) Four positions of a dice are given, find out which digit will be opposite to 3?



$$\Rightarrow 3 = \cancel{1}, \cancel{2}, \boxed{4}, \cancel{5}, \cancel{6}$$

4 is opposite to 3

(12.)  $\begin{smallmatrix} 4 \\ 3 \end{smallmatrix} \begin{smallmatrix} 2 \end{smallmatrix}$      $\begin{smallmatrix} 4^3 \\ 5 \end{smallmatrix}$      $\begin{smallmatrix} 4^6 \\ 3 \end{smallmatrix}$      $\begin{smallmatrix} 3^2 \\ 1 \end{smallmatrix}$     which digit will be opposite to 4?

$\Rightarrow$  Which digit can't be opposite to 4

$$4 = 5, 6, 3, 2$$

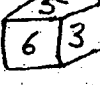
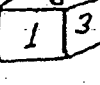
$\therefore 1$  is opposite to 4.

(13.)  $\begin{smallmatrix} 4^5 \\ 6 \end{smallmatrix}$      $\begin{smallmatrix} 4 \\ 1 \end{smallmatrix} \begin{smallmatrix} 5 \end{smallmatrix}$      $\begin{smallmatrix} 6^3 \\ 5 \end{smallmatrix}$      $\begin{smallmatrix} 1^2 \\ 4 \end{smallmatrix}$     which digit will be opposite to 3?

$\Rightarrow$  Which digit can't be opposite to 4.

$$4 = 5, 6, 1, 2$$

$\therefore 3$  is opposite to 4.

(14.)     Which digit will be opposite to 6?

$$\Rightarrow \begin{smallmatrix} 3 \\ 4 \end{smallmatrix} \begin{smallmatrix} 5 \end{smallmatrix} \quad \begin{smallmatrix} 4 \\ 2 \end{smallmatrix} \begin{smallmatrix} 5 \end{smallmatrix} \quad \begin{smallmatrix} 5 \\ 6 \end{smallmatrix} \begin{smallmatrix} 3 \end{smallmatrix} \quad \begin{smallmatrix} 6 \\ 1 \end{smallmatrix} \begin{smallmatrix} 3 \end{smallmatrix}$$

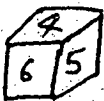
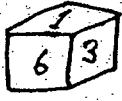
$$6 = \cancel{1} \cancel{2} \cancel{3} \cancel{4} \Rightarrow \boxed{6 = 5}$$

(15.)    Which is opposite to A?

$$\Rightarrow \begin{smallmatrix} A \\ B \end{smallmatrix} \begin{smallmatrix} C \end{smallmatrix} \quad \begin{smallmatrix} C \\ D \end{smallmatrix} \begin{smallmatrix} E \end{smallmatrix} \quad \begin{smallmatrix} E \\ B \end{smallmatrix} \begin{smallmatrix} F \end{smallmatrix}$$

$$E = \cancel{F} \cancel{D} \cancel{C} \neq A \Rightarrow \boxed{A = E}$$



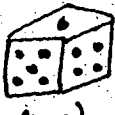
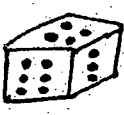
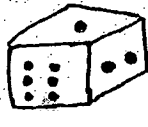
(16.)   Which digit is opposite to 3?

$\Rightarrow$ 

|   |   |   |
|---|---|---|
| 6 | 4 | 5 |
| 6 | 1 | 3 |

 5 is opposite to 3.

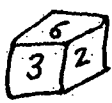
(17.) Which is standard dice in the following given dice?

(A)  (B)  (C)  (D) 

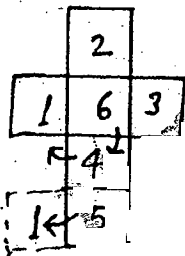
$\Rightarrow$  option (A) is correct.

Note  $\Rightarrow$  No two adjacent sides can give sum of 7.

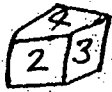
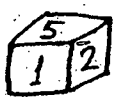
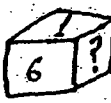
Type-3:

(18.)    = 1

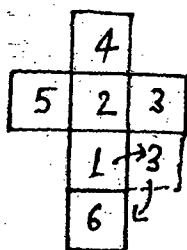
$\Rightarrow$



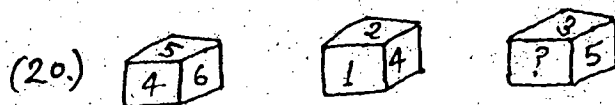
Note  $\Rightarrow$  Take a common number and move clock-wise.

(19.)   

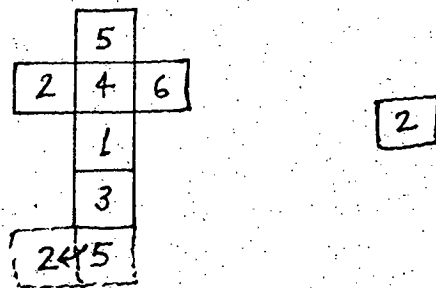
$\Rightarrow$



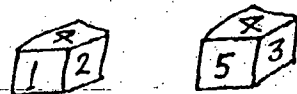
3



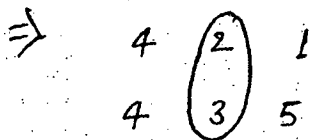
⇒



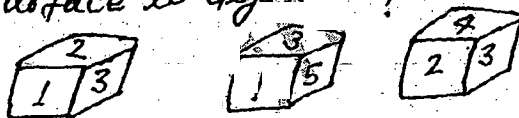
(21.) Two positions of a dice are shown below. If 3 is bottom then which number will be at top? (CGL-2013)



(A) 4 (B) 5 (C) 2 (✓) (D) 1



(22.) In the below given dice which digit will be on the opposite surface to digit 2? (CGL-2014)

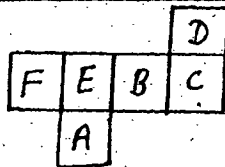


(A) 1 (B) 5 (✓) (C) 4 (D) 5

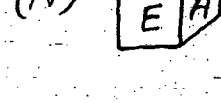
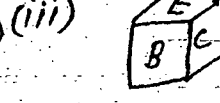
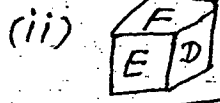
⇒ 3 = 1, 2, 4, 6

6 is opposite to 3

(23.)



(CGL-2015)



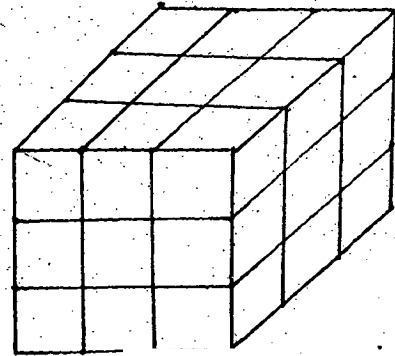
⇒ D is opposite to F, C is opposite to E and A is opposite to B.

## \* CUBES AND CUBOID \*

1. CUBE : A cube is a three dimensional object whose length, breadth and height are all of equal measures.

Properties :-

- (i) A cube will always have six surfaces.
- (ii) A cube will always have twelve edges.
- (iii) A cube will always have eight corners.



2. CUBOID : A cuboid is such a three dimensional object whose length, breadth and height will always be of unequal or different measures.  
A cuboid has other properties are similar to a cube.

⊗ Total No. of smaller cube =  $[\text{Layer}]^3$

$$[27] = [3]^3$$

⇒ (i) 3 Layers

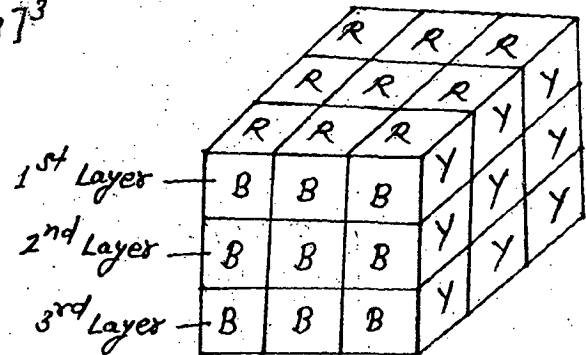
(ii) 27 smaller cube = 3 Layer

216 smaller cube = 6 Layer

(iii)  $15 \div 5 = 3 \text{ Layer}$

$20 \div 4 = 5 \text{ Layer}$

(iv)  $4 \times 4 \times 4 \div 1 \times 1 \times 1 = 4 \div 1 = 4 \text{ Layer.}$



(1) 3 side painted :

$$\text{No. of cubes} = [8]$$

(3) 1 side painted :

$$\begin{aligned} \text{No. of cubes} &= 6(n-2)^2 \\ &= 6(3-2)^2 = [6] \end{aligned}$$

$$n = \text{no. of Layer}$$

(2) 2 side painted :

$$\begin{aligned} \text{No. of cubes} &= 12(n-2) \\ &= 12(3-2) \\ &= 12 \times 1 = [12] \end{aligned}$$

(4) No side painted :

$$\begin{aligned} \text{No. of cubes} &= (n-2)^3 = (3-2)^3 \\ &= [1] \end{aligned}$$

\* How many cubes are there which is —

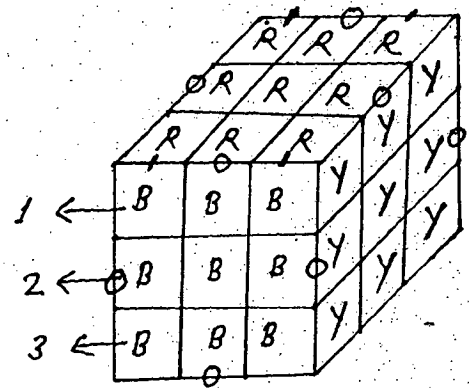
(1) Three Sides painted: I. 4

II. X

III. 4

8

$$I = III$$



Layer = 3

(2) Two sides painted: I. 4

II. 4

III. 4

12

(3) One side painted: I. 1

II. 4

III. 1

6

(4) No side painted: I. X

II. 1

III. X

1

(5) How many sides which are coloured by only Red and Blue?

I. 2

II. X

III. 2

4

RBXXXX

Question:- A solide cube of sides 8 cm. has been painted red blue and black on the pair of opposite faces. It is cut into smaller cube in 2 cm. So, —

(1) How many cubes are there which is three face painted?

(2) How many cubes which is two face painted?

(3) one face painted?

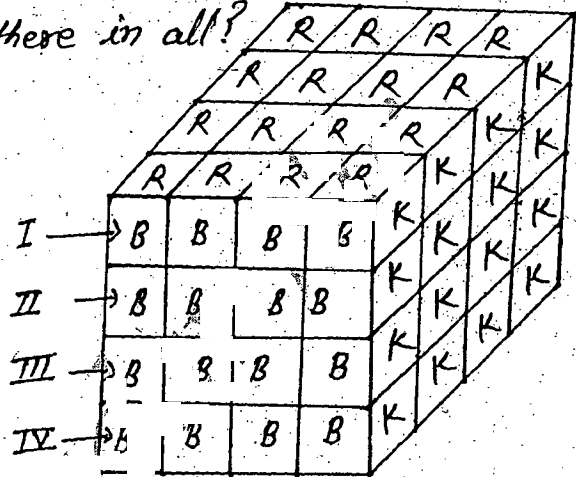
- (4.) No face painted ?  
 (5.) How many cubes which one face red and all other unpainted?  
 (6.) How many cubes which one face blue; one red and all other are unpainted?  
 (7.) Two face painted black ?  
 (8.) One face red and one blue other may be painted or not?  
 (9.) How many cubes are there in all?

$\Rightarrow$  Total Layer =  $8 \div 2$   
 $n = 4$  Layer.

(1.) Three face Painted -:

$$\begin{array}{l} \text{I} - 4 \\ \text{II} - x \\ \text{III} - x \\ \text{IV} - 4 \end{array} \quad \begin{array}{l} \text{first layer} \\ \& \\ \text{Last layer are} \\ \text{similar} \end{array}$$

$$\boxed{8}$$



(2.) Two face Painted -:

$$\begin{array}{l} \text{I} - 0 \\ \text{II} - 4 \\ \text{III} - 4 \\ \text{IV} - 0 \end{array} \quad \begin{array}{l} \text{OR} \\ 12(n-2) \\ = 12(4-2) = 12 \times 2 \\ = \boxed{24} \end{array}$$

$$\boxed{24}$$

(3.) One face Painted -:

$$\begin{array}{l} \text{I} - 4 \\ \text{II} - 8 \\ \text{III} - 8 \\ \text{IV} - 4 \end{array} \quad \begin{array}{l} \text{OR} \\ 6(n-2)^2 \\ = 6(4-2)^2 = 6(2)^2 \\ = 6 \times 4 = \boxed{24} \end{array}$$

$$\boxed{24}$$

(4.) No face Painted :  $\text{OR} \quad (n-2)^3 = (4-2)^3$

$$\begin{array}{l} \text{I} - x \\ \text{II} - 4 \\ \text{III} - 4 \\ \text{IV} - x \end{array} \quad \begin{array}{l} = (2)^3 = \boxed{8} \end{array}$$

$$\boxed{8}$$

(5) RXXXXX

(one face red and all other unpainted)

I - 4

II-X

III-X

IV-4

④

(6.)  $BRXXXX$

(one face blue, one red and all other unpainted)

I - 4

II-X

III-X

IV-4

8

(7) Two face painted block

KKXXXX)

$\Rightarrow$  NIL

④ R B L L L L

RB X X X X

I-8

II-X

**III-X**

IV - 8

16

{ one face Red & one blue other }  
{ may be painted or not }

(9) Total Number of smaller cube =  $(n)^3$

$$= (4)^3 = 4 \times 4 \times 4$$
$$= \boxed{64}$$

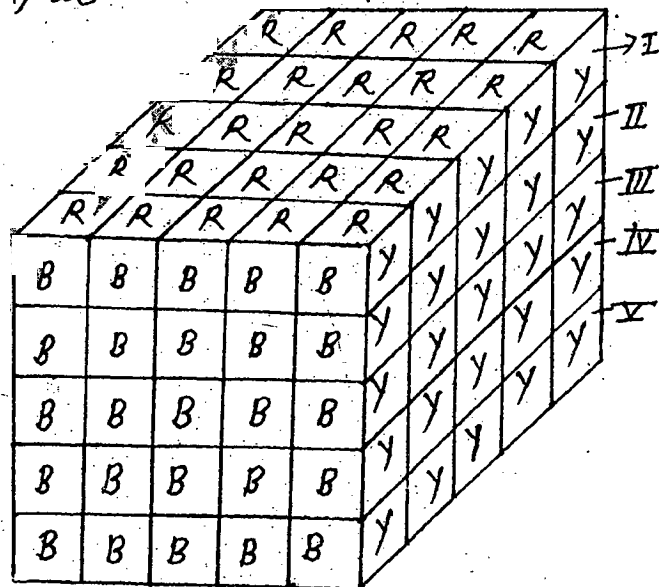


Question: A cube is coloured one face is red and green is opposite face and a face is blue and yellow on face adjacent to blue side and other face unpainted then it is cut into 125 cubes.

- (1.) How many cubes are there which is three face painted?
- (2.) How many cubes which is two face painted?
- (3.) How many cubes which is one face painted?
- (4.) No face painted?
- (5.) How many cubes which one face red and all other unpainted?
- (6.) How many cubes are there which have at least two coloured faces?
- (7.) How many cubes which one face blue and one face red or green and other face no painted? (SBI-PO-2012)

⇒ Total cubes = 125

$$\therefore \text{Layer} = \sqrt[3]{125} = 5$$



|          |          |
|----------|----------|
| (1) I- 1 | (2) I- 8 |
| II- X    | II- 1    |
| III- X   | III- 1   |
| IV- X    | IV- 1    |
| V- 1     | V- 8     |
| <u>2</u> | <u>9</u> |

|           |           |
|-----------|-----------|
| (3) I- 16 | (4) I- X  |
| II- 8     | II- 16    |
| III- 8    | III- 16   |
| IV- 8     | IV- 16    |
| V- 16     | V- X      |
| <u>56</u> | <u>48</u> |

|           |
|-----------|
| (5) RXXXX |
| I- 16     |
| II- X     |
| III- X    |
| IV- X     |
| V- X      |
| <u>16</u> |

|           |             |
|-----------|-------------|
| (6) I- 9  | (7) B GXXXX |
| II- 1     | B RXXXX     |
| III- 1    | I- 4        |
| IV- 1     | II- X       |
| V- 9      | III- X      |
| <u>21</u> | IV- X       |
|           | V- 4        |
|           | <u>8</u>    |



Directions (8-14): Read the informations given below and answer the questions based on it.

- (i) A dice have 6 sides and each side has a number (1 to 6) and a colour.
- (ii) 3, 6 lies correspondingly opposite to 1, 5.
- (iii) When you see number 2 and Red surface then you find that the green colour and number 4 are exactly opposite to it and the white colour and number 1 lies on the top.
- (iv) yellow and Blue are in front of each other.
- (v) When you observe the pink colour in front of you, you find number 2 on the top, number 3 on your right and Blue colour on your left number? (CAT-2013)
- (8.) Which of these following four colours touch the blue coloured surface?
- (1) Red, green, Pink, Yellow
- (2) Green, Pink, yellow, White
- (3) Red, White, Pink, Green. (✓)
- (4) Can't be determined (5) None of these.
- 

(9.) When digit 1 is in front of you then which of the colours definitely lies at the bottom?

- (1) Green (2) Blue (3) Yellow
- (4) Can't be determined (5) None of these. (✓)
- 

(10.) Which colour is represented by digit 5?

- (1) Blue (2) yellow (3) Pink (✓)
- (4) Can't be determined. (5) None of these.

(11.) Which of the following colour is represented by digit 3?

(1.) Blue (2.) Yellow (✓) (3.) Pink

(4.) Can't be determined (5.) None of these.

---

(12.) When green colour is in front of you then which number is definitely on the top?

(1.) 1 (2.) 5 (✓) (3.) 6

(4.) can't be determined (5.) None of these

---

(13.) When the number 4 is at the top then which colour would be in front of you?

(1.) Either Blue or Yellow

(2.) Either Pink or Red.

(3.) Either Pink or white (✓)

(4.) Either white or green.

(5.) Either white or Red.

---

(14.) If digit 6 is in front of you then which colour would be definitely on the opposite surface to this surface?

(i) Green

(ii) Red

(iii) Blue

(iv) can't be determined (v) None of these. (✓)

---

=> 6 (Blue) is in opposite to the 3 (yellow)

⇒ Direction (Q-14): Position for cube will be as follows:

| Surface      | No. | Colour | Surface       | No. | Colour |
|--------------|-----|--------|---------------|-----|--------|
| Top          | 1   | White  | Top           | 2   | Red    |
| Bottom       | 5   | Pink   | Bottom        | 4   | Green  |
| Front        | 2   | Red    | Front         | 5   | Pink   |
| Back         | 4   | Green  | Back          | 1   | White  |
| Left         | 3   | Yellow | Left          | 6   | Blue   |
| Right        | 6   | Blue   | Right         | 3   | Yellow |
| Position - I |     |        | Position - II |     |        |

(15.) A rectangular block with dimensions  $6 \times 4 \times 2$  cms, has been cut and changed into smaller cubes of dimension 1 cm. each. calculate how many total number of smaller cubes would be obtained? (CGL-2014)

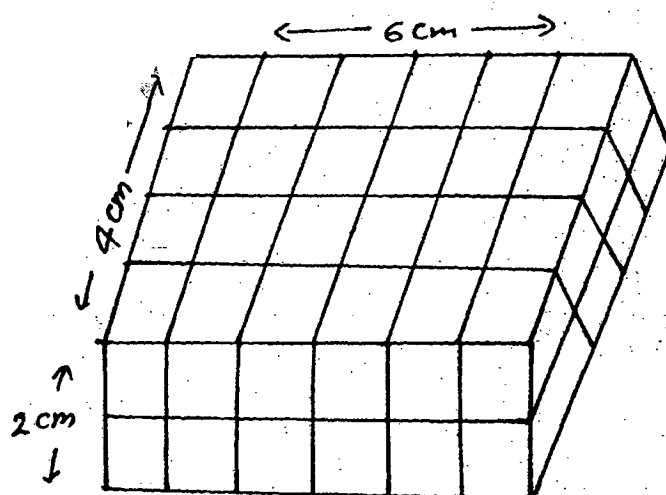
(i) 64

(ii) 48 (✓)

(iii) 32

(iv) 96

(v) None of these.



From the above figure it is clear that the each layer there are 24 cubes.

## \* — CALENDAR — \*

(\*) 1 Normal Year = 365 days

OR

52 Weeks + 1 odd day

(\*) 1 Leap Year = 366 days

OR

52 Weeks + 2 odd days

Note  $\Rightarrow$  (1.) How many leap years in 400 years?  
 $\Rightarrow$  97 times

(2.) How many leap years in a hundred year?  
 $\Rightarrow$  Minimum 24 and Maximum 25 times.

|                               |    |                                   |          |
|-------------------------------|----|-----------------------------------|----------|
| (*) 1801 - 1900 $\rightarrow$ | 24 | $\xrightarrow{400 \text{ Years}}$ | 97 times |
| 1901 - 2000 $\rightarrow$     | 25 |                                   |          |
| 2001 - 2100 $\rightarrow$     | 24 |                                   |          |
| 2101 - 2200 $\rightarrow$     | 24 |                                   |          |
| 2201 - 2300 $\rightarrow$     | 24 |                                   |          |
| 2301 - 2400 $\rightarrow$     | 25 |                                   |          |

(\*) (1.) How many leap years in the year 1952 to 1951?  
 $\Rightarrow 24 + 0 = 24$

(2.) How many leap years in the year 1952 to 2051?  
 $\Rightarrow 24 + 1 = 25$

Note  $\Rightarrow$  If century year is a leap year then in a hundred year leap year comes 25 times or century year is a normal year then leap year comes only 24 times.

(\*) In a normal year -

First day of the year = Last day of the year

Example :

01-01-2017 — Sunday

31-12-2017 — Sunday

(\*) In a Leap Year :

01-01-2016 — Friday

31-12-2016 — Saturday

(\*) In the following years which is not leap year ?

(i) 1998 (x)

(ii) 1974 (x)

(iii) 1978 (x)

(iv) 1994 (x)

Answer - None of these.

TYPE-1 : (Same date and same month)

(1) 11-04-1717 — FRI

11-04-1721 — ?

$$\Rightarrow 1721 - 1717 = 4$$

Leap year = 1

Total odd days = 5

$\therefore 11-04-1721 = \boxed{\text{Wednesday}}$

11-04-1717 — FRI

1718 — SAT

1719 — SUN

1720 — TUE\* (L.Y.)

**1721 — WED**

1722 — THU

1723 — FRI

1724 — SUN\* (L.Y.)

1725 — MON

(2) if 23-07-1921  $\rightarrow$  Wednesday then 23-07-1941 = ?

$$\Rightarrow 1941 - 1921 = 20$$

Leap year = +5

25

$$\rightarrow \frac{25}{7} = \boxed{4} \text{ — odd days (Remainder)}$$

$\therefore 23-07-1941 = \boxed{\text{Sunday}}$

(3.) Given that 19th November 1735 is Saturday, what day is 19th November 1782?

$$\Rightarrow 1782 - 1735 = 47$$

$$\begin{array}{r} \text{Leap Year} = 12 \\ + \\ \hline 59 \end{array}$$

$$\text{Now, } 59 \div 7 = \boxed{3} \text{ odd days (Remainder)}$$

$$\text{Saturday} + 3 = \text{Tuesday}$$

$$\therefore \boxed{19-11-1782 = \text{Tuesday}}$$

---

(4.) If 18 September 1315 is Friday then what day is 18 September 1320?

$$\Rightarrow 1320 - 1315 = 5$$

$$\begin{array}{r} \text{Leap Year} = 2 \\ \hline 7 = \text{odd days} \end{array}$$

$$7 \div 7 = 0 \rightarrow \text{Friday} + 0 = \text{Friday}$$

$$\therefore \boxed{18-09-1320 = \text{Friday}}$$

---

(5.) If 23 July 1897 is Sunday then what day is 23 July 1901?

$$\Rightarrow 1901 - 1897 = 4$$

$$\begin{array}{r} \text{Leap Year} = 0 \\ \hline 4 \end{array}$$

$$\text{Now, } 4 \div 7 = \boxed{4} \text{ Remainder}$$

$$\text{Sunday} + 4 = \text{Thursday}$$

$$\therefore \boxed{23-07-1901 = \text{Thursday}}$$



TYPE-2 : (Date, Month, Year change)

(1.) IF 15 March 1941 is Thursday then what day is 10 September 1947?

$$\Rightarrow 1947 - 1941 = 6$$

$$\text{Leap Year} = 1$$

$$* \text{March} = 31 - 15 = 16$$

$$\text{April} = 30$$

$$\text{May} = 31$$

$$\text{June} = 30$$

$$\text{July} = 31$$

$$\text{August} = 31$$

$$* \text{September} = 10$$

$$\text{Total} = 186$$

$$\text{Now } 186 \div 7 = 4 \text{ (Remainder)}$$

So, Thursday + 4 = Monday

$$\therefore \boxed{10-09-1947 = \text{Monday}}$$

II<sup>nd</sup> Method:- Total odd days =  $(6+1) + 2+2+3+2+3+3+3$   
 $= 25$

$$25 \div 7 = 4 \text{ Remainder}$$

$$\boxed{10-09-1947 = \text{Monday}}$$

(2) 27-04-1738  $\rightarrow$  Sunday

16-11-1750  $\rightarrow$  ?

$$\Rightarrow 1750 - 1738 = 12$$

$$\text{Leap Year} = 3$$

$$\text{April}(30-27) = 3$$

$$\text{November} = 16 = 16 \div 7 = 2$$

$$\text{May to Oct} = 3+2+3+3+2+3 = 16$$

$$\text{Total odd days} = 12+3+3+2+16 = 36 \div 7$$

= 1 Remainder

$$\therefore \boxed{16-11-1750 = \text{Monday}}$$



(3) IF 10 February 2004 is Friday then what day is 01-10-2040 = ?

⇒ Number of odd days -

$$2040 - 2004 = 36$$

$$\text{Leap year} = 9$$

$$\text{In february} = 29 - 10 = 19 \div 7 = 5$$

$$\text{In october} = 1$$

$$\text{March to september} = 3 + 2 + 3 + 2 + 3 + 3 + 2 = 18$$

$$\text{Total odd days} = 36 + 9 + 5 + 1 + 18 = 69$$

$$\text{Now, } 69 \div 7 = 9 \frac{6}{7} \text{ (Remainder 6)}$$

$$\therefore \text{Friday} + 6 = \text{Thursday}$$

$$\text{So, } \boxed{01-10-2040 \rightarrow \text{Thursday}}$$

$$\text{Note} \Rightarrow \boxed{01-01-0001 \rightarrow \text{Monday}}$$

Number of odd days in a hundred year = 5

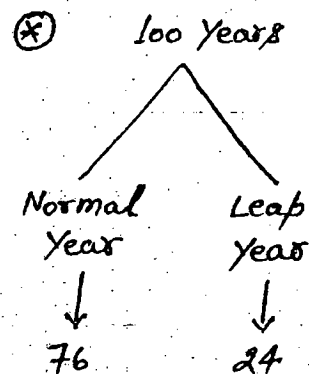
No. of odd days in 200 years = 3

300 years = 1

400 years = 0

800 years = 0

1200 years = 0



$$\text{Now, } 1 \times 76 = 76$$

$$2 \times 24 = 48$$

$$\text{Total odd days} = 124$$

$$\therefore 124 \div 7 = 17 \frac{5}{7} \rightarrow \text{Remainder 5}$$

$$\text{So, } \boxed{\text{No. of odd days in 100 years} = 5}$$

(4.) What was the day on 15 August 1947?

⇒ 15-08-1947

1946-07-15

[Month-1 and Year-1]

[OR]

[Subtract 1 from month and year.]

|                    |      |    |
|--------------------|------|----|
| 1900+46            | 31   | 15 |
|                    | (28) |    |
|                    | 31   |    |
|                    | 30   |    |
|                    | 31   |    |
|                    | 30   |    |
|                    | 31   |    |
| odd days = 1+46+11 | 212  | 15 |
| by                 |      |    |

Total odd days = 285

$$\therefore 285 \div 7 = 40 \frac{5}{7} \rightarrow \text{Remainder}$$

∴ 15 August 1947 → Friday

(5.) What was the day on 11 April 1987?

⇒ 11-04-1987

|    |     |           |
|----|-----|-----------|
|    | ↓-1 | ↓-1       |
| 11 | 03  | 1986      |
|    | 31  | 1900+86   |
|    | 28  |           |
|    | 31  |           |
| 11 | 90  | 1+86+21   |
|    |     | Leap year |

{ Subtract 1 from month and year }

= 209 = No. of odd days

$$209 \div 7 = 29 \frac{6}{7} \rightarrow \text{Remainder}$$

∴ 11-04-1987 → Saturday

(6) Which two months in a year have the same calendar?

(i) June - Oct

(ii) April - Nov

(iii) April - July (✓)

(iv) Oct - Dec

⇒ April + May + June

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ 2 & + 3 & + 2 = \frac{7}{7} = 0 \text{ (odd days)} \end{array}$$

So the next month calendar (July) is the same to April.

(7) In which year the calendar 2015 repeat itself?

(i) 2019

(ii) 2024

(iii) 2025

(iv) 2026 (✓)

(8) Which day will never become the first day of the century year?

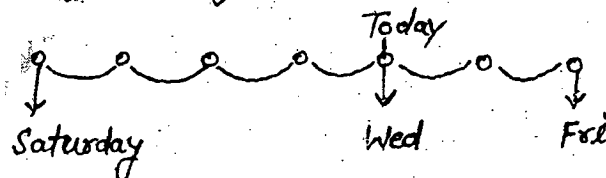
⇒ Tuesday, Thursday, and Saturday

(9) Which day will never become the first day of the century year?

⇒ Tuesday, Thursday and Sunday.

(10) If the day after tomorrow is Friday then what day was three days before yesterday?

⇒ Saturday



(11) Monika went to movies nine days ago. She goes to the movies only Thursday. So what day of the week is today?

⇒ Saturday

9 days ago → Thursday

2 days ago → Thursday

So, Today is → Thursday + 2 = Saturday.

⊗ When a calendar is repeated itself

- (i) If Leap year passed before 1 year from current year than  
current year + 6 year = Same calendar
- (ii) if leap year passed before 2 years than +11 = Same calendar
- (iii) if leap year passed before 3 years than +11 = Same calendar
- (iv) if leap year passed before 4 years than +28 = Same calendar

⊛ 2015 → 2026 (Same calendar) [Hints: Question No. 7]

Leap year passed before 3 years from 2015.

So  $2015 + 11 = 2026$

OR

$$2026 - 2015 = 11 + 3 \quad (\text{Leap year before 3 years from 2015})$$

$$= 14 \div 7 = 0 \quad (\text{Remainder})$$

Note ⇒  $2001 + 6 = 2007$  (Same Calendar)

$$2002 + 11 = 2013$$

$$2003 + 11 = 2014$$

$$2004 + 28 = 2032$$

⊛ Number of odd days in a hundred year = 5 days

|              |          |                    |
|--------------|----------|--------------------|
| In 200 years | = 3 days | ⇒ 31-12-2000 → SUN |
| In 300 years | = 1 day  | 31-12-2100 → FRI   |
| In 400 years | = 0 days | 31-12-2200 → WED   |
| In 500 years | = 5 days | 31-12-2300 → MON   |
| 600 years    | = 3 days | 31-12-2400 → SUN   |
| 700 years    | = 1      |                    |
| 800 years    | = 0      |                    |

A leap year of a century year mean a century year, which is a leap year have no odd days.  
So its last day of year is Sunday.

So,  
 01-01-2000  $\longrightarrow$  Saturday  
 01-01-2100  $\longrightarrow$  Friday  
 01-01-2200  $\longrightarrow$  Wednesday  
 01-01-2300  $\longrightarrow$  Monday  
 01-01-2400  $\longrightarrow$  Saturday

$\Rightarrow$  So, Tuesday, Thursday, and Sunday will never become the first day of a century year.

(1) IF February 29<sup>th</sup> falls on a Monday, What would be the 11<sup>th</sup> day of the month? (SSC CGL 26-10-2014)

$\Rightarrow$  29 February  $\longrightarrow$  Monday  
 22 Feb  $\longrightarrow$  Monday  
 15 Feb  $\longrightarrow$  Monday  
 $\therefore$  11 February  $\longrightarrow$  Thursday

(2) If in a year, which happens to be a leap year, 1<sup>st</sup> January falls on a Sunday, after how many years will the year begin on Sunday? (CGL-2014)

$\Rightarrow$  Leap year  $\longrightarrow$  1 January — Sunday  
 Next year  $\longrightarrow$  1 January — Tuesday  
 Next year  $\longrightarrow$  1 January — Wednesday  
 Next year  $\longrightarrow$  1 January — Thursday  
 Next year  $\longrightarrow$  1 January — Friday  
 Next year  $\longrightarrow$  1 January — Sunday

After 5 years next Sunday will fall

(3) Ram was born on 29<sup>th</sup> February. He celebrated his birthday falling on exactly 29<sup>th</sup> February 2008 for the fourth time. In which year he was born? (CGL-2013)

⇒ Ram celebrated his birthday on 29<sup>th</sup> February 2008, 4<sup>th</sup> time, it means he was born in a leap year.

Ram celebrated his birthday for third time in 2004.  
2<sup>nd</sup> time — 2000

So, He was born in 1996.

(4) Suresh was born on 4<sup>th</sup> October 1999. Shashikant was born 6 days before Suresh. The Independence Day of that year fell on Sunday. Which day was Shashikant born? (CGL-2011)

⇒ 4<sup>th</sup> October → Suresh birthday

29<sup>th</sup> September → Shashikant birthday

15<sup>th</sup> August 1999 → Sunday

15 August to 29<sup>th</sup> September = 16 + 29 = 45 days

$45 \div 7 = 6 \text{ Week, } 3 \text{ odd days}$

So, 29<sup>th</sup> September falls on Sunday + 3 = Wednesday.

(5) Mrs. Susheela celebrated her wedding anniversary on Tuesday, 30<sup>th</sup> September 1997. When will she celebrate her next wedding anniversary on the same day? (Bank PO - 2012)

⇒ 30<sup>th</sup> September 1997 → Tuesday

30<sup>th</sup> September 1998 → Wednesday

30<sup>th</sup> September 1999 → Thursday

30<sup>th</sup> September 2000 → Saturday.

It is so, because 2000 is a leap year and there is one extra day in the month of February.

30<sup>th</sup> September 2001 → Sunday

30<sup>th</sup> September 2002 → Monday

30<sup>th</sup> September 2003 → Tuesday

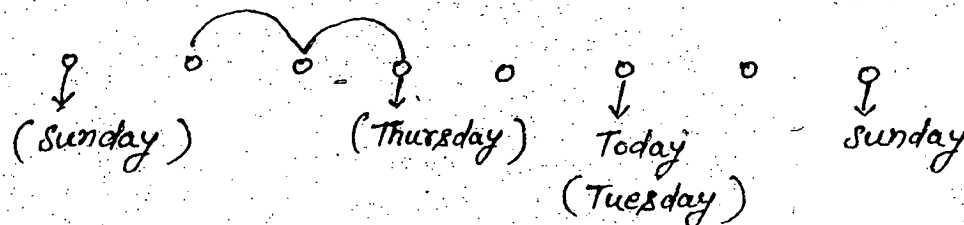


(6) If the day before yesterday was Sunday, What day will it be three days after the day after tomorrow? (CGL-2010)

$\Rightarrow$  Day before yesterday = Sunday

Today = Tuesday

3 days after the day after tomorrow will be Sunday



(7) IF Republic Day 1980 falls on Saturday, then on what day of the week Independence Day celebrated in the same year?

$\Rightarrow$  Republic day = 26 January (LDC-2012)

Independence day = 15 August

Odd days in - Jan = 05

Feb = 01 (Leap year)

March = 03

April = 2

May = 3

June = 2

July = 3

1 Aug to 15 Aug = 1

Total odd days = 20 days

Required odd days =  $\frac{20}{7} = 6 \rightarrow$  Remainder

Saturday + 6 = Friday

(8) How many days are there in  $x$  week  $x$  days? (DP-2012)

$\Rightarrow x$  week +  $x$  days

$\Rightarrow 7$  days = 1 week

$\therefore 7x$  days +  $x$  days =  $(7x + x)$

= 8x days



(9) How many odd days till 28 May 2003? (LDC-2012)

$\Rightarrow$  (I) 2000 — 0 (odd days)

(II) 2001 to 2002  $\rightarrow$  2 odd days

(III) 01-01-2003 to 28-05-2003  $\rightarrow 3+0+3+2 = 8$

$$\text{Total odd days} = \frac{10}{7} = 3 \text{ (Remainder)}$$

$\therefore$  28 May 2003  $\rightarrow$  Wednesday

(10) Satish remembers that his brother's birthday is after fifteenth but before eighteenth of February where as his sister Kajal remembers that her brother's birthday is after sixteenth but before nineteenth of February. On which day of February is Satish's brother's birthday? (CPO-2013)

$\Rightarrow$  According to Satish,

His brother's birthday = 16th or 17th

According to Kajal

Her brother's birthday = 17th or 18th

From all the given information we can say his brother's birthday is on 17th February.

(11) IF February 1, 1996 was Wednesday, what day was March 3, 1996? (CPO-2009)

$\Rightarrow$  01 February 1996 = Wednesday

Since, 1996 is a leap year. So, in February there will be 29 days.

Rest Wednesdays of February, 8, 15, 22, 29

Hence, March 1 will be Thursday

$\therefore$  March 3, 1996 will be Saturday.

## \* NUMBER-SERIES \*

Note  $\Rightarrow$  In the number series, numbers can increase or decrease by addition, subtraction, multiplication, division, arithmetic or geometric progression etc.

(1.) 2, 4, 6, 8, 10

(2.) 2, 3, 5, 7, 11  $\rightarrow$  (Prime number series)

(3.) 1, 3, 5, 7, 9

(4.) 4, 9, 16, 25, 36, 49

(5.) 2, 3, 10, 15, 26, 35

$\Rightarrow 1^2+1=2, 2^2-1=3, 3^2+1=10, 4^2-1=15, 5^2+1=26, 6^2-1=35$

(6.) 64, 125, 216, 343, 512, +29

$\Rightarrow 4^3=64, 5^3=125, 6^3=216, 7^3=343, 8^3=512, 9^3=729$

(7.) 62, 127, 214, 345, 510

$\Rightarrow 4^3-2=62, 5^3+2=127, 6^3-2=214, 7^3+2=345, 8^3-2=510$

(8.) 2, 3, 8, 63, 3968

(a) 3971

(b) 3958

(c) 3968 (✓) (d) 3981

$\Rightarrow$

$$\begin{array}{ccccccc}
 2 & & 3 & & 8 & & 63 & & 3968 \\
 \downarrow & \nearrow & \downarrow & \nearrow & \downarrow & \nearrow & \downarrow & \nearrow \\
 2^2-1 & & 3^2-1 & & 8^2-1 & & 63^2-1 & & 
 \end{array}$$

(9.) 6, 11, 21, 36, 56, 81

$\Rightarrow$

$$\begin{array}{ccccccc}
 6 & 11 & 21 & 36 & 56 & & 81 \\
 \underbrace{+5} & \underbrace{+10} & \underbrace{+15} & \underbrace{+20} & \underbrace{+25} & & 
 \end{array}$$

$$(10.) \quad 0.5, \quad 0.55, \quad 0.65, \quad 0.8, \quad \underline{1.00}$$

$$\Rightarrow \quad \begin{array}{ccccccc} 0.50 & & 0.55 & & 0.65 & & 0.80 & & 1.00 \\ & \underbrace{+0.5} & & \underbrace{+0.10} & & \underbrace{+0.15} & & \underbrace{+0.20} & \end{array}$$


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$$(11.) \quad 0.5, \quad 1.5, \quad 4.5, \quad 13.5, \quad \underline{40.5}$$

$$\Rightarrow \quad \begin{array}{ccccccc} 0.5 & & 1.5 & & 4.5 & & 13.5 & & 40.5 \\ & \underbrace{\times 3} & & \underbrace{\times 3} & & \underbrace{\times 3} & & \underbrace{\times 3} & \end{array}$$


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$$(12.) \quad \frac{1}{2}, \quad \frac{3}{4}, \quad \frac{5}{8}, \quad \frac{7}{16}, \quad \underline{\frac{9}{32}}$$

$$\Rightarrow \quad \begin{array}{ccccccc} \frac{1}{2} & & \frac{3}{4} & & \frac{5}{8} & & \frac{7}{16} & & \boxed{\frac{9}{32}} \\ & \underbrace{+2} & & \underbrace{+2} & & \underbrace{+2} & & \underbrace{+2} & \\ & \underbrace{\times 2} & & \underbrace{\times 2} & & \underbrace{\times 2} & & \underbrace{\times 2} & \end{array}$$


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$$(13.) \quad 1, \quad 3, \quad 7, \quad 15, \quad 31, \quad 63, \quad \underline{127}$$

$$\Rightarrow \quad \begin{array}{ccccccc} 1 & & 3 & & 7 & & 15 & & 31 & & 63 & & 127 \\ & \underbrace{\times 2+1} & & \underbrace{\times 2+1} & & \underbrace{\times 2+1} & & \underbrace{\times 2+1} & & \underbrace{\times 2+1} & & \underbrace{\times 2+1} & \end{array}$$


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$$(14.) \quad 9, \quad 11, \quad \underline{20}, \quad 31, \quad 51, \quad 82.$$

$$\Rightarrow \quad 82 - 51 = 31$$

$$\therefore 51 - 31 = \boxed{20}$$

$$31 - 20 = 11$$

$$20 - 11 = 9$$


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$$(15.) \quad 1, \quad 3, \quad 4, \quad 8, \quad 15, \quad 27, \quad \underline{50}$$

$$\Rightarrow \quad 1 + 3 + 4 = 8$$

$$3 + 4 + 8 = 15$$

$$4 + 8 + 15 = 27$$

$$8 + 15 + 27 = \boxed{50}$$

$$(16) \quad 1, 2, 5, 12, 27, 58, \underline{121}$$

$$\Rightarrow 1, 2, 5, 12, 27, 58, \boxed{121}$$

$\underbrace{\quad \times 2 + 0 \quad} \underbrace{\quad \times 2 + 1 \quad} \underbrace{\quad \times 2 + 2 \quad} \underbrace{\quad \times 2 + 3 \quad} \underbrace{\quad \times 2 + 4 \quad} \underbrace{\quad \times 2 + 5 \quad}$

$$(17) \quad 0, 6, 24, 60, 120, 210, \underline{336}$$

$$\Rightarrow 1^3 - 1 = 0, 2^3 - 2 = 6, 3^3 - 3 = 24, 4^3 - 4 = 60, \\ 5^3 - 5 = 120, 6^3 - 6 = 216, 7^3 - 7 = \boxed{336}$$

$$(18) \quad 840, 168, 42, 14, 7, \underline{7}$$

$$\Rightarrow 840, 168, 42, 14, 7, \boxed{7}$$

$\underbrace{\quad \times 5 \quad} \underbrace{\quad \times 4 \quad} \underbrace{\quad \times 3 \quad} \underbrace{\quad \times 2 \quad} \underbrace{\quad \times 1 \quad}$

$$(19) \quad 240, \underline{240}, 120, 40, 10, 2$$

$$\Rightarrow 240, \underline{240}, 120, 40, 10, 2$$

$\underbrace{\quad \times 1 \quad} \underbrace{\quad \times 2 \quad} \underbrace{\quad \times 3 \quad} \underbrace{\quad \times 4 \quad} \underbrace{\quad \times 5 \quad}$

$$(20) \quad 10, 100, 200, 310, \underline{430}$$

$$\Rightarrow 10, 100, 200, 310, \boxed{430}$$

$\underbrace{\quad + 90 \quad} \underbrace{\quad + 100 \quad} \underbrace{\quad + 110 \quad} \underbrace{\quad + 120 \quad}$

$$(21) \quad 4, 10, \underline{28}, 82, 244, 730$$

$$\Rightarrow 4, 10, \boxed{28}, 82, 244, 730$$

$\underbrace{\quad \times 3 - 2 \quad} \underbrace{\quad \times 3 - 2 \quad} \underbrace{\quad \times 3 - 2 \quad} \underbrace{\quad \times 3 - 2 \quad} \underbrace{\quad \times 3 - 2 \quad}$

$$(22) \quad 23, 35, 57, 711, \underline{1113}$$

$$\Rightarrow (23), (35), (57), (711), (1113) \quad \left[ \text{Prime Number Series} \right]$$

$\boxed{OR}$

$$\Rightarrow \underline{23}, \underline{57}, \underline{1113}, 17, 19$$

(23.) 4, 2, 2, 3, 6, 15

$\Rightarrow$  4, 2, 2, 3, 6, 15

$\times \frac{1}{2}$     $\times 1$     $\times 1\frac{1}{2}$     $\times 2$     $\times 2\frac{1}{2}$

(24.) 6, 9, 18, 45, 135

$\Rightarrow$  6, 9, 18, 45, 135

$\times 1\frac{1}{2}$     $\times 2$     $\times 2\frac{1}{2}$     $\times 3$

(25.) 7, 7, 10.5, 21, 52.5

$\times 1$     $\times 1\frac{1}{2}$     $\times 2$     $\times 2\frac{1}{2}$

⊛ There is a wrong Number in this series which of the following alternative given below should be replaced in place of the wrong Number.

(16) 8, 16, 24, 40, 62, 104, 168.

$\Rightarrow 8 + 16 = 24 \Rightarrow 24 + 16 = 40 \Rightarrow 40 + 24 = \boxed{64} \Rightarrow 64 + 40 = 104$   
 So 64 is replaced in place of wrong number 62.

(27.) 8, 13, 21, 32, 47, 63, 83.

$\Rightarrow$  8, 13, 21, 32, 46, 63, 83

$+5$     $+8$     $+11$     $+14$     $+17$     $+20$   
 $+3$     $+3$     $+3$     $+3$     $+3$

Replaced No. = 46  
Wrong No. = 47

(28.) 7, 28, 63, 124, 215, 342, 511

$\Rightarrow$  7, 28<sup>26</sup>, 63, 124, 215, 342, 511

$2^3 - 1 = 7, 3^3 + 1 = 28, 4^3 - 1 = 63, 5^3 - 1 = 124, 6^3 - 1 = 215, 7^3 - 1 = 342$   
 $8^3 - 1 = 511.$  Wrong No. = 28, Replaced No. = 26

(29) 15, 16, 22, 29, 45, 70

$$\Rightarrow \begin{array}{cccccc} 15 & 16 & 22 & 29 & 45 & 70 \\ \underbrace{+} & \underbrace{+} & \underbrace{+} & \underbrace{+} & \underbrace{+} & \underbrace{+} \\ 1^2=1 & 2^2=4 & 3^2=9 & 4^2=16 & 5^2=25 & \end{array}$$

Wrong No. = 22, Replaced No. = 20

(30) 2, 5, 10, 50, 500, 5000

$$\Rightarrow 2 \times 5 = 10$$

$$10 \times 5 = 50$$

$$50 \times 10 = 500$$

$$500 \times 50 = \boxed{25000}$$

Wrong No. = 5000

Replaced correct no. = 25000

(31) 5, 10, 40, 80, 320, 550, 2560

$$\Rightarrow \begin{array}{ccccccccc} 5 & 10 & 40 & 80 & 320 & 550 & 2560 \\ \underbrace{\times 2} & \underbrace{\times 4} & \underbrace{\times 2} & \underbrace{\times 4} & \underbrace{\times 2} & \underbrace{\times 4} & \end{array}$$

(32) 8, 14, 26, 48, 98, 194, 386

$$\Rightarrow \begin{array}{ccccccc} 8 & 14 & 26 & 48 & 98 & 194 & 386 \\ \underbrace{\times 2 - 2} & \underbrace{\times 2 - 2} & \underbrace{\times 2 - 2} & \underbrace{\times 2 - 2} & \underbrace{\times 2 - 2} & \underbrace{\times 2 - 2} & \end{array}$$

$$8 \times 2 - 2 = 14$$

$$14 \times 2 - 2 = 26$$

$$26 \times 2 - 2 = \boxed{50}$$

$$50 \times 2 - 2 = 98$$

$$98 \times 2 - 2 = 194$$

$$194 \times 2 - 2 = 386$$

(33) 8, 27, 125, 343, 1331

$$\Rightarrow 2^3 = 8, 3^3 = 27, 5^3 = 125, 7^3 = 343, 11^3 = 1331$$

(Prime number series.) (No Error)

(34) 3, 8, 15, 24, 34, 48, 63

$$\Rightarrow \begin{array}{ccccccc} 3 & 8 & 15 & 24 & 34 & 48 & 63 \\ \downarrow & \downarrow & & & & & \\ 2^2-1 & 3^2-1 & 4^2-1 & 5^2-1 & 6^2-1 & 7^2-1 & \end{array}$$

$$2^2-1, 3^2-1, 4^2-1=15, 5^2-1=24, 6^2-1=35, 7^2-1=48$$

$$8^2-1=63$$

Wrong Number = 34

Replaced No. = 35.



(35) B, E, J, Q, Z.

$\Rightarrow$   $\begin{matrix} 2 \\ B \end{matrix}$ ,  $\begin{matrix} 5 \\ E \end{matrix}$ ,  $\begin{matrix} 10 \\ J \end{matrix}$ ,  $\begin{matrix} 17 \\ Q \end{matrix}$ ,  $\begin{matrix} 26 \\ Z \end{matrix}$

$$\begin{aligned} 1^2+1 &= 2 & 3^2+1 &= 10 & 5^2+1 &= 26 \\ 2^2+1 &= 5 & 4^2+1 &= 17 \end{aligned}$$

(36) BDF, CFI, DHL, EJO

**EJO**

$\Rightarrow$   $\begin{matrix} B & D & F \\ +1 & & \\ & +3 & \end{matrix}$   $\begin{matrix} C & F & I \\ +1 & & \\ & +3 & \end{matrix}$   $\begin{matrix} D & H & L \\ +1 & & \\ & +3 & \end{matrix}$   $\begin{matrix} E & J & O \\ +1 & & \\ & +3 & \end{matrix}$

(37) YEB, WFD, UHG, SKI, QOL

$\Rightarrow$   $\begin{matrix} 25 & 5 & 2 \\ Y & E & B \end{matrix}$   $\begin{matrix} 23 & 6 & 4 \\ W & F & D \end{matrix}$   $\begin{matrix} 21 & 8 & 7 \\ U & H & G \end{matrix}$   $\begin{matrix} 19 & 11 & 9 \\ S & K & I \end{matrix}$   $\begin{matrix} 17 & 15 & 12 \\ Q & O & L \end{matrix}$

$\begin{matrix} Y & -2 & W & -2 & U & -2 & S & -2 & Q \\ E & +1 & F & +2 & H & +3 & K & +4 & O \\ B & +2 & D & +3 & G & +4 & I & +5 & L \end{matrix}$

**QOL**

(38) CXG, JTH, QPT, XLM, EHQ

$\Rightarrow$   $\begin{matrix} 3 & +7 & 10 & +7 & 17 & +7 & 24 & +7 & 31 \\ C & & J & & Q & & X & & E \end{matrix}$   $\begin{matrix} 24 & -4 & 20 & -4 & 16 & -4 & 12 & -4 & 8 \\ X & & T & & P & & L & & H \end{matrix}$   $\begin{matrix} 7 & +1 & 8 & +2 & 10 & +3 & 13 & +4 & 17 \\ G & & H & & J & & M & & Q \end{matrix}$

**EHQ**

(39) Z1A, X2D, V6G, T21J, R88M, P445P, N2676S

$\Rightarrow$   $\begin{matrix} 26 & -2 & 24 & -2 & 22 & -2 & 20 & -2 & 18 & -2 & 16 & -2 & 14 \\ Z & & X & & V & & T & & R & & P & & N \end{matrix}$

$\begin{matrix} 1 \times 1 + 1 & 2 \times 2 + 2 & 6 \times 3 + 3 & 21 \times 4 + 4 & 88 \times 5 + 5 & 445 \times 6 + 6 & 2676 \end{matrix}$

$\begin{matrix} 1 & +3 & 4 & +3 & 7 & +3 & 10 & +3 & 13 & +3 & 16 & +3 & 19 \\ A & & D & & G & & J & & M & & P & & S \end{matrix}$

= **N2676S**

(UPPCS-2015)



\* Find the missing letter/number of series.

(40.) 16, 61, 25, 52, 36, 63, 49, ? (SSC LDC-2015)

(A) 36 (B) 94 (✓) (C) 72 (D) 46

⇒  $\begin{array}{cc} 16 & 61 \\ \text{Reverse} \end{array}$  Similarly,  $\begin{array}{cc} 49 & 94 \end{array}$

(41.)  $7\frac{1}{7}$ ,  $8\frac{2}{6}$ ,  $9\frac{5}{5}$ ,  $12\frac{2}{4}$ ,  $16\frac{2}{3}$ , ? (CGL-2015)

(A)  $\frac{50}{2}$  (✓) (B)  $15\frac{2}{4}$  (C)  $16\frac{4}{4}$  (D) 35

⇒  $7\frac{1}{7} = \frac{50}{7}$ ,  $8\frac{2}{6} = \frac{50}{6}$ ,  $9\frac{5}{5} = \frac{50}{5}$ ,  $12\frac{2}{4} = \frac{50}{4}$ ,  $16\frac{2}{3} = \frac{50}{3}$ ,

⇒  $\frac{50}{7}$ ,  $\frac{50}{6}$ ,  $\frac{50}{5}$ ,  $\frac{50}{4}$ ,  $\frac{50}{3}$ ,  $\frac{50}{2}$

(42.) 0, 6, 24, 60, 120, 210, ? (CPO-2015)

(A) 240 (B) 504 (C) 336 (✓) (D) 290

⇒  $\begin{array}{ccccccc} 0 & 6 & 24 & 60 & 120 & 210 & 336 \\ \hline & 6 & 18 & 36 & 60 & 90 & 126 \\ \hline & & 12 & 18 & 24 & 30 & 36 \\ \hline & & & 6 & 6 & 6 & 6 \end{array}$

(43.) 2, 15, 41, 80, ? (CPO-2015)

(A) 120 (B) 121 (C) 132 (✓) (D) 111

⇒  $\begin{array}{ccccccc} 2 & 15 & 41 & 80 & 132 \\ \hline & +13 & +26 & +39 & +52 \end{array}$

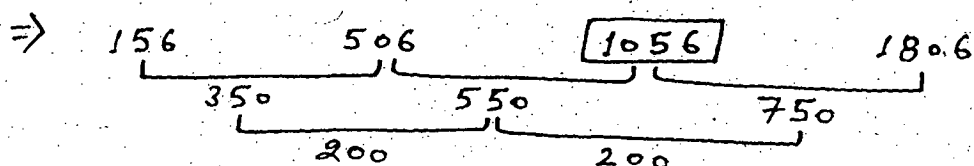
(44.) 3, 15, 4, 16, 5, 17, 6, ? , 7. (LDC-2014)

(A) 12 (B) 13 (C) 15 (D) 18 (✓)

⇒  $\begin{array}{ccccccccccc} & & +1 & & +1 & & +1 & & & & \\ & & \swarrow & & \swarrow & & \swarrow & & & & \\ 3 & 15 & 4 & 16 & 5 & 17 & 6 & \boxed{18} & & & 7 \end{array}$

(45.) 156, 506, ?, 1806. (CGL-2014)

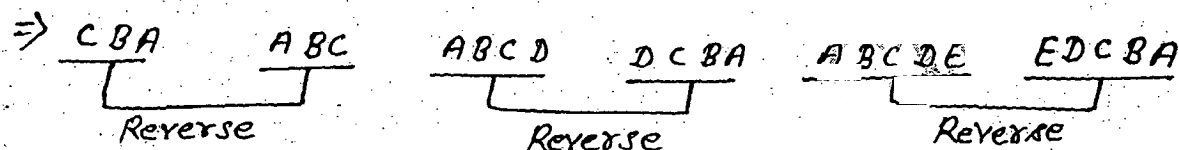
(A) 1056 (✓) (B) 856 (C) 1456 (D) 1506



(46.) CBA, ABC, ABCD, DCBA, ABCDE, ? (MTS-2012)

(A) EDCBA (✓) (B) DBAC

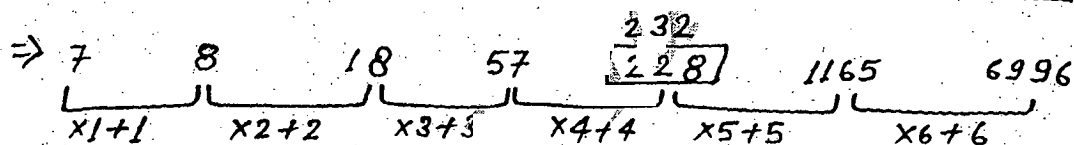
(C) CABD (D) BACD



\* Find the wrong number in series.

(47.) 7, 8, 18, 57, 228, 1165, 6996 (CGL-2014)

(A) 8 (B) 18 (C) 41 (D) 228 (✓)



228 is wrong number which should be replaced by 232.

(48.) 701, 348, ~~173~~, 85, 41, 19, 8 (FCI-2011)

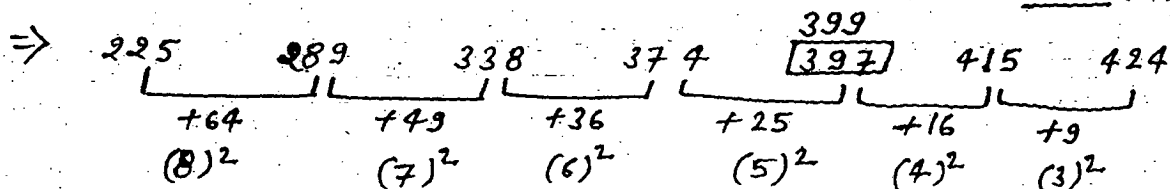
(A) 173 (B) 41 (C) 19 (D) 348 (✓)

⇒ Series follows the pattern from the right end as  $8 \times 2 + 3 = 19$   
 $19 \times 2 + 3 = 41$ ,  $41 \times 2 + 3 = 85$ ,  $85 \times 2 + 3 = 173$ ,  $173 \times 2 + 3 = 349$ ,  
 $349 \times 2 + 3 = 701$ .

Hence, the number 348 should be replaced by 349.

(49.) 225, 289, 338, 374, 397, 415, 424. (CGL-2016)

(A) 415 (B) 338 (C) 374 (D) 397 (✓)



Hence, number 397 is wrong and should be replaced by 399.

\*

## CODED EQUATION

\*

TYPE-1: (1.)  $>$  denotes  $+$  $<$  denotes  $-$  $+$  denotes  $\div$  $-$  denotes  $=$  $=$  denotes Less than $\times$  denotes greater than

Find the correct statement:

(a)  $3+2 > 4 = 9+3 < 2$

(b)  $3 > 2 > 4 = 18+3 < 1$

(c)  $3 > 2 < 4 \times 8 + 4 < 2$  (✓)

(d)  $3+2 < 4 \times 9 + 3 < 3$

(e) N.O.T

$\Rightarrow 3 > 2 < 4 \times 8 + 4 < 2$

$\Rightarrow 3+2-4 > 8 \div 4 - 2$

$= \boxed{1 > 0}$

TYPE-2: Which two symbol interchange to correct the statement?

(2.)  $12 \div 2 - 6 \times 3 + 8 = 16$

(a)  $\times, +$

(b)  $+, -$  (✓)

(c)  $\div, \times$

(d)  $\div, +$

$\Rightarrow 12 \div 2 + 6 \times 3 - 8 = 16$

$6 + 6 \times 3 - 8 = 16$

$6 + 18 - 8 = 16$

$\boxed{16 = 16}$

(3) Which No. is interchange to correct the statement.

$$7 \times 2 - 3 + 8 \div 4 = 5 + 6 \times 2 - 24 \div 3$$

(a) 7, 6 (✓)

(b) 6, 5

(c) 2, 6

(d) 3, 24

(e) N.O.T

$$\Rightarrow 6 \times 2 - 3 + 8 \div 4 = 5 + 7 \times 2 - 24 \div 3$$

$$12 - 3 + 2 = 5 + 14 - 8$$

$$14 - 3 = 19 - 8$$

$$\boxed{11 = 11}$$

TYPE-3 :

(4)  $24 * 4 * 5 * 4 * 8$

(a) +, ÷, ×, =

(b) -, =, +, ×

(c) -, ×, +, = (✓)

(d) +, -, =, ×

(e) N.O.T

$$\Rightarrow 24 - 4 \times 5 + 4 = 8$$

$$24 - 20 + 4 = 8$$

$$28 - 20 = 8$$

$$\boxed{8 = 8}$$

(5)  $12 * 4 * 3 * 16 * 9 * 1$

(a) -, +, =, ÷, ×

(b) ÷, +, =, -, ×

(c) +, ×, =, +, - (✓)

(d) ×, ÷, +, -, =

(e) N.O.T

$$\Rightarrow 12 + 4 \times 3 = 16 + 9 - 1$$

$$12 + 12 = 25 - 1$$

$$\boxed{24 = 24}$$

TYPE-4 : Find the Value of following Expression without using BODMAS.

(6)  $8 - 3 + 4 \div 3 \times 2 = ?$

(a) 3

(b) 4

(c) 5

(d) 6 (✓)

$$\Rightarrow 8 - 3 + 4 \div 3 \times 2$$

$$= 5 + 4 \div 3 \times 2$$

$$9 \div 3 \times 2$$

$$3 \times 2 = \boxed{6}$$

(7.) Select the set of symbols which can be fitted correctly in the equation. (LDC-2015)

$$5 \_ 3 \_ 8 \_ 4 \_ 2 = 21$$

(A)  $-, \times, +, \div$

(B)  $\times, +, -, \div$  (✓)

(C)  $+, \times, -, \div$

(D)  $+, \times, +, \div$

$$\Rightarrow 5 \times 3 + 8 - 4 \div 2 = 21$$

$$15 + 8 - 2 = 21$$

$$\boxed{21 = 21}$$

(8.) If,  $\div$  stands for division,  $\times$  stands for addition,  $-$  stands for multiplication,  $+$  stands for subtraction, which of the following is correct? (CGL-2015)

(a)  $46 \times 6 \div 4 - 5 + 3 = 74$  (b)  $46 \div 6 + 4 \times 5 \div 3 = 71$

(c)  $46 \div 6 \times 4 - 5 + 3 = 75.5$  (d)  $46 \times 6 - 4 + 5 \div 3 = 70.1$

(A) d

$$\Rightarrow 46 \div 6 + 4 \times 5 \div 3 = 71$$

(B) b (✓)

$$46 \times 6 \div 4 + 5 - 3 = 71$$

(C) a

$$46 \times \frac{3}{2} + 2 = 71$$

(D) c

$$23 \times 3 + 2 = 71$$

$$\boxed{71 = 71}$$

(9.) If  $A = \div$ ,  $B = +$ ,  $C = -$  and  $D = \times$ , then which of the following is correct? (UPPSC-2014)

(A)  $24 A 6 B 10 = 5 D 6 C 12$

(B)  $6 A 4 D 6 = 4 B 2 D 6$

(C)  $30 D 4 A 12 = 30 A 12 D 4 A 5$

(D)  $10 B C 72 = 78 C 42$  (✓)

$$\Rightarrow 108 C 72 = 78 C 42$$

$$108 - 72 = 78 - 42$$

$$\boxed{36 = 36}$$

(10.) Which one of the four interchanges in signs and numbers would make the given equation correct?  
(CPO-2015)

$$6 \times 4 + 2 = 16$$

(A) + and  $\times$  and 4

(B) + and  $\times$ , 2 and 4

(C) + and  $\times$ , 4 and 6 (✓)

(D) None of these.

$$\Rightarrow 6 \times 4 + 2 = 16$$

$$4 + 6 \times 2 = 16$$

$$4 + 12 = 16$$

$$\boxed{16 = 16}$$

(11.) If I means  $\times$ , You means  $\div$ , We means  $-$ ,

He means  $+$ , then  $8 \text{ I } 12 \text{ He } 16 \text{ You } 2 \text{ We } 10$  is -  
(FCI-2011)

(A) 45

(B) 94 (✓)

(C) 96

(D) 112

$$\Rightarrow \text{I} \Rightarrow \times \quad \text{You} \Rightarrow \div$$

$$\text{We} \Rightarrow - \quad \text{He} \Rightarrow +$$

$$\Rightarrow 8 \text{ I } 12 \text{ He } 16 \text{ You } 2 \text{ We } 10$$

$$8 \times 12 + 16 \div 2 - 10$$

$$96 + 8 - 10 = 94$$

(12.) Select the correct combination of Mathematical signs to replace \* signs and to balance the given equation.  
(CGL-2014)

$$8 * 6 * 96 * 2 = 0$$

(A)  $\times \div -$

(B)  $\times - \div$

(C)  $- \times \div$

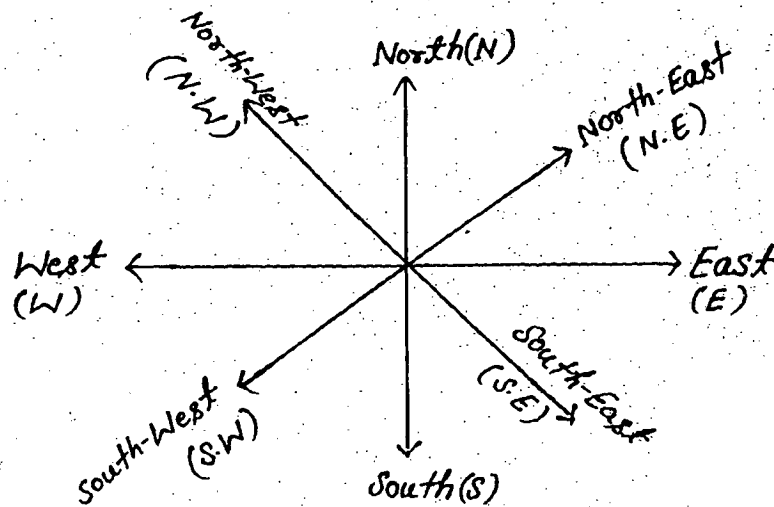
(D)  $\div - \times$

$$\Rightarrow 8 \times 6 \times 96 \times 2 = 0$$

$$8 \times 6 - 96 \div 2 = 0$$

$$48 - 48 = 0$$

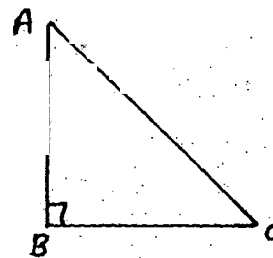
## \* DIRECTION-TEST \*



\* Pythagoras Theorem :

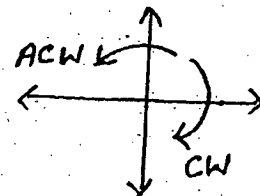
$$\Rightarrow AC^2 = AB^2 + BC^2$$

$$\therefore AC = \sqrt{AB^2 + BC^2}$$



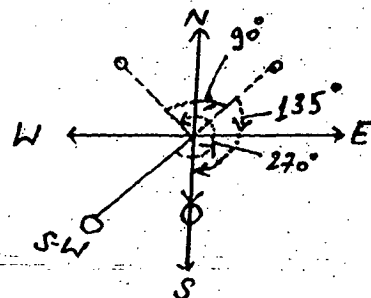
Note  $\Rightarrow$  CW  $\rightarrow$  Clockwise

ACW  $\rightarrow$  Anti-Clockwise



(1.) A Person is standing in South-West direction. There he moves  $270^\circ$  anti clockwise and  $90^\circ$  in clockwise and further he moves  $135^\circ$  in same direction. So in which direction is he now?

$\Rightarrow$  South.



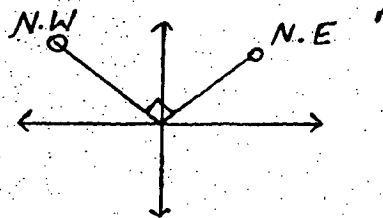
OR  $270^\circ \text{ ACW} + 90^\circ \text{ CW} + 135^\circ \text{ CW} = 45^\circ \text{ ACW}$  From the original position.



- (2) A Person is standing in North-West. From there he moves  $540^\circ$  clockwise direction and then  $315^\circ$  anti-clockwise and further  $270^\circ$  in same direction, then  $135^\circ$  clockwise. In which direction is he now?

$\Rightarrow$

| CW                  | ACW         |
|---------------------|-------------|
| $540^\circ$         | $315^\circ$ |
| $135^\circ$         | $270^\circ$ |
| Total = $675^\circ$ | $585^\circ$ |



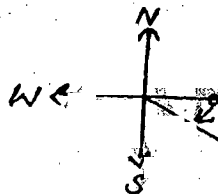
$$\therefore \text{CW} \sim \text{ACW} = 675 - 585 = 90^\circ \text{CW}$$

North-East

- (3) A Person is standing in East direction. From there he moves  $100^\circ$  anti-clockwise and then  $135^\circ$  clockwise. In which direction is he now?

$\Rightarrow$

| CW  | ACW |
|-----|-----|
| 135 | 100 |

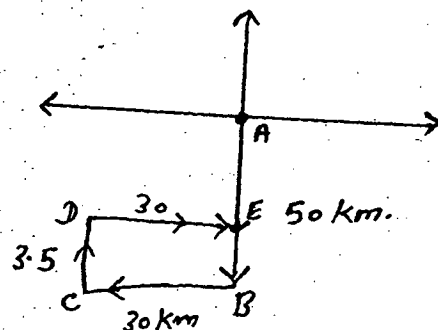


- (i) S.E  
 (ii) N.E  
 (iii) S.W  
 (iv) E-SE (✓)

$$\text{Now, CW} \sim \text{ACW} = 135 \sim 100 \\ = 35^\circ \text{CW}$$

- (4) A man walks 50 km. to South and then he takes right turn, he covers 30 km. and he again takes right turn and walk 3.5 km. then he takes right turn and covers 30 km. So how far and in which direction is he now from starting point?

$\Rightarrow$  46.5 km., South.

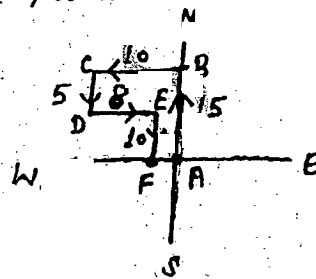


$$\therefore \text{AE} = \text{AB} - \text{EB} \\ = 50 - 3.5 = 46.5$$

- ⊗ Left → Anti-clockwise  
 Right → Clockwise

- (5.) A man walks 15 km. in north direction and then he takes a left turn and cover 10 km. He again takes a left turn and cover 5 km. then turn into East, he cover 8 km. Then he takes a right turn and he cover 10 km. How and in which direction is he now from starting point.

⇒ 2 km., West.

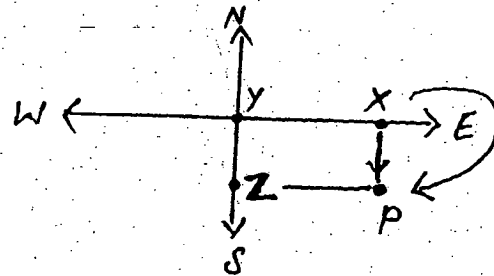


AF = 2 km.

- (6.) X is in the East of Y which is in the North of Z.

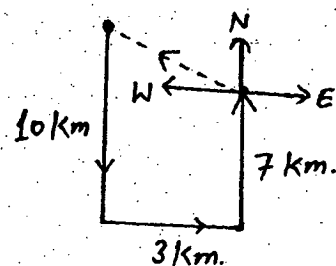
If P is East of Z, then which direction of X is P?

⇒ South.



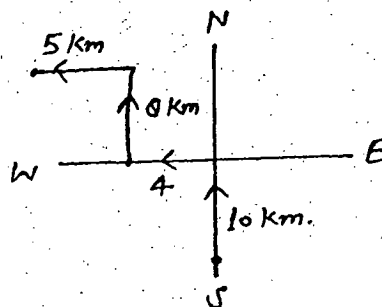
- (7.) Rahul left home towards South and cover 10 km. and he takes a left turn and cover 3 km. Then he turning North and cover 7 km. So in which direction is his house from this point?

⇒ North-West.



(8) After moving 10 km. Rahul takes a left turn and cover 4 km. Then he takes a right turn and cover 8 km. Then turning to left he covers 5 km. In the end he was moving towards south. So in which direction did he start his Journey?

⇒ West.



Trick:-

|                 | START | END   |
|-----------------|-------|-------|
| As Per figure   | North | West  |
| As per Question | West  | South |

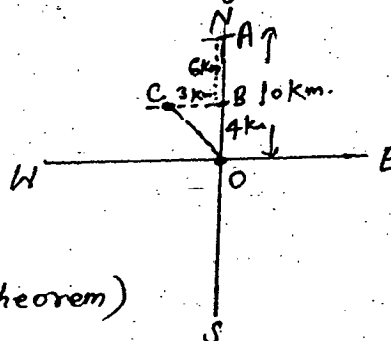
(i) North : West :: West : South

(ii) North : West :: South : East

(9) A man walks 10 km. in North direction and then 6 km. towards to south, then he takes a right turn and cover 3 km. So how far and which direction from his starting point?

⇒ 5 km., (North-West)

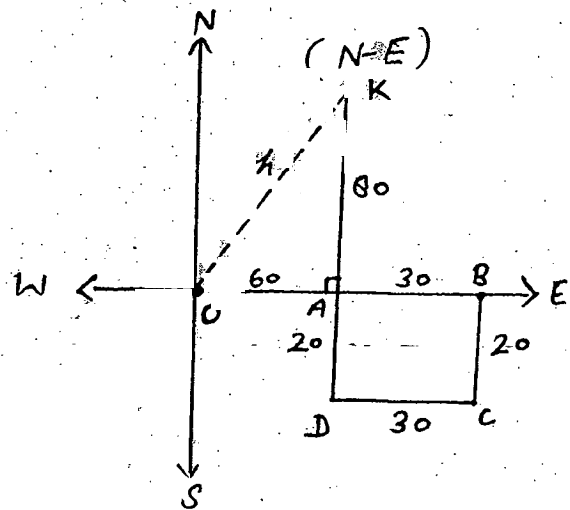
$$\begin{aligned}
 &\left\{ \begin{aligned} OA &= 10 \text{ km} \\ AB &= 6 \text{ km} \\ \therefore OB &= 10 - 6 = 4 \\ BC &= 3 \text{ km} \\ \therefore OC &= 5 \text{ km. (Pythagoras theorem)} \end{aligned} \right.
 \end{aligned}$$



- (10.) A child is looking for his father. He went 90 mtr. in East before turning to his right. He went 20 mtr. before turning to his right again to look for his father at his uncle's place. Which is 30 mtr. from this point but his father was not there. From there he went 100 mtr. towards North where he meets his father in a street so how far did the son meet his father from starting point?

$\Rightarrow$  100 mtr. [N-E]

$$\begin{aligned} h &= \sqrt{80^2 + 60^2} \\ &= \sqrt{6400 + 3600} \\ &= \sqrt{10000} \\ h &= 100 \text{ (N-E)} \end{aligned}$$



- (11.) A child moves  $\sqrt{2}x$  km. towards East from his house and then  $x$  km. towards North and then  $x$  km. towards North-East and then he turns  $90^\circ$  towards right and cover  $x$  km. and reaches his school. so what is shortest distance between his school and house?

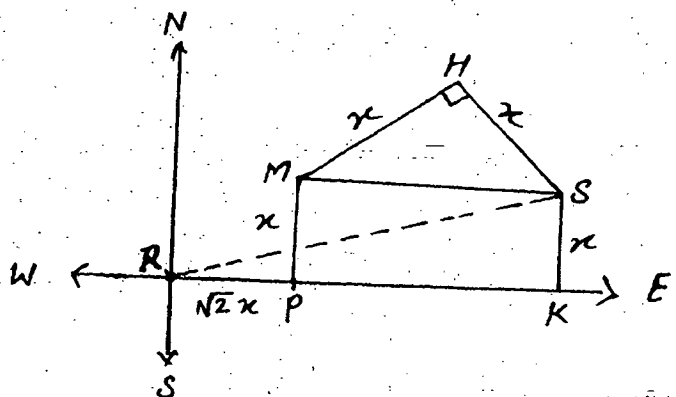
$\Rightarrow$  3x km Ans.

or  $\Delta KRS$

$$\begin{aligned} RS^2 &= SK^2 + RK^2 \\ &= (x)^2 + (KP + RP)^2 \\ &= x^2 + (\sqrt{2}x + \sqrt{2}x)^2 \\ &= x^2 + (2\sqrt{2}x)^2 \\ &= x^2 + 8x^2 \end{aligned}$$

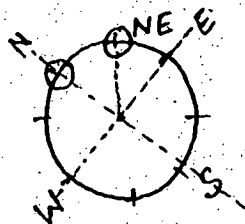
$$RS^2 = 9x^2$$

$$\boxed{RS = 3x}$$



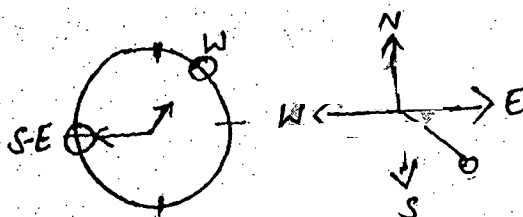
- (12.) It is 3 o'clock in a watch. If minute hands points towards North-East then in which direction the hour hand will be at 10:30 PM.

⇒ North



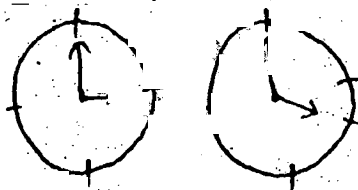
- (13.) It is 4:45 in a watch. If minute hands point towards S-E. Then which direction hour hand will be at 1:30?

⇒ West



- (\*) By how much time the minute hand gains over an hour hand?

⇒ 55 minute



In One hour -

Minute hand = 60 min

Hour hand = 5

55 Min

55 Min gain → 60 Min

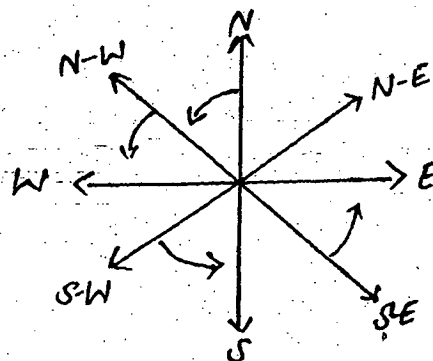
So, 1 Min gain →  $\frac{60}{55}$  Min

$= \frac{12}{11}$  min

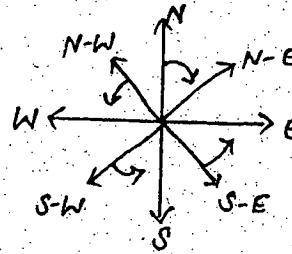
25/35/45 gain →  $\frac{12}{11} \times 25/35/45$

- (14.) IF South-East is called East. South-West is called South. North-West is called west, then what will North be called?

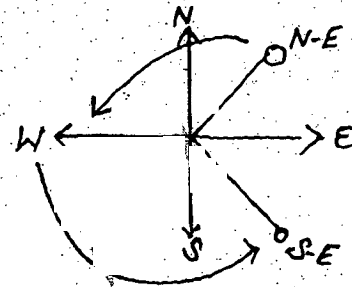
⇒ (North-West)



- (15.) IF South-East is called East. South-West is called South. North-West is called West. Then which direction will become North?  
 $\Rightarrow$  (North-East)



- (16.) If West is called S-E then what will North-East be called?  
 $\Rightarrow$  West



- (17.) One morning after sunrise Rahul and Kavita is talking face to face. If Kavita's shadow fall exactly to right of Rahul then in which direction Kavita's face?  
 $\Rightarrow$  North.

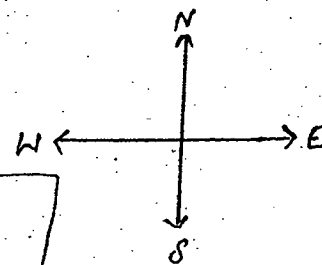
Shadow — West  
 Sunrise — East  
 Rahul's face — South.

- (18.) One morning after sunrise Rahul and Kavita is talking. If Kavita's shadow fall exactly to right of Rahul then in which direction Kavita's face?  
 $\Rightarrow$  Can not be determine.

- (19.) A man is performing yoga with his hands down and legs up, if his face is to West then in which direction his left hand will be?

$\Rightarrow$  North.

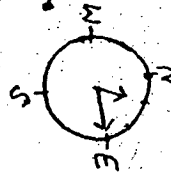
face — West  
 then Right hand — South  
 Left hand — North





- (20.) A watch reads 4:30. If the minute hand points East in which direction will the hour hand point? (LDC-2011)

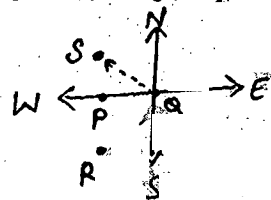
⇒ North-East.



if at 4:30 minute hand points East it means direction is rotated by  $90^\circ$  anti-clockwise in this way, At 4:30 hour hand points towards North-East.

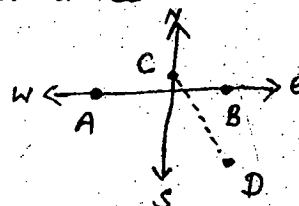
- (21.) Four Players P, Q, R and S are standing a play field in such a way that Q is to East of P, R is to the South of P and S is to the North of P. In which direction of Q is S standing? (SSC CGL-2014)

⇒ North-West



- (22.) A is located to the West of B. C is located at North in between A and B. D is exactly to the South of B and also in line with P. In which direction of C is D located? (CGL-2014)

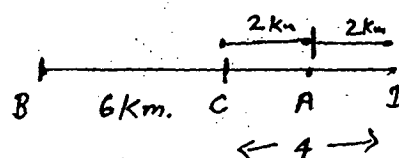
⇒ South-East



D is in South-East wrt C.

- (23.) Four villages A, B, C and D lie in a straight line. D is 10 kms. from B. A is exactly between D and C, and C from B is 2 kms. more than it is from D. How far is C from B? (CPO-2013)

⇒ 6 Km.

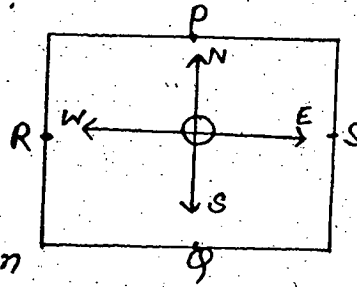


Distance between B and C = 6 Km.



- (24.) P, Q, R and S are playing a game of carom. P, R and S, Q are partners. S is the right of R who is facing West. Then Q is facing in which direction? (CPO-2012)

⇒ North

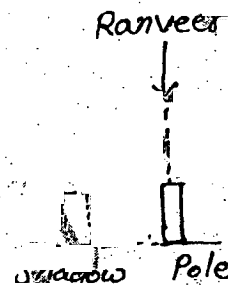


Q is facing in north direction

- (25.) One morning after sunrise, Ranveer was walking facing a Pole. The shadow of the pole fell exactly to his right, which direction was he facing? (CONL 20-4-2014)

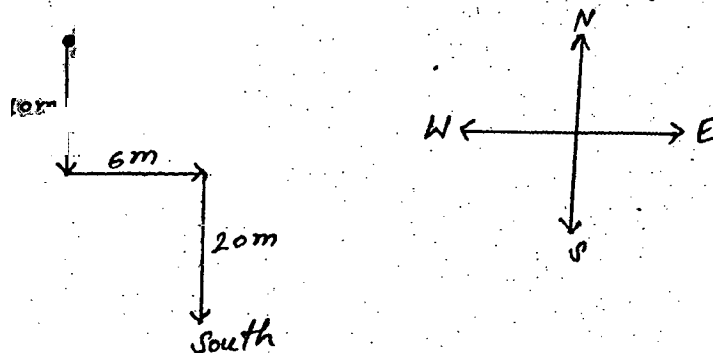
⇒ South

He was facing in south.



- (26.) After walking 10m, Shankar turned left and covered a distance of 6 m., then turned right and covered a distance of 20m. In the end, he was moving towards the south. In which direction did Shankar start his Journey? (CPO-2011)

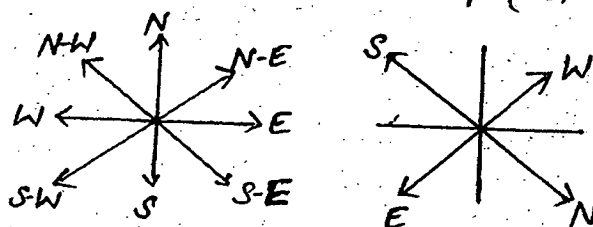
⇒ North



It is clear that Shankar moves from North to South.

- (27.) If South-East become North, North-East become West and so on. Which direction will become South? (CLL-2013)

⇒ North-West



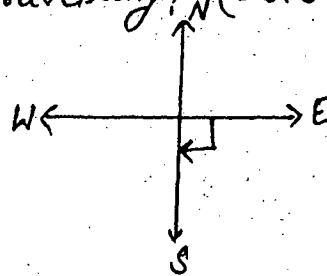
It is clear that North-West will become South.

- (20.) A direction pole was situated on the road crossing. Due to an accident, the pole turned in such a manner that the pointer which was showing East, started showing South. Sita, a traveller went to the wrong direction thinking it to be West. In what direction actually she was travelling? (IBPS-2013)

⇒ **North**

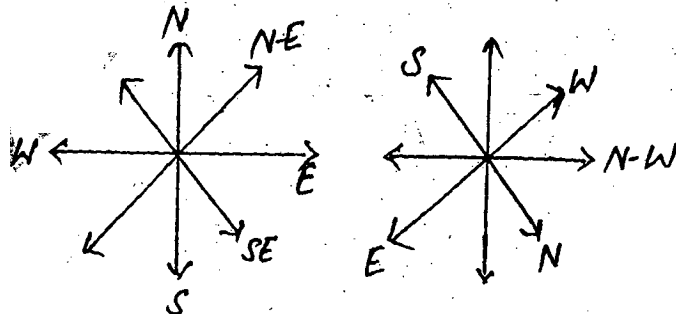
— Due to an accident, the pointer showing East, started showing South.

It means, the pole has been rotated through  $90^\circ$  clockwise. So, when she was travelling towards West, actually it was North.



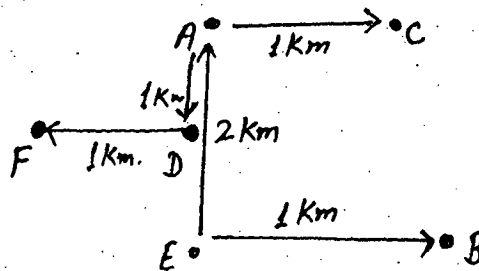
- (29.) If South-East becomes North and North-East becomes West and all directions change accordingly, then what will be the new direction for East? (SSC CGL 20-07-2014)

⇒ **North-West**



- (30.) There are six villages A, B, C, D, E and F. F is 1 km. to the West of D. B is 1 km to the East of E. A is 2 km to the North of E. C is 1 km to the East of A. D is 1 km to the South of A. Which three villages are in a line? (CSAT-2014)

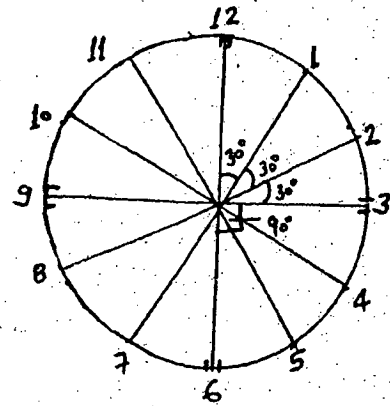
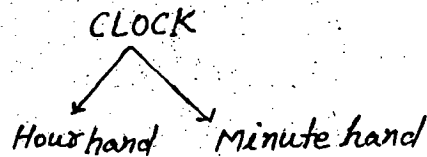
⇒ (A, D, E)



A, D, E are three villages which are in a line.

# \* — CLOCK — \*

(\*)



For Second hand →  $60 \text{ Sec} = 360^\circ$

$1 \text{ Sec} = 6^\circ$

$5 \text{ Sec} = 30^\circ$

For Minute hand :-  $60 \text{ Min} = 360^\circ$

$1 \text{ Min} = 6^\circ$

Hour hand :  $12 \text{ hours} = 360^\circ$

$1 \text{ hour} = 30^\circ$

| Both hands                  | Required Angle                  | Number of times it happens in 12 hours |
|-----------------------------|---------------------------------|----------------------------------------|
| to be coincident            | $0^\circ$                       | 11                                     |
| to be at right angle        | $90^\circ$                      | 22                                     |
| to be in opposite direction | $180^\circ$                     | 11                                     |
| to be in straight line      | $0^\circ \text{ or } 180^\circ$ | 22                                     |

|                                                                                                                                                                          |                                                                                                                                                 |                                                                                                                                                 |                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| $\left. \begin{array}{l} 12:00 \\ 01:05 \\ 02:11 \\ 03:16 \\ 04:22 \\ 05:27 \\ 06:32 \end{array} \right\} \begin{array}{l} 65 \text{ min} \\ 65 \text{ min} \end{array}$ | Approximately<br>Co-incide                                                                                                                      | Approximately<br>opposite                                                                                                                       | $\left. \begin{array}{l} 6:00 \\ 7:05 \\ 8:11 \\ 9:16 \\ 10:22 \\ 11:27 \end{array} \right\} \begin{array}{l} 65 \text{ min} \end{array}$ |
|                                                                                                                                                                          | $1 \text{ hour} \rightarrow 1 \text{ time}$<br>$12 \text{ hours} \rightarrow 11 \text{ time}$<br>$24 \text{ hours} \rightarrow 22 \text{ time}$ | $1 \text{ hour} \rightarrow 1 \text{ time}$<br>$12 \text{ hours} \rightarrow 11 \text{ time}$<br>$24 \text{ hours} \rightarrow 22 \text{ time}$ |                                                                                                                                           |

Approximately right angle after every 33 Mins

TYPE-1

Formula —

$$30H - \frac{11M}{2}$$

↓ Hour time
 ↓ Minute time

(1.) What will be the angle between two hands of a clock at 7:20 A.M.?

$$\Rightarrow 30 \times 7 - \frac{11}{2} \times 20$$

$$= 210 - 110 = \boxed{100^\circ}$$



\* Reflex Angle =  $360^\circ - 100^\circ = 260^\circ$

| (i) 7:20    |             | (ii) 7:20   |             | (iii) 7:20   |  |
|-------------|-------------|-------------|-------------|--------------|--|
| Angle       | R. Angle    | Angle       | R. Angle    | Larger Angle |  |
| (a) 130     | (a) 130     | (a) 130     | (a) 130     | (a) 130      |  |
| (b) 160     | (b) 160     | (b) 160     | (b) 166     | (b) 160      |  |
| (c) 260 (✓) | (c) 360     | (c) 260     | (c) 260 (✓) | (c) 260 (✓)  |  |
| (d) 360     | (d) 100 (✓) | (d) 100 (✓) | (d) 100     | (d) 100      |  |

(2.) What will be the angle between two hands of a clock at 3:40 ?

$$\Rightarrow 30 \times 3 - \frac{11}{2} \times 40$$

$$= 90 - 220$$

$$= \boxed{-130^\circ}$$

- (a)  $120^\circ$   
 (b)  $130^\circ$  (✓)  
 (c)  $220^\circ$   
 (d)  $230^\circ$

Trick -

①

|       |              |
|-------|--------------|
| 7:35  | $17.5^\circ$ |
| 10:50 | $25^\circ$   |
| 9:45  | $22.5^\circ$ |
| 5:25  | $12.5^\circ$ |

If Hour : Minute = 1 : 5  
 then Angle is =  $\frac{\text{Minute}}{2}$

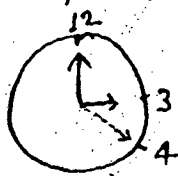
②

|       |            |
|-------|------------|
| 9:36  | $72^\circ$ |
| 4:16  | $32^\circ$ |
| 11:44 | $88^\circ$ |
| 3:12  | $24^\circ$ |

If Hour : Minute = 1 : 4  
 then Angle is =  $2 \times (\text{Minute})$

TYPE-2: (3) At what time between 3 and 4, the hands of clock co-incide?

⇒



Move / gain

$$\begin{aligned} & \frac{12}{11} \times 15 \quad \leftarrow \begin{array}{l} \text{(a) Minute to be gain} \\ \text{(b) Gap between HH \& MH.} \end{array} \\ & = \frac{180}{11} \\ & = 16 \frac{9}{11} \text{ Min} \end{aligned}$$

$$\begin{array}{r} 3 \\ \times 5 \\ \hline 15 \end{array}$$

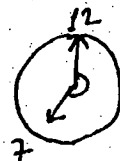
Approximately time for co-incide =  $3 \text{ hrs } 16 \frac{9}{11} \text{ Min}$

(4) At what time between 7 & 8 the hands of clock meet together.

⇒

$$\begin{aligned} \frac{12}{11} \times 35 &= \frac{420}{11} \\ &= 38 \frac{2}{11} \text{ Min} \end{aligned}$$

$$\begin{array}{r} 7 \\ \times 5 \\ \hline 35 \end{array}$$

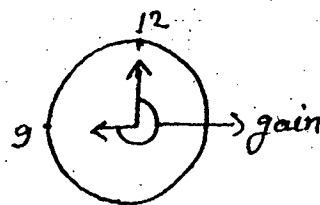


Time =  $7 \text{ hrs } 38 \frac{2}{11} \text{ Min}$

$$\left\{ \frac{12}{11} \times \text{Minute to be gain} \right\}$$

(5) At what time between 9 & 10 the hands of the clock are straight line but not opposite to each other.

$$\begin{aligned} \frac{12}{11} \times 45 &= \frac{540}{11} \\ &= 49 \frac{1}{11} \end{aligned}$$



Time =  $9 \text{ hrs } 49 \frac{1}{11} \text{ Min}$

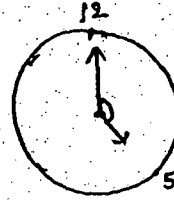
(6) At what time between 5 & 6 the hands of a clock are 3 mins apart from each other.

⇒

$$\frac{12}{11} \times 25 = \frac{300}{11}$$

$$= 27 \frac{3}{11} \text{ Mins}$$

$$\begin{array}{r} 5-6 \\ \downarrow \\ \times 5 \\ \hline 25 \end{array}$$



For 3 min gain →  $\frac{12}{11} \times 3 = \frac{36}{11} = 3 \frac{3}{11} \text{ Min}$

A.T.Q.

$$\boxed{27 \frac{3}{11} - 3 \frac{3}{11}} = 24 \text{ Min}$$

∴ Time = 5 hrs 24 Min

$$\boxed{\text{OR}} \quad 27 \frac{3}{11} + 3 \frac{3}{11}$$

$$= 30 \frac{6}{11}$$

Time = 5 hrs 30  $\frac{6}{11}$  Min

Imp

| First × 5                               | (3-4)<br>Co-inside        | Meet together<br>(7-8)    | (9-10)<br>Co-inside<br>S.L. < [opposite] × | 5-6                       |                           |
|-----------------------------------------|---------------------------|---------------------------|--------------------------------------------|---------------------------|---------------------------|
| Gross Value                             | 15                        | 35                        | 45                                         | 25                        | 25                        |
| Condition                               | - 0                       | - 0                       | - 0                                        | - 3                       | + 3                       |
| Net Value                               | 15                        | 35                        | 45                                         | 22                        | 28                        |
| $\frac{12}{11} \times \text{Net Value}$ | $\frac{12}{11} \times 15$ | $\frac{12}{11} \times 35$ | $\frac{12}{11} \times 45$                  | $\frac{12}{11} \times 22$ | $\frac{12}{11} \times 28$ |

Condition :

co-inside = 0

opposite = 30 Min

Right Angle = 15 Min

(7) At what time between 7 am. and 8 am. the hands of clock are in opposite direction.

⇒

Net Value = Gross Value - Condition

$$= 35 - 30 = 5$$

$$\begin{array}{l} \text{Gross Value} \\ = 7 \times 5 \\ = 35 \end{array}$$

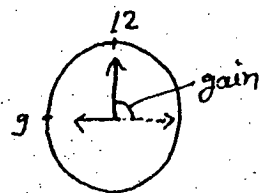
$$\therefore \frac{12}{11} \times 5 = 5 \frac{5}{11} \text{ Min}$$

$$\therefore \boxed{\text{Time} = 7 \text{ hrs } 5 \frac{5}{11} \text{ Mins}}$$



(8) At what time between 9 am. and 10 am. the hands of clock are in straight line but does not co-incide?

$$\Rightarrow \frac{12}{11} \times 15 = \frac{180}{11} = 16 \frac{4}{11}$$



$$\text{Time} = \boxed{9 \text{ hrs } 16 \frac{4}{11} \text{ Mins}}$$

(9) At what time between 4 am and 5 am the hand of clock are in opposite direction?

$$\Rightarrow \frac{12}{11} \times 50 = \frac{600}{11} = 54 \frac{6}{11} \text{ Min}$$

if Gross Value < Condition  
then gross value + Condition

$$\text{Time} = \boxed{4 \text{ hrs } 54 \frac{6}{11} \text{ Min}}$$

(10) At what time between 4 & 5 the hands of clock are in right angle.

$$\Rightarrow \text{Gross Value} = 4 \times 5 = 20$$

$$\text{Condition} = 15$$

$$\text{Net Value} = 20 - 15 = 5$$

$$\frac{12}{11} \times 5 = \frac{60}{11} = 5 \frac{5}{11} \text{ Min}$$

$$\text{Time} = \boxed{4 \text{ hrs } 5 \frac{5}{11} \text{ Mins}}$$

(11) At what time between 6-7 the hands of clock are at right angle.

$$\Rightarrow \frac{12}{11} \times 15 = \frac{180}{11} = 16 \frac{4}{11} \text{ Min} \quad \left| \quad \frac{12}{11} \times 45 = \frac{540}{11} = 49 \frac{1}{11} \text{ Min}$$

$$\text{Time} = \boxed{6 \text{ hrs } 16 \frac{4}{11} \text{ Mins}}$$

$$\text{Time} = \boxed{6 \text{ hrs } 49 \frac{1}{11} \text{ Mins}}$$



(12) At what time between 1 & 2, the hands of clock are at right angle.

⇒

1-2

$$\text{Gross Value} = 1 \times 5 = 5$$

$$\text{Condition} = -15$$

$$\text{Net Value} = -10$$

$$\text{So, Net Value} = 60 - 10 = 50$$

$$\frac{12}{11} \times 50 = \frac{600}{11} = 54 \frac{6}{11} \text{ Min}$$

$$\text{Time} = \boxed{1 \text{ hour } 54 \frac{6}{11} \text{ Min}}$$

OR

$$\text{Gross Value} = 1 \times 5 = 5$$

$$\text{Condition Value} = 15$$

$$\text{Net Value} = 20$$

$$\therefore \frac{12}{11} \times 20 = \frac{240}{11} = 21 \frac{9}{11} \text{ Min}$$

$$\text{Time} = \boxed{1 \text{ hour } 21 \frac{9}{11} \text{ Min}}$$

(13) (10-11) - Right Angle

$$\begin{array}{r} 50 \\ -15 \\ \hline 35 \end{array}$$

$$\frac{12}{11} \times 35 = \frac{420}{11} = 38 \frac{2}{11} \text{ Min}$$

$$\text{Time} = \boxed{10 \text{ hrs } 38 \frac{2}{11} \text{ Min}}$$

$$\begin{array}{r} 50 \\ +15 \\ \hline 65 \end{array}$$

$$65 - 60 = 5$$

$$\frac{12}{11} \times 5 = \frac{60}{11} = 5 \frac{5}{11}$$

$$\text{Time} = \boxed{10 \text{ hrs } 5 \frac{5}{11} \text{ Min}}$$

(14.) At what time between 5:30 & 6:00, the hands of clock are at right angle.

⇒

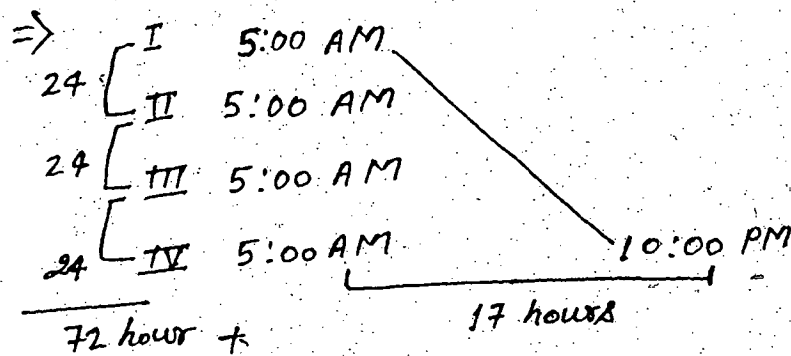
$$\begin{array}{r} 25 \\ +15 \\ \hline 40 \end{array} \quad (\checkmark)$$

$$\begin{array}{r} 25 \\ -15 \\ \hline 10 \end{array} \quad \times$$

$$= \frac{12}{11} \times 40 = \frac{480}{11} = 43 \frac{7}{11} \text{ Mins}$$

$$\text{Time} = \boxed{5 \text{ hrs } 43 \frac{7}{11} \text{ Min}}$$

TYPE-3: (15.) A clock loses 16 min in every 24 hours. if it is set right at 5 in the morning. Then what will be the true time when clock shows 10 PM the fourth day.



Total hours = 89 hours

|               |               |
|---------------|---------------|
| Wrong         | Correct/clock |
| 23 hrs 44 min | = 24 hrs      |

↓

$$23\frac{44}{60} \text{ hrs} = 24 \text{ hrs}$$

↓

$$\frac{356}{15} \text{ hrs} = 24 \text{ hrs}$$

1 hrs =  $\frac{24 \times 15}{356}$

$$\therefore 89 \text{ hrs} = \frac{24 \times 15 \times 89}{356} = 90 \text{ hrs} \quad \left[ \because 89 \text{ hrs} = 10:00 \text{ PM} \right]$$

$$\therefore \boxed{\text{True time} = 11:00 \text{ PM}}$$

Formula ⇒ ①  $\frac{C}{W} \times T.T$  (Total time)

$$\left[ \begin{array}{l} C = \text{Clock Time} \\ W = \text{Wrong time} \end{array} \right]$$

$$= \frac{24}{23 \text{ hrs } 44 \text{ min}} \times 89$$

$$= \frac{24}{356} \times 15 \times 89 = 90 \text{ hrs}$$

$$= \boxed{11:00 \text{ PM}}$$

II<sup>nd</sup> Formula :-  $\boxed{\frac{L/G}{W} \times T.T}$   $\begin{pmatrix} L - \text{Late} \\ G - \text{Gain} \end{pmatrix}$

$$\frac{16 \text{ Min}}{23 \text{ hrs } 44 \text{ Min}} \times 89 = \frac{16^4}{356} \times 15 \times 89 = 60 \text{ Min}$$

True Time = 11:00 PM

(16.) A clock gain 10 minute in every 24 hours. If it is set right 8:00 in the morning then what will be the true time when clock shows 1 PM the next day.

$$\Rightarrow \frac{10}{24 \text{ h } 10 \text{ Min}} \times 29 \quad \Rightarrow 24 \frac{10}{60} = \frac{145}{6}$$

$$\therefore \frac{10^2}{145} \times 6 \times 29 = 12 \text{ Min}$$

$\therefore \text{True Time} = 12:48 \text{ PM}$

(17.) A clock is 2 min low at 12 PM on Monday and its 4 min 48 sec fast at 2 PM on next Monday so when was its correct time.

$$\Rightarrow \begin{array}{l} \text{C} \rightarrow \begin{array}{c} 12:00 \text{ PM} \text{ (Monday)} \quad \xrightarrow{170 \text{ hrs}} \quad 2:00 \text{ PM (Monday)} \\ \text{W} \rightarrow \begin{array}{c} M \quad M \\ 11:58 \quad 2:04:48 \end{array} \\ \quad \quad \quad 170 \text{ hrs } 6 \text{ Min } 48 \text{ Sec} \end{array} \end{array}$$

$$\begin{array}{l} \Rightarrow 6 \text{ min } 48 \text{ sec} \xrightarrow{\text{gain}} 170 \text{ hrs} \\ 6 \frac{48^4}{605} \text{ Min} \rightarrow 170 \text{ hrs} \\ 1 \text{ Min} \rightarrow \frac{170 \times 5}{34} \\ \therefore 2 \text{ Min} \rightarrow \frac{170 \times 5}{34} \times 2 \\ \quad \quad \quad = 50 \text{ hrs} \end{array}$$

So, After 50 hrs its show correct time.

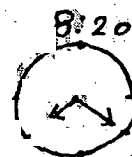
Trick :-  $\frac{L}{L+F} \times T.T.$

II<sup>nd</sup> Method :-  $\frac{2}{2+4.48} \times 170$

$= \frac{2}{6 \text{ Min } 48 \text{ Sec}} \times 170 = \frac{2}{34} \times 5 \times 170 = \boxed{50 \text{ hrs}}$

(18.) It is 8:20 in the clock. what is its reflected image of clock?

$\Rightarrow \begin{array}{r} 23:60 \\ - 8:20 \\ \hline 15:40 \end{array} \text{ or } \boxed{3:40}$



(19.) What is the reflected image of clock at 2:29?

$\Rightarrow \begin{array}{r} 23:60 \\ - 2:29 \\ \hline 21:31 \end{array} \text{ or } \boxed{9:31}$

(20.) Through how many degrees does hour hand in a clock move as the time changes from 3 hours 12 minutes to 6 hours?

$\Rightarrow$  Hour hand moves  $1/2^\circ$  in 1 minute. (UP PCS-2015)

$\therefore$  Required time in minute from 3:12 to 6:00

$= 48 + 60 + 60 = 168 \text{ Min}$

$= \boxed{84^\circ}$

(21.) How many times in a day are the hands of a clock overlap each other? (CGLE-2014)

$\Rightarrow$  (A) 21 (B) 24 (C) 22 (D) 23

In 24 hour, hands will overlap each other  $= \frac{24 \times 11}{12}$

$= \boxed{22 \text{ times}}$

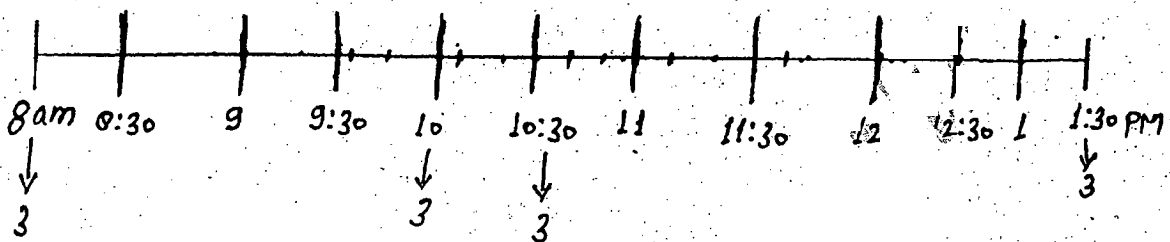
(22) In a school, the bell is rung once after each half an hour. The school starts at 8:00 a.m. and closes at 1:30 p.m. The bell is rung 3 times continuously, at the time of beginning, at the time of lunch break at 10:00 and 10:30 am and at the end. How many times is the bell rung every day?

(A) 21

(B) 22

(C) 19

(D) 20 (✓)



∴ Total No. of rings = 20 (CGL-2014)

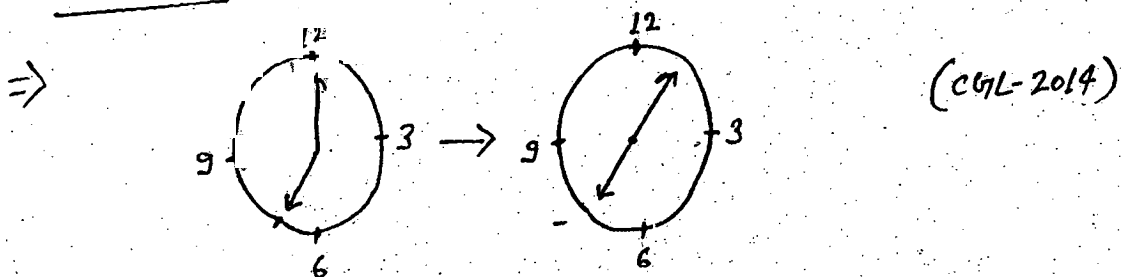
(23) After 7 o'clock at what time between 7 am and 8 am will the hour and minutes hand of a clock point in opposite direction?

(A)  $7:5\frac{5}{11}$  (✓)

(B) 7:16

(C) 7:6

(D)  $7:16\frac{4}{11}$



Minute hand has to travel 5 minutes space.

Now, 55 minute space is gained in 60 minute.

∴ 5 minute space is gained in  $\frac{60}{55} \times 5 = 5\frac{5}{11}$

∴ at  $7:5\frac{5}{11}$  am two hands will be in opposite direction.

(24.) What will be the angle between two hands of a clock at 8:30 am? (CGL-2014)

- (A)  $35^\circ$  (B)  $75^\circ$  (✓) (C)  $60^\circ$  (D)  $45^\circ$

$$\Rightarrow \left| 30H - \frac{11}{2}M \right| = 30 \times 8 - \frac{11}{2} \times 30^{15} = 240 - 165 = \boxed{75^\circ}$$

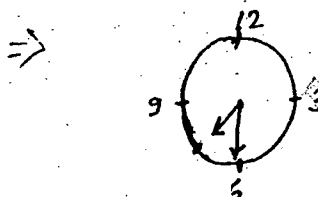
(25) Train to delhi leaves every 6 hours. At the information counter. I learnt that the train has left the station 35 min before. If the time now is 11:45 am. What is the time for next train? (CGL-2015)

- (A) 5:45 pm (B) 4:45 pm (C) 4:10 pm (D) 5:10 pm (✓)

$\Rightarrow$  1<sup>st</sup> train has left at 11:10 am next train will be after 6 hour i.e. at 5:10 pm.

(26.) A watch reads 7:30. if the minute hand points West, then in which direction will the hour hand point? (CGL-2013)

- (A) North (B) North-East (C) North-west (✓) (D) South-East.



if the minute hand point in west, the hour hand will be in-

North - West.

(27.) By looking in a mirror it appears that it is 6:30 in the clock. What is the real time? (CGL-2010)

- (A) 6:30 (B) 5:30 (✓) (C) 6:00 (D) 5:00

$\Rightarrow$

|    |                                                              |    |      |
|----|--------------------------------------------------------------|----|------|
| OR | $\begin{array}{r} 23:60 \\ 6:30 \\ \hline 17:30 \end{array}$ | or | 5:30 |
|----|--------------------------------------------------------------|----|------|

When it appears 6:30, real time will be 5:30.

(28.) The chairman of the Selection Committee arrived at the interview room for conducting an interview at 10 min to 12:30 hrs. He was earlier by twenty minutes than the other members of the board, who arrived late 30 min. At what time were the interview scheduled? (CGL-2012)

(A) 12:10 (✓) (B) 12:20 (C) 12:30 (D) 12:40

⇒ Chairman arrived at → 12:20  
other members → 12:40

∴ Scheduled time of interview = 12:10

(29.) A clock gains five minutes every hour. What will be the angle traversed by the second hand in one minute? (CGL-2011)

(A)  $360^\circ$  (B)  $360.5^\circ$  (C)  $390^\circ$  (✓) (D)  $380^\circ$

⇒ Each second space =  $1^\circ$

A clock gains 5 minute every hour it means the clock gains  $\frac{5}{60}$  minute in one minute =  $\frac{5}{60} \times 360 = 30^\circ$

∴ The second hand will traverse  $360 + 30 = \boxed{390^\circ}$  in one minute.

(30.) After 9 o'clock at what time between 9 pm and 10 pm will the hour and minute hands of a clock point in opposite direction? (LDC-2014)

$$\Rightarrow \frac{12}{11} (5x - t)$$

$$= \frac{12}{11} (5 \times 9 - 30) \quad \text{here } x = 9, t = 30^\circ (\because \text{opposite direction})$$

$$= \frac{12}{11} \times 15 = 16 \frac{4}{11} \text{ min}$$

∴ Hands will be in opposite direction at  $9:16 \frac{4}{11}$

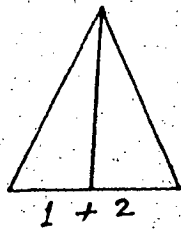
$$= \boxed{9 \text{ hrs } 16 \frac{4}{11} \text{ min}}$$



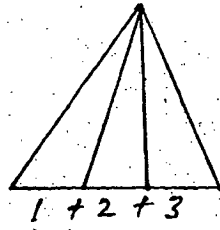
\* ——— Counting of figure ——— \*

(\*) How many triangle there in the figure?

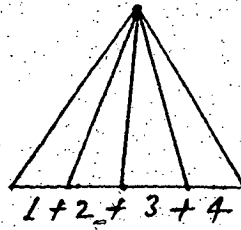
(1.)



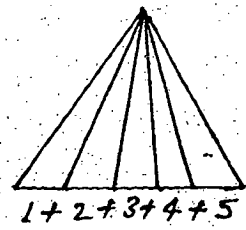
$$1 + 2 = 3$$



$$1 + 2 + 3 = 6$$

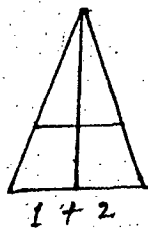


$$1 + 2 + 3 + 4 = 10$$

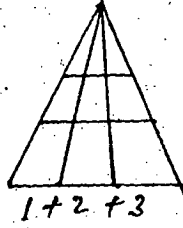


$$1 + 2 + 3 + 4 + 5 = 15$$

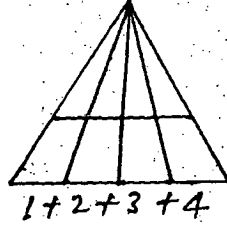
(2.)



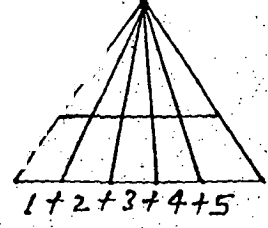
$$1 + 2 = 3 \times 2 = 6$$



$$1 + 2 + 3 = 6 \times 3 = 18$$

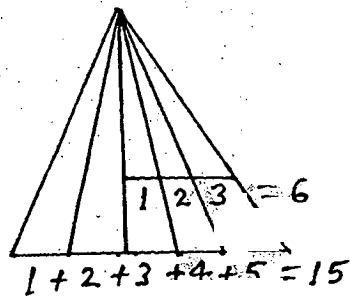


$$1 + 2 + 3 + 4 = 10 \times 2 = 20$$



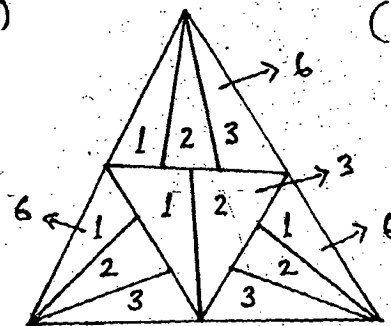
$$1 + 2 + 3 + 4 + 5 = 15 \times 2 = 30$$

(3.)



$$\text{Total} = 15 + 6 = 21$$

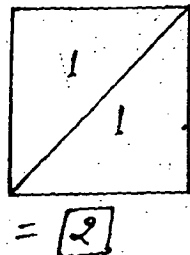
(4.)



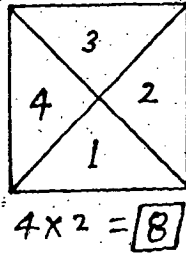
$$\text{Total} = 6 + 6 + 6 + 3 + 1 = 22$$

(CGL-2014, 15)

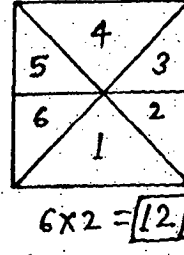
(5.)



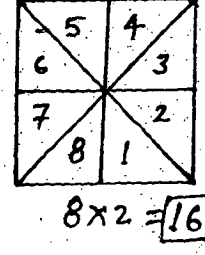
$$= 2$$



$$4 \times 2 = 8$$



$$6 \times 2 = 12$$

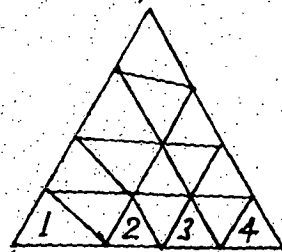


$$8 \times 2 = 16$$

TRICK/Note  $\Rightarrow$

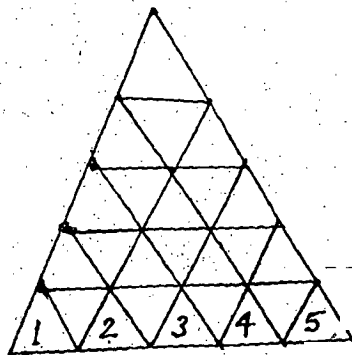
$$\text{Big Number} \times 2 = \text{Total Triangle}$$

How many Triangle in this figure!  
(6.)



$$\begin{aligned} (n=4) \\ \text{Even triangle of the base-} \\ &= \frac{n(n+2)(2n+1)}{8} \\ &= \frac{4 \times 6 \times 9}{8} = \boxed{27} \end{aligned}$$

(7.)

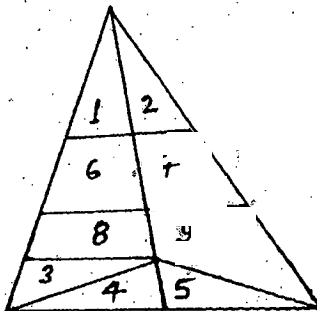


$$\begin{aligned} (n=5) \text{ (odd)} \\ &= \frac{[n(n+2)(2n+1)]-1}{8} \\ &= \frac{[5 \times 7 \times 11]-1}{8} \\ &= \frac{385-1}{8} = \boxed{48} \end{aligned}$$

(CGL-2014)

(8.) Find out triangle in the figure ?

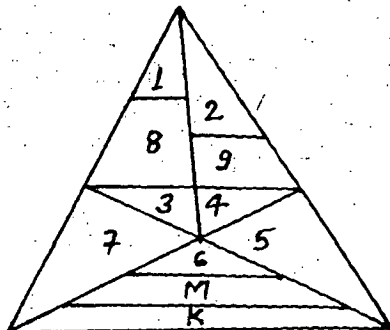
(CPO-2014)



$$\begin{aligned} \Rightarrow 1 \text{ to } 5, 1 \text{ complete,} \\ \underline{16, 168, 1683, 16834,} \\ \underline{12, 1276, 27, 279, 2795} \\ \underline{45} \\ = \boxed{16} \end{aligned}$$

(9.)

(CPO-2013)



$$\begin{aligned} \Rightarrow 1 \text{ to } 7, \text{ complete} \\ \underline{18, 183, 1837, 1892, 1298347} \\ \underline{1298345, 29, 294, 2945} \\ \underline{34, 345, 347, 56MK} \\ \underline{76MK, 6M, 6MK} = \boxed{24} \end{aligned}$$

\* How many square and Rectangle in the figure?

(10.)

|   |   |   |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 9 |

square:  $\frac{1 \text{ to } 9, \text{ complete}}{1254, 2365, 6589}$   
 $\frac{5478}{5478} = \boxed{14}$

(CGL-2013)

Rectangle:  $\frac{12, 123, 23, 45, 456, 56, 78, 789}{89, 14, 147, 47, 25, 258, 58, 36}$

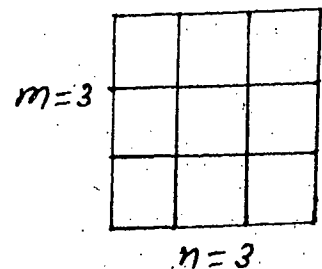
$\frac{369, 69, 123654, 456789, 125478, 236589}{369, 69, 123654, 456789, 125478, 236589} = \boxed{22}$

∴ All square are rectangle.

∴ Total Rectangle =  $14 + 22 = \boxed{36}$

TRICK: Square =  $\frac{n(n+1)(2n+1)}{6}$

$= \frac{3 \times 4 \times 7}{6} = \boxed{14}$



quadrilateral:  $\left[ \frac{n(n+1)}{2} \right]^2$

$= \left( \frac{3 \times 4}{2} \right)^2 = (6)^2 = \boxed{36}$

Rectangle:  $36 - 14 = \boxed{22}$

Trick:-

square:  $\frac{1^2 \rightarrow 1}{2^2 \rightarrow 4}{3^2 \rightarrow 9}$   
 $\frac{14}{14}$

quadrilateral:

$\frac{1^3 \rightarrow 1}{2^3 \rightarrow 8}{3^3 \rightarrow 27}$   
 $\frac{36}{36}$

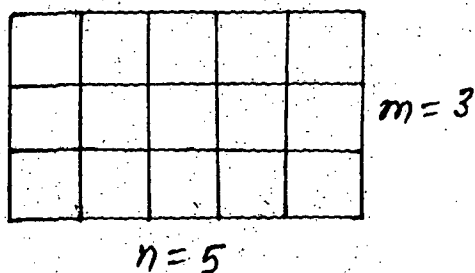
(10.) How many square in a chess board?

- (i) 63      (ii) 64      (iii) 65      (iv) N.O.T (✓)

$$\Rightarrow \text{Square} = \frac{n(n+1)(2n+1)}{6} = \frac{\cancel{4} \times \cancel{3} \times 17}{\cancel{6}_2}$$

$$= 12 \times 17 = \boxed{204}$$

(11.) How many square in the figure?



square:  $5 \times 5 = 15$   
 $4 \times 2 = 8$   
 $3 \times 1 = 3$   
26

Quadrilateral:

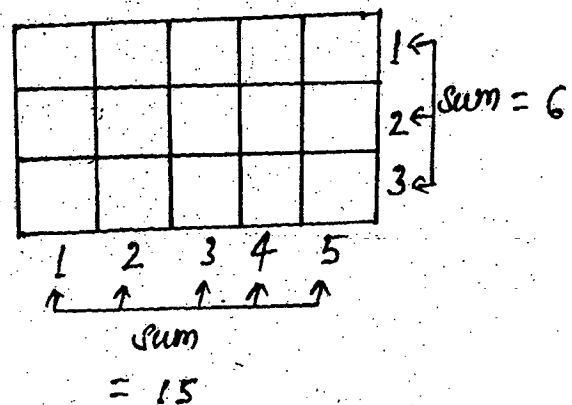
$$\frac{n(n+1)}{2} \times \frac{m(m+1)}{2}$$

$$\frac{5(5+1)}{2} \times \frac{3(3+1)}{2} = 15 \times 6 = \boxed{90}$$

Rectangle:  $90 - 26 = 64$  or  $90$  (As per option)

Trick: Quadrilateral:

$$\Rightarrow 15 \times 6 = \boxed{90}$$



(12.) Find out the number of triangle in the given figure?

(CGL-2011)

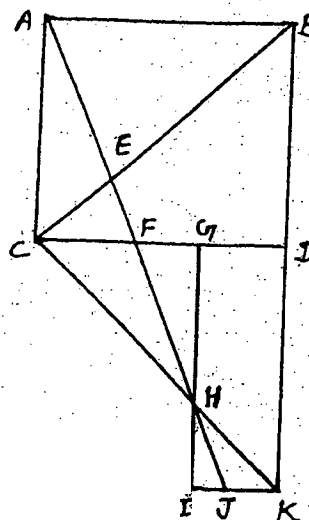
(i) 12

(ii) 11

(iii) 15 (✓)

(iv) 13

⇒ Triangle:  $\triangle ABE$ ,  $\triangle BCK$ ,  $\triangle FGH$   
 $\triangle ACE$ ,  $\triangle CEF$ ,  $\triangle HIJ$   
 $\triangle ABC$ ,  $\triangle CEH$ ,  $\triangle HIK$   
 $\triangle ACH$ ,  $\triangle CDK$ ,  $\triangle HJK$   
 $\triangle ACF$ ,  $\triangle CGH$ ,  $\triangle BCD$



Total Triangle = 15

(13.) Find out the number of triangle in the given figure?

(LDC-2012)

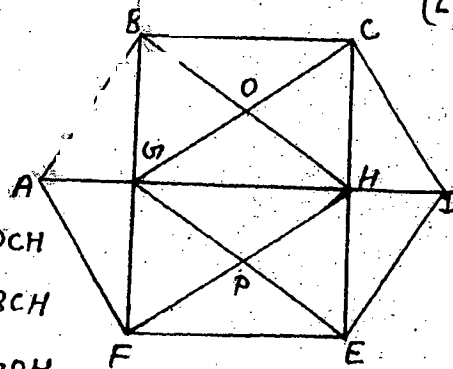
(i) 26

(ii) 28 (✓)

(iii) 32

(iv) 34

⇒ Triangle:  $\triangle ABG$ ,  $\triangle AGF$ ,  $\triangle AFE$ ,  $\triangle DCH$   
 $\triangle DHE$ ,  $\triangle DCE$ ,  $\triangle BGH$ ,  $\triangle BCH$   
 $\triangle CHG$ ,  $\triangle GCA$ ,  $\triangle ROG$ ,  $\triangle GOH$   
 $\triangle OHC$ ,  $\triangle BOC$ ,  $\triangle GCD$ ,  $\triangle ABH$ ,  $\triangle GHE$ ,  $\triangle GEF$ ,  $\triangle GDE$   
 $\triangle CGE$ ,  $\triangle AHE$ ,  $\triangle BHF$ ,  $\triangle GHF$ ,  $\triangle HFE$ ,  $\triangle GHP$ ,  $\triangle HEP$ ,  $\triangle EPF$ ,  $\triangle FGP$ ;  
 Now Total triangle = 28



(14.) How many triangle in the given figure? (CGL-2014)

(i) 11

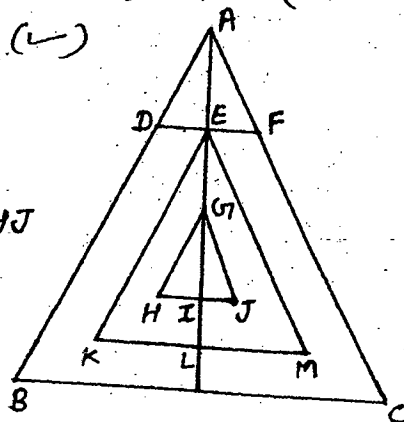
(ii) 12 or more than (✓)

(iii) 9

(iv) 10

⇒ Triangle:  $\triangle ADF$ ,  $\triangle ADE$ ,  $\triangle AFE$   
 $\triangle EKM$ ,  $\triangle ELK$ ,  $\triangle ELM$ ,  $\triangle GHJ$   
 $\triangle GHI$ ,  $\triangle GIJ$ ,  $\triangle ABC$ ,  $\triangle ANB$ ,  $\triangle ANC$

No. of Triangle = 12



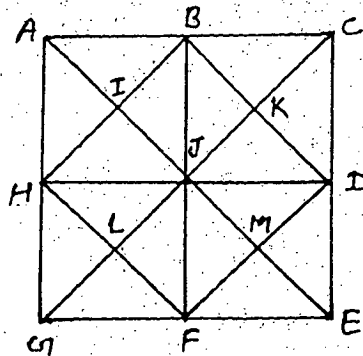
(15.) How many triangle in the given figure? (CGL-2013)

(i) 40 or More than (✓)

(ii) 16

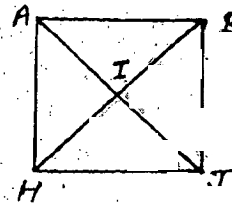
(iii) 18

(iv) 25



⇒ Firstly calculate the no. of triangle in 1<sup>st</sup> part.

Triangle:  $\triangle AIB, \triangle IJB, \triangle IAH, \triangle IHT$   
 $\triangle AHB, \triangle ABT, \triangle AHT, \triangle BTH$



Because no. of total part is 4.

So, no. of triangle =  $4 \times 8 = 32$

⇒ Now, calculate the triangle which is more than one part.

$\triangle JAC, \triangle JAG, \triangle JCE, \triangle JGE, \triangle BHD, \triangle DBF, \triangle FDH, \triangle HBF$   
 $\triangle AGE, \triangle CEG, \triangle CAE, \triangle AGC$

So, total triangle = 44

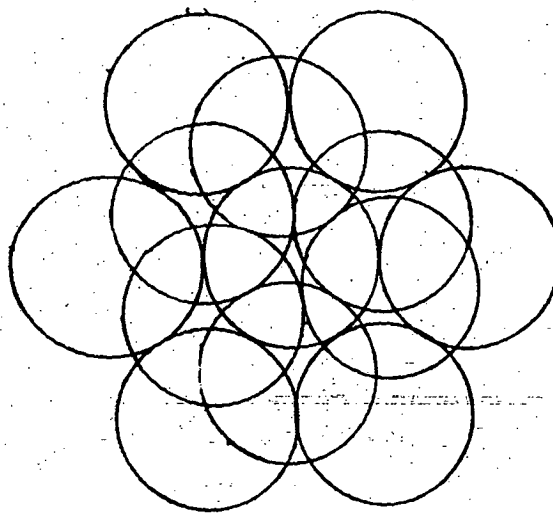
(16.) How many circle in the figure? (CGL-2013)

(i) 12

(ii) 13 (✓)

(iii) 14

(iv) 11



No. of Circle = 13



## \* PROBLEMS BASED ON ALPHABET \*

⊗ Some important facts to solve the problems based on Alphabet -:

- (1.)  $A \rightarrow Z$  = English alphabet Series
- (2.)  $Z \rightarrow A$  = Reverse order of English alphabet.
- (3.)  $A \rightarrow M$  = First half of English alphabet series.
- (4.)  $M \rightarrow A$  = Reverse order of first half of english alphabet series.
- (5.)  $N \rightarrow Z$  = Second half of English alphabets
- (6.)  $Z \rightarrow N$  = Reverse order of Second half of English alphabets.
- (7.) Important facts on 'left-right' position of english alphabets.

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

→ (Left)

(Right) ←

TYPE-1 : (1) How many pairs of the letters are there in the word 'CREATIVE' which have as many letters between them as in the English alphabet series.

⇒ Three Pairs

CREATIVE  
3 18 5 1 20 9 22 5

(2) How many pairs of the letters are there in the word 'NECESSARY' which have as many letters between them as in the english alphabet series. (SSC-2012)

⇒ 1/2

NECESSARY

1/2 → (SSC) Ans.

(3) PARADISE = 3

(4) DONATE = 2

(5) SEQUENTIAL = 4

(6) CLANGOUR = 5

(7) ABCDE = 10

(8) SEQUENTIAL = 2



TYPE-2: (9) How many meaningful word can be made from the 'LAPAROSCOPY' without changing order of the letter and used each letter only once.

⇒ LAPAROSCOPY → LAP - COPY  
LAP - COP  
PAR - COPY  
PAR - COP } (2)

only two words can be made from these.

\* How many meaningful word can be made under given word without changing order of the letter and used each letter only once.

(10) ASTOUNDER - 3

AS, UNDER, TO = [3]

(11) DETERMINATION

TERM, IN, AT, ON = [4]

(12) HEARTLESS

HE, ART, LESS = [3]

(13) TODAY

TO, DAY = [2]

TYPE-3 (A) How many meaningful word can be made using letters (T, C, A) using each letter only one in each word.

(i) 1 (ii) 2 (iii) 3 (iv) More than 3

⇒ [CAT, ACT]

(15) A, E, R

⇒ ARE, EAR, ERA

~~REAR~~ = [3]

(16) I, T, D, E = [4]

⇒ TIDE, DIET

TIED, EDIT

(17) O, T, P, S = [5]

⇒ TOPS, POST, SPOT

STOP, POTS

TYPE-4: (Arrangement of Alphabets order.)

(18) ① INTELLIGENT, ② INTEND, ③ INTEGUMENT, ④ INTENSE, ⑤ INTELLECT.

(i) 41523

(ii) 53241

(iii) \*34152

(iv) 35124 (✓)

(19.) SPORT, SQUASH, SPROUT, SPOUSE, SPORADIC.

⇒ 51432

(20.) PROLONG, PROHIBIT, PROMISE, PROGRAMME, PROMINENT.

⇒ 42153

TYPE-5:

(21.) 'COMPREHENSION' (Which word can not be framed.)

(i) COMPRISE

(ii) ONION

(iii) PENSION

(iv) PREACH (✓)

(22.) 'MULTIPLICATION' (Which word can be framed)

(i) MATUAL

(ii) APPLICATION

(iii) LIMITATION (✓)

(iv) NOTION

(23.) 'MULTIPLICATION' (can't be framed)

(i) LIMIT

(ii) LIMITATION

(iii) MUTUAL

(iv) APPLICATIONS. (✓)

TYPE-6: (24) In the word 'DOCUMENTATION' if the

position of third and tenth position are interchange and likewise second and sixth and as fourth & seventh then which letter is eleventh from right hand.

⇒ <sup>1 2 3 4 5 6 7 8 9 10 11 12 13</sup>  
DOCUMENTATION  
→ DETNMOU<sup>3</sup>TAC<sup>10</sup>ION  
= T

OR →

Total Letters = 13

⇒  $(13 - 11 + 1) = 3$

3<sup>rd</sup> is interchanged to 10<sup>th</sup>

so = T

(25.) In the word 'DECLARATION' if the position 3 & 9 of the word interchange and like wise 11 & 4 and then 5 & 10 are interchange then which letter is 8<sup>th</sup> from right hand.

$$\Rightarrow 11 - 8 + 1$$

$$= 4$$

$$= \boxed{N} \quad (\text{because } 11 \text{ \& } 4 \text{ are interchanged})$$

TYPE-7 : (26.) The position of how many letters in the word 'BRAKES' remains unchanged when they are arranged in alphabetical order?

$\Rightarrow$

|                                                       |              |              |              |              |              |              |
|-------------------------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|
| position of letters in the given word                 | B            | R            | A            | K            | E            | S            |
|                                                       | $\uparrow$   | $\uparrow$   | $\uparrow$   | $\uparrow$   | $\uparrow$   | $\uparrow$   |
| New position of letters according to English alphabet | A            | B            | E            | K            | R            | S            |
|                                                       | $\downarrow$ | $\downarrow$ | $\downarrow$ | $\downarrow$ | $\downarrow$ | $\downarrow$ |

Therefore, the position of letter K and letter S remain unchanged.

(27.) 'DECLARATION'

$\Rightarrow$  position of letters in the given word  $\rightarrow$  DECLARATION

New position according to alphabet  $\rightarrow$  AACDEILNORT

Therefore, the position of letter C remain unchanged.

(28.) 'DOCUMENTATION'

$\Rightarrow$

|  |   |   |   |   |   |   |   |   |   |   |   |   |   |
|--|---|---|---|---|---|---|---|---|---|---|---|---|---|
|  | D | O | C | U | M | E | N | T | A | T | I | O | N |
|  |   |   |   |   |   |   |   |   |   |   |   |   |   |
|  | A | C | D | E | I | M | N | O | O | T | T | U |   |

Therefore, the position of letter N remain unchanged.

TYPE-8 : (29) Which single letter is to be prefixed in given words to make them all a new meaningful word -

EAT      AT      OUR      IS      AS

(i) S      (ii) H (✓)      (iii) B      (iv) C

⇒ HEAT, HAT, HOUR, HIS, HAS

(30) Which single letter is to be deleted in given words to make them all a new meaningful word -  
(CGL-2011)

COST      POST      HOST      LOST      TOPS

(i) S (✓)      (ii) T      (iii) O      (iv) N.O.T.

⇒ COT, POT, HOT, LOT, TOP

TYPE-9 : (31) From the word 'TRIFU' make a meaningful word using each letter only once?

⇒ <sup>1 2 3 4 5</sup> T R I F U → F R U I T

(i) 31254      (ii) 51243      (iii) 21534      (iv) 42 531 (✓)

|                      |                        |
|----------------------|------------------------|
| (32) TLEMNE → MENTAL | (33) TIRBHG → BRIGHT   |
| (34) RUSGA → SUGAR   | (35) IPELOC → POLICE   |
| (36) HNRCA → BRANCH  | (37) RMNBUE → NUMBER   |
| (38) INLASG → SIGNAL | (39) DIFERN → FRIEND   |
| (40) RTAOUH → AUTHOR | (41) RTANUE → NATURE   |
| (42) INETOC → NOTICE | (43) JCOPTER → PROJECT |

TYPE-10 : (44) IF it is possible to make a meaningful word with 2<sup>nd</sup>, 5<sup>th</sup> and 8<sup>th</sup> letter of "CARETAKER". Then third letter of that word is your answer. IF no such word can be made then give 'X' is Answer. If more than one answer then give 'M' is your Answer.

(i) 'X'      (ii) M (✓)      (iii) A      (iv) E      (v) T

⇒ CARETAKER → A, T, E  
⇒ ATE, EAT, TEA

More than one meaningful word so answer is 'M'

(45) CONTRACT - (1, 2, 4, 5, 6)

Taken word - 1, 2, 4, 5, 6 [C O, T, R, A]

COTRA → ACTOR = T

Because, from these letters we can make only one meaningful word. So according to question its third letter is answer

(46) COUNTERACT , Taken letter → (4, 8, 10) (CPO-2011)

⇒ N, A, T → ANT, TAN = M

More than one meaningful word so answer is 'M'.

(47) TRADEMARK , Taken letter → (1, 3, 5, 6)

⇒ T, A, E, M → TEAM, MEAT, MATE, TAME = M

more than one meaningful word so answer is 'M'.

(48) REASONING , Taken letter → (1, 2, 4, 5, 6)

⇒ R, E, S, O, N → SNORE = O

only one meaningful word so answer is 3<sup>rd</sup> letter.

(49) Which word cannot be formed using the letters of the given word. (LDC-2015)

'RAILWAYSTATION'

(A) STAIR (B) STORY (C) NOTARY (D) OSTITIS (✓)

⇒ There is only one 'S' in word 'RAILWAYSTATION'.

(50) Arrange the following words as per the reverse order in a dictionary. (LDC-2015)

(1) multitude (2) multinational (3) multiplier  
(4) multinomial (5) multilingual.

(A) 13245 (B) 45321 (C) 54321 (D) 13425 (✓)

(51) Which of the following Jumbled words is not an animal?

(A) TICRECK (B) TAR (C) FEFEOC (✓) (D) LATHPEEN

⇒ TICRECK — CRICKET  
TAR — RAT  
LATHPEEN — ELEPHANT

} There are different kind of animals.

FEFEOC — COFFEE — is a beverage.

(CGL-2015)

(52) Name a single letter which can be prefixed to the following words in order to obtain entirely new words?

TILL TABLE SPILE TAB SPRING (CGL-2015)

(A) S (✓) (B) H (C) B (D) C

⇒ STILL, STABLE, SPILE, STAB, SPRING are meaningful words.

(53) Unscramble the following letters to frame a meaningful word. Then find out the correct numerical position of the letters.

B C U S M E L R N A  
1 2 3 4 5 6 7 8 9 10

(CPO-2014)

(A) 61432587910

(B) 31571042698

(C) 39428105176 (✓)

(D) 21346897510

⇒ BCUSMELRNA → UNSCRAMBLE



(54) Decoded the coded word by using these code and key.

(FCI-2013)

Code : L X P Z J Y Q M N B

Key : b a e s p r h i g t

coded word : ZBYXMNB

(A) height (B) struggle (C) straight (✓) (D) strength.

⇒ ZBYXMNB → straight

Direction (55-59): Arrange the following words as per order in the dictionary.

(CGL-2015)

(55) (1) Hale (2) Hake (3) Ham (4) Halt (5) Hamlet

⇒ (A) 2,1,3,5,4 (B) 2,1,3,4,5 (C) 2,1,5,3,4 (D) 2,1,4,3,5 (✓)

(56) (1) Fenestration (2) Feather (3) Feed head

(4) Feature

(5) feminine.

(CPO-2015)

⇒ (A) 5,1,4,2,3 (B) 2,4,1,5,3 (C) 4,2,3,5,1 (D) 2,4,3,5,1 (✓)

(57) (1) LEAF (2) LEARNED (3) LEAVED (4) LEAK (5) LEADEN

⇒ (A) 5,1,4,2,3 (✓) (B) 5,1,4,3,2 (C) 3,5,1,4,2 (D) 1,4,2,3,5

(58) 1. Fish

2. Fan

3. Flesh

4. Feast

5. Fraud.

(SSC LDC-2015)

⇒ (A) 2,4,1,3,5 (✓)

(B) 3,5,4,2,1

(C) 2,4,3,1,5

(D) 4,2,1,5,3

(59) (1) Euphrasy

(2) Eupesya

(3) Euphonic

(4) Eugenic

(5) Euphony.

(LDC-2014)

⇒ (A) 4,3,2,1,5

(B) 3,4,1,2,5

(C) 4,2,3,5,1 (✓)

(D) 3,5,2,4,1



(60) Which letter is exactly midway between G and Q in the given alphabets?

- (A) K (B) L (✓) (C) M (D) N (UPSC-2015)

⇒ Position Value of G = 7

Position Value of Q = 17

$$\text{Mid Value} = \frac{17+7}{2} = 12 = L$$

(61) Some letters are given with numbers from 2 to 9. Select the sequence of numbers which arranges the letters into a meaningful word. (CGL-2014)

N A E X I M O T N A I  
9 2 3 4 5 6 7 8 9 2 5

- (A) 3, 4, 6, 2, 9, 5, 8, 5, 2, 9, 7 (B) 3, 6, 2, 4, 2, 9, 5, 7, 5, 9, 8  
(C) 3, 4, 2, 2, 6, 5, 9, 7, 8, 9, 5 (D) 3, 4, 2, 6, 5, 9, 2, 8, 5, 7, 9 (✓)

⇒ N A E X I M O T N A I → EXAMINATION

(62) In the following question from the given alternative words, select the word which cannot be formed using the letters of given word. (CPO-2015)

'ADMINISTRATION'

- (A) MIND (B) RATION  
(C) MINISTER (✓) (D) STATION

⇒ 'E' is not present in given word.

(63) In the following question from the given alternative words, select the word which can be formed using the letters of the given word. (CGL-2015)

'OMNIPRESENT'

- (A) PRESIDENT (B) ERASER  
(C) TENSION (✓) (D) ESCORT

⇒ TENSION can be formed.

(64.) IF the first and second letter in the word 'COMMUNICATION' were interchanged, also the third and the fourth letters, the 5th and 6th letters and so on, which letter would be the tenth letter counting from your right? (CGL-2013)

- (A) N (B) U (C) M (✓) (D) T

⇒  $\begin{array}{ccccccc} \text{CO} & \text{MM} & \text{UN} & \text{IC} & \text{AT} & \text{IO} & \text{N} \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ \text{OC} & \text{MM} & \text{NU} & \text{CI} & \text{TA} & \text{OI} & \text{N} \end{array}$

M is 10th letter from right.

OR Total letters = 13  
So,  $(13 - 10 + 1) = 4 = \boxed{M}$   
Because 'M' is on fourth.

(65.) In the following question, the number of letters skipped in between adjacent letters in the series is successive even numbers. Which of the following series observes this rule? (SBI Clerk-2015)

- (A) ADGJM (B) BEJQZ (✓) (C) BDGKQ (D) FINUZ

⇒  $\begin{array}{ccccc} \text{B} & \text{E} & \text{J} & \text{Q} & \text{Z} \\ \text{+2} & \text{+4} & \text{+6} & \text{+8} & \end{array}$

(66.) IF the first four letters of a term 'HIPNOWADIASM' are written in reverse order, the next five letters are written without changing their order and the remaining letters are again written in reverse order, then which letter is in the middle of the word?

- (A) O (B) W (✓) (C) A (D) I  
(CGL-2012)

⇒  $\begin{array}{ccc} \text{HIPP} & \text{NOWAD} & \text{IASM} \\ \downarrow & \downarrow & \downarrow \\ \text{reverse} & \text{No change} & \text{reverse} \\ \downarrow & \downarrow & \downarrow \\ \text{PPIH} & \text{NOWAD} & \text{MSAI} \end{array}$

so, Middle letter is 'W'.

## \* SEQUENCE-TEST \*

(1) 35 95 45 53 58 45 67 35 75 54 52 35 10.

How many 5s are there in the series which are immediately preceded by 3 or 4 but not immediately followed by 0 or 9?

$$\Rightarrow \begin{array}{ccccccccccccccc} \underline{3} & \underline{5} & 95 & \underline{4} & \underline{5} & \underline{5} & \underline{3} & \underline{5} & \underline{8} & \underline{4} & \underline{5} & 67 & \underline{3} & \underline{5} & 75 & \underline{5} & \underline{4} & \underline{5} & \underline{2} & \underline{3} & \underline{5} & 10 \\ & \times & & \checkmark & & \times & & \checkmark & & \checkmark & & & \checkmark & & \checkmark & & \checkmark & & \checkmark & & \checkmark & & \checkmark \end{array}$$

$$= \boxed{5}$$

(2) 64 12 2 8 7 4 2 1 5 3 8 6 2 1 7 1 4 1 3 2 8 6.

How many pairs of successive numbers there in the series which difference of 2 each?

$$\Rightarrow \underline{6} \underline{4} \underline{1} \underline{2} \underline{2} \underline{8} \underline{7} \underline{4} \underline{2} \underline{1} \underline{5} \underline{3} \underline{8} \underline{6} \underline{2} \underline{1} \underline{7} \underline{1} \underline{4} \underline{1} \underline{3} \underline{2} \underline{8} \underline{6}$$

$$= \boxed{6}$$

(3) 6 4 12 2 8 7 4 2 1 5 3 8 6 2 1 7 1 4 1 3 2 8 6.

How many pairs of alternate number in this series which difference of 2 each?

$$\Rightarrow \underline{6} \underline{4} \underline{1} \underline{2} \underline{2} \underline{8} \underline{7} \underline{4} \underline{2} \underline{1} \underline{5} \underline{3} \underline{8} \underline{6} \underline{2} \underline{1} \underline{7} \underline{1} \underline{4} \underline{1} \underline{3} \underline{2} \underline{8} \underline{6}$$

$$= \boxed{2}$$

(4) 5 1 4 7 3 9 8 5 7 2 6 3 15 8 6 3 8 5 2 2 4 6 7 4 9 6.

How many even no. there are in this series which is immediately preceded by odd no. but immediately followed by even no.?

$$\Rightarrow \underline{5} \underline{1} \underline{4} \underline{7} \underline{3} \underline{9} \underline{8} \underline{5} \underline{7} \underline{2} \underline{6} \underline{3} \underline{15} \underline{8} \underline{6} \underline{3} \underline{8} \underline{5} \underline{2} \underline{2} \underline{4} \underline{6} \underline{7} \underline{4} \underline{9} \underline{6}$$

$$= \boxed{3}$$

(5) 5 1 4 7 3 9 8 5 7 2 6 3 15 8 6 3 8 5 2 2 4 6 7 4 9 6

How many odd no. are there in the series which are immediately preceded and as well as followed by even no.?

$$\Rightarrow \underline{5} \underline{1} \underline{4} \underline{7} \underline{3} \underline{9} \underline{8} \underline{5} \underline{7} \underline{2} \underline{6} \underline{3} \underline{15} \underline{8} \underline{6} \underline{3} \underline{8} \underline{5} \underline{2} \underline{2} \underline{4} \underline{6} \underline{7} \underline{4} \underline{9} \underline{6}$$

$$= \boxed{4}$$

(6) 5 1 4 7 3 9 8 5 7 2 6 3 1 5 8 6 3 8 5 2 2 4 6 7 4 9 6.

How many prime numbers are there in the series which is immediately preceded by odd no. but immediately followed by even number?

⇒ 5 1 4 7 3 9 8 5 <sup>odd</sup> 7 <sup>even</sup> 2 6 3 1 <sup>odd</sup> 5 <sup>even</sup> 8 6 3 8 5 <sup>odd</sup> 2 <sup>even</sup> 2 4 6 7 4 9 6  
 = 4  
 Prime no.

(7) \$ = B F 2 \* K S 7 5 # \$ P L V 8 @ M U E 6 ↑ Q G © 9 3 Δ T Z ®

How many letters in the series which are either preceded by a symbol or followed by a number but not both?

(i) 9 (ii) 1 (iii) 6 (iv) 8 (v) 10 (vi) 7

⇒ \$ = B F 2 \* <sup>1</sup>K <sup>2</sup>S <sup>3</sup>7 <sup>4</sup>5 # <sup>5</sup>\$ <sup>6</sup>P <sup>7</sup>L <sup>8</sup>V <sup>9</sup>8 @ <sup>10</sup>M <sup>11</sup>U <sup>12</sup>E <sup>13</sup>6 <sup>14</sup>↑ <sup>15</sup>Q <sup>16</sup>G <sup>17</sup>© <sup>18</sup>9 <sup>19</sup>3 <sup>20</sup>Δ <sup>21</sup>T <sup>22</sup>Z <sup>23</sup>®

SL or LN  
 SLN X

Symbol letter (✓)  
 letter number (✓)  
 symbol letter number (X)

(8) \$ = B F 2 \* K S 7 5 # \$ P L V 8 @ M U E 6 ↑ Q G © 9 3 Δ T Z ®

How many consonant are there in the series which are immediately followed by a consonant but not immediately preceded by a symbol?

⇒ \$ = B F 2 \* K S 7 5 # \$ P L V 8 @ M U E 6 ↑ Q G © 9 3 Δ T Z ®  
 = 1

(9) How many 8s are there in the following sequence which are immediately preceded by 6 but not immediately followed by 5?  
 (CGL-2014)

6, 8, 5, 7, 8, 5, 4, 3, 6, 8, 1, 9, 8, 5, 4, 6, 8, 2, 9, 6, 8, 1, 3, 6, 8, 5, 3, 6

(A) Three (✓) (B) Four (C) one (D) Two

⇒ 6, 8, 5, 7, 8, 5, 4, 3, 6, 8, 1, 9, 8, 5, 4, 6, 8, 2, 9, 6, 8, 1, 3, 6, 8, 5, 3, 6.

(10.) In the following series how many G, K, R occur in such a way that K in the middle of G & R? (CGL-2014)

DSGKUMGRDPGRNKGDKTNVQSPQD

(A) 4 (B) 1 (✓) (C) 3 (D) 2

⇒ DSGKUMGRDPGRNKGDKTNVQSPQD

only one K lies in middle of G and R.

(11.) In the given series how many 5's are followed by a cube number and preceded by square number? (CGL-2014)

958417451635892359675

(A) 1 (✓) (B) 2 (C) 3 (D) 4

⇒ 958417451635892359675

only one 5 which is followed by a cube number and preceded by a square number.

(12.) 4536467365423453642857836732123536673673641254436.

From the given number series, how many possible numbers of two digit can be formed in which one digit is 8?

8, 5, 2, 1, 7, 6

(CGL-2011)

(A) 9 (B) 11 (C) 10 (✓) (D) 12

⇒ 85, 82, 81, 87, 86, 58, 28, 18, 78, 68.

(13.) In the given number series, how many times 7 is followed by 4, and it preceded by 1? (CPO-2011)

847217414714767141474714729417

(A) 4 (✓) (B) 5  
(C) 6 (D) None of these.

⇒ 847217414714767141474714729417

Four times 7 is followed by 4 and is preceded by 1.

(14.) 1 2 3 5 4 5 6 7 9 4 8 4 5 5 4 8.

How many numbers are followed by odd numbers but is not preceded by odd numbers? (CGL-2012)

(A) 1 (B) 2 (C) 3 (D) 4 (✓)

⇒ 1 2 3 5 4 5 6 7 9 4 8 4 5 5 4 8

There are 4 numbers which are followed by odd numbers but is not preceded by odd numbers.

(15.) In the given series 1 2 3 5 4 5 6 7 9 4 8 4 5 5 4 8 (CGL-2010)

How many times an even number is followed by odd number?

(A) 5 (B) 2 (C) 3 (D) 4 (✓)

⇒ 1 2 3 5 4 5 6 7 9 4 8 4 5 5 4 8

Four times an even number is followed by odd number.

(16.) How many 5's are there in the following sequence such that the sum of the two immediately following digits is greater than the sum of the two immediately preceding digits? (CGL-2014)

3 7 6 5 8 3 2 4 5 5 4 8 7 9 1 5 3 4 8 7 5 9 8 7 6 4

(A) one (B) Two (C) Three (✓) (D) None of these

⇒ 3 7 6 5 8 3 2 4 5 5 4 8 7 9 1 5 3 4 8 7 5 9 8 7 6 4

There are three 5's in the sequence such that the sum of the two immediately following digit is greater than the sum of the two immediately preceding digits.

(17.) How many times 5 is followed by 2 and is immediately preceded by 4? (LDC-2010)

7 6 3 2 5 4 3 5 2 7 6 4 5 2 3 6 2 7 5 2 3 2 3 6 5 2 6 3 7 4 2 5

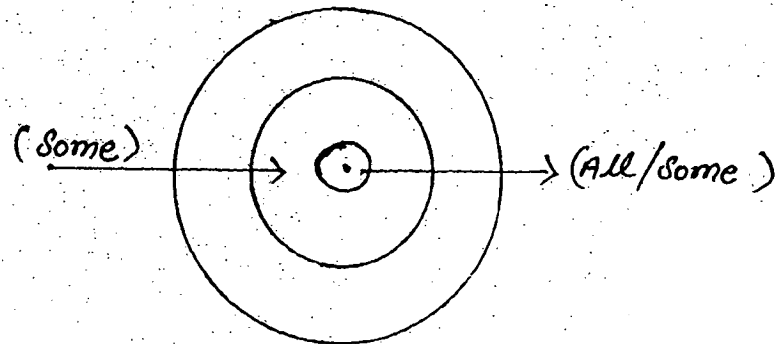
(A) 1 (✓) (B) 2 (C) 4 (D) 7

⇒ 7 6 3 2 5 4 3 5 2 7 6 4 5 2 3 6 2 7 5 2 3 2 3 6 5 2 6 3 7 4 2 5



## \* Syllogism (Statement & conclusion) \*

Note  $\Rightarrow$  Some is include in all but all is not include in some.



(\*)

| <u>Universal Positive</u> | <u>Universal Negative</u> | <u>Partial Positive</u> | <u>Partial Negative</u> |
|---------------------------|---------------------------|-------------------------|-------------------------|
| [A]                       | [E]                       | [I]                     | [O]                     |
| all A are B               | No A are B                | Some A are B            | Some A are not B        |

Table (\*)

|              |              |               |              |
|--------------|--------------|---------------|--------------|
| $A + A = A$  | $I + A = I$  | $E + A = O^*$ | $O + A = No$ |
| $A + E = E$  | $I + E = O$  | $E + E = No$  | $O + E = No$ |
| $A + I = No$ | $I + I = No$ | $E + I = O^*$ | $O + I = No$ |
| $A + O = No$ | $I + O = No$ | $E + O = No$  | $O + O = No$ |

|             |             |               |
|-------------|-------------|---------------|
| $A + A = A$ | $I + A = I$ | $E + A = O^*$ |
| $A + E = E$ | $I + E = O$ | $E + I = O^*$ |

Ex. -  $O$  = Some dogs are not cat.

$O^*$  = Some cats are not dog.



### STEP-1:

$$(1.) \text{ all } A \text{ are } \cancel{B} = A$$

$$\text{all } \cancel{B} \text{ are } C = A$$

$$\text{all } A \text{ are } C = A$$

$$(3.) \text{ Some } C \text{ are } \cancel{A} = I$$

$$\text{All } \cancel{A} \text{ are } B = A$$

$$\text{Some } C \text{ are } B = I$$

$$(5.) \text{ Some } C \text{ are } \cancel{A} = I$$

$$\text{No } \cancel{A} \text{ are } B = E$$

$$\text{Some } C \text{ are not } B = O$$

$$(2.) \text{ all } B \text{ are } \cancel{C} = A$$

$$\text{No } \cancel{C} \text{ are } A = E$$

$$\text{No } B \text{ are } A = E$$

$$(4.) \text{ Some } A \text{ are } \cancel{B} = I$$

$$\text{No } \cancel{B} \text{ are } C = E$$

$$\text{Some } A \text{ are not } C = O$$

$$(6.) \text{ No } A \text{ are } \cancel{C} = E$$

$$\text{All } \cancel{C} \text{ are } B = A$$

$$\text{Some } B \text{ are not } A = O^*$$

$$(7.) \text{ No } A \text{ are } \cancel{B} = E$$

$$\text{All } \cancel{B} \text{ are } C = A$$

$$\text{Some } C \text{ are not } A = O^*$$

### STEP-2: (1.) All $\cancel{A}$ are $B = A$

$$\text{all } C \text{ are } \cancel{A} = A$$

$$\text{All } C \text{ are } B = A$$

### (2.) All $A$ are $C = A$

$$\text{No } B \text{ are } A = E$$

$$\Rightarrow \text{No } B \text{ are } \cancel{A} = E$$

$$\text{All } \cancel{A} \text{ are } C = A$$

$$\text{Some } C \text{ are not } B = O^*$$

### (3.) All $B$ are $A = A$

$$\text{Some } C \text{ are } B = I$$

$$\Rightarrow \text{Some } C \text{ are } B = I$$

$$\text{All } B \text{ are } A = A$$

$$\text{Some } C \text{ are } A = I$$

### STEP-3 :

(1.) All A are B = A — (conversion to A)

All A are C = A

⇒ Some B are A = I

All A are C = A

Some B are C = I

Note ⇒ : Conversion :

A → I

E → E

I → I

O → No

(2.) Some B are C = I → (conversion)

No B are A = E

⇒ Some C are B = I

No B are A = E

Some C are not A = O

Note : Complementary Pairs :

|   |   |
|---|---|
| A | O |
| I | E |
| I | E |

Subject, Predicate

Subject, Predicate

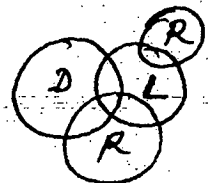
Same

(1) Statement: Some dogs are Lion. — I

Some Lions are Rat. — I

Conclusion : I. Some Dogs are Rat. — I

II. Some Dogs are not Rat. — O



⇒ [I. or II. conclusion is correct.]

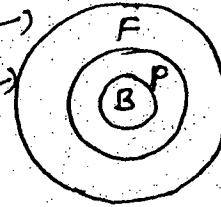
(2) Statement: All birds are peacocks. (CGL-2011)

All peacocks are farmers.

Conclusion: I. Some farmers are birds. (✓)

II. Some peacocks are birds. (✓)

$\Rightarrow$  [All conclusions are correct.]



(3) Statement: All bells are hammers. — A

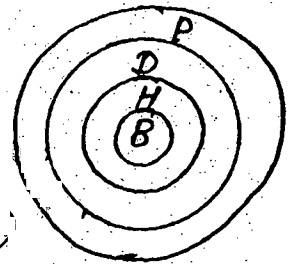
All hammers are dogs. — A

All dogs are packets. — A

Conclusion: I. Some packets are hammers. —

II. Some dogs are bells. (✓)

$\Rightarrow$  [All conclusion are correct]



(CPO-2016)

(4) Statement: Some days are nights. — I (CPO-2016)

All nights are stars. — A

All stars are cloud. — A

Conclusion: I. Some clouds are days. (✓)

II. Some stars are days. (✓)

$\Rightarrow$  [All conclusion are correct.]

(5) Statement: All cards are boxes. — A (CGL-2013)

No box is slates — E

Some slates are tiles. — I

Conclusion: I. No slate is card. (✓)

II. Some tiles are boxes. (X)

$\Rightarrow$  (I)  $A + E = E \Rightarrow$  No card is slate. — E

Now, conversion of E = No slate is card.

(II)  $E + I = O^* \Rightarrow$  Some tiles are not boxes.

[Only I. conclusion is correct.]

(6) Statement : All jeeps are car. — A

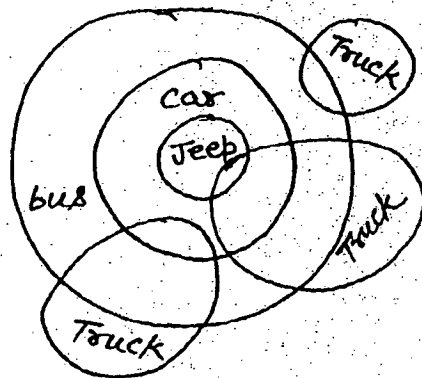
(CPO-2010)

All cars are buses. — A

Some buses are truck. — I

Conclusion : I. Some jeeps are truck. (X)

II. All jeeps are buses. (✓)



⇒ Only II. conclusion is correct.

(7) Statement : Some balls are rackets. — I

(CPO-2012)

Some rackets are bats. — I

All bats are nets. — A

Conclusion : I. No net is ball. (X)

II. All rackets are net. (X)

⇒ Both I and II conclusion is incorrect (wrong).

(8) Statement : All computers are printer. — A

(CGL-2011)

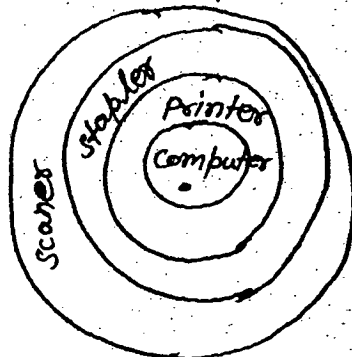
All printer are stapler. — A

All staplers are scanners. — A

Conclusion : I. All printers are scanner. (✓)

II. Some staplers are computer. (✓)

⇒



⇒ (All conclusion is correct.)

(9) Statement : No drum is guitar. — E (CGL-2011)

All guitar are violins. — A

Some violins are flutes. — I

Conclusion : I. Some violins are guitar. (✓)

II. Some drums are flutes. (X)

⇒ Conversion of A — 'Some violins are guitar.

{ only I. conclusion is correct. }

---

(10) Statement : All guns are cannons. — A (CGL-2014)

All arrows are cannons. — A

Some arrows are bows. — I

Conclusion : I. Some guns are arrows. (X)

II. Some arrows are bows. (✓)

⇒ [ only II conclusion is correct ]

---

(11) Statement : Illiterate people are poor. (CSAT-2015)

Conclusion : I. All poor people are illiterate. (X)

II. Some poor people are illiterate. (✓)

⇒ Conclusion (II) is the converse of given statement.

---

(12) Statement : Ministers arrived at the public function in their cars. (UPSI-2013)

Conclusion : I. Ministers have cars. (✓)

II. Minister came to the public function. (✓)

⇒ [ Both (I) & (II) follow ]

---

(13) Statement : All wealthy people are happy in life. (FCI-2011)

Conclusion : I. Wealth and happiness are directly related. (✓)

II. Wealth leads to happiness in life. (✓)

⇒ [ Both (I) & (II) conclusion follow ]

(14.) Statement: All keys are locks.

(SSC-LDC-2013)

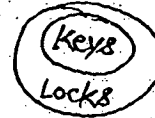
No lock is toy.

All bags are toys.

Conclusion: I. No bag is key.

II. Some bags are keys.

III. Some toys are keys.



Conclusion: I. ✓

II. X

III. X

[only I follows.]

(15.) Statements: Some days are nights.

(CGL-2013)

Some nights are month.

Some months are years.

Conclusion:

I. Some years are nights.

II. Some months are days.

III. No year is night.



Conclusion: I. (—)

II. (—)

III. (—)

⇒ [only either I or III follows.]

(16.) Statements: Some books are bags. —(I)

(CGL-2015)

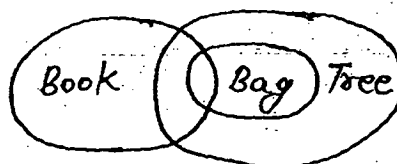
All bags are trees. —(A)

Conclusion: I. Some books are trees.

II. Some trees are books.

⇒  $I + A = I$  = Some books are trees.

Now, Conversion of I = Some trees are book.



Conclusion:

I. ✓

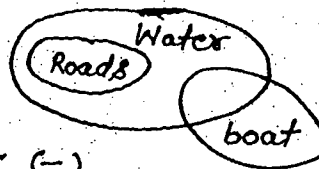
II. ✓

[Both conclusion I & II follow.]



(17.) Statements: All roads are water. — A (CGL-2014)  
Some water is boat. — I

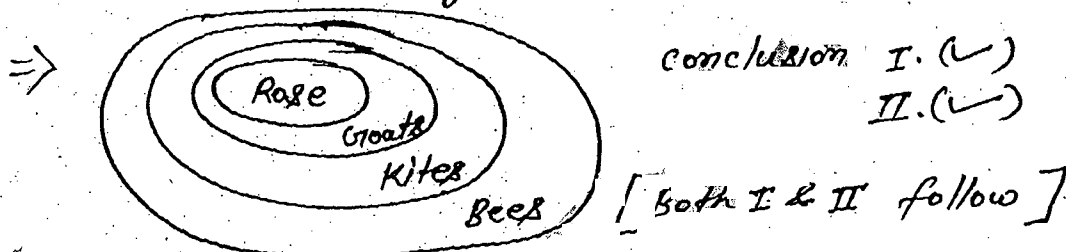
Conclusion: I. Some boats are roads.  
II. All waters are boats.



⇒ [Neither I nor II follows.] I. (—)  
II. (—)

(18.) Statements: All goats are kites. (CGL-2015)  
All roses are goats.  
All kites are bees.

Conclusion: I. Some kites are roses.  
II. Some goats are bees.



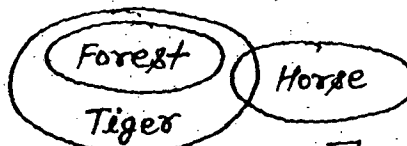
Conclusion I. (✓)  
II. (✓)

[Both I & II follow]

(19.) Statements: All forests are tigers. (CPO-12)  
Some tigers are horses.

Conclusion: I. Some horses are forests.  
II. No horse is forest.

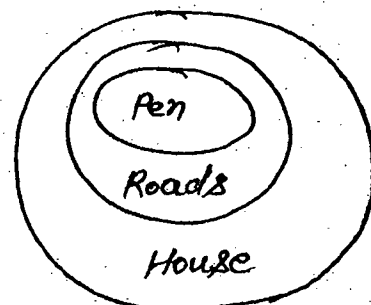
⇒ I. (—)  
II. (—)



[Either conclusion I nor II follows.]

(20.) Statements: All pens are roads. (CGL-2015)  
All roads are houses.

Conclusion: I. All houses are pens  
II. Some houses are pens.



⇒ I. (—)  
II. (✓)

[Only II conclusion follows.]



## \* WATER IMAGE \*

The Reflection of an object as appears in water is known as water image.

In water image the image of upper part of the object is seen downward and the lower part of the object is seen upward.

### \* Water Images of numbers:

Number : 0 1 2 3 4 5 6 7 8 9  
 Water Image : 0 1 2 3 4 5 6 7 8 9

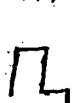
### \* Water Image of the capital letters.

| Letter | Water Image | Letter | Water Image |
|--------|-------------|--------|-------------|
| A      | A           | Q      | Q           |
| B      | B           | R      | R           |
| C      | C           | S      | S           |
| D      | D           | T      | T           |
| E      | E           | U      | U           |
| F      | F           | V      | V           |
| G      | G           | W      | W           |
| H      | H           | X      | X           |
| I      | I           | Y      | Y           |
| J      | J           | Z      | Z           |
| K      | K           |        |             |
| L      | L           |        |             |
| M      | M           |        |             |
| N      | N           |        |             |
| O      | O           |        |             |
| P      | P           |        |             |

Note  $\Rightarrow$  The objects given below are such objects, the image of which is identical to the object.

C, D, E, H, I, K, O, X

### ⊗ Water-Images of some Symbols.

| Symbol                                                                              | Water Image                                                                         | Symbol                                                                              | Water Image                                                                           |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|    |    |    |    |
|    |    |    |    |
|    |    |    |    |
|   |   |    |    |
|  |  |  |  |
|  |  |  |  |
|  |  |  |  |

\* What is the correct water Image in the following word.

(1) FAMILY

(A) ELIMAF

(B) ELIMAF

(C) ELIMAF

(D) ELIMAF (✓)

(2) WATER

(A) RETAW (✓)

(B) WATER

(C) RETAW

(D) WATER

## \* MIRROR IMAGES \*

The Image of an object as seen in a mirror is known as mirror image.

### \*) Mirror images of numbers

| Number | Mirror Image |
|--------|--------------|
| 1      | 1            |
| 2      | 5            |
| 3      | E            |
| 4      | 4            |
| 5      | 2            |
| 6      | 9            |
| 7      | F            |
| 8      | 8            |
| 9      | 6            |
| 0      | 0            |

### \*) Mirror images of capital letters.

| Letters | Mirror Image | Letters | Mirror Image |
|---------|--------------|---------|--------------|
| A       | A            | N       | N            |
| B       | B            | O       | O            |
| C       | C            | P       | q            |
| D       | D            | Q       | Q            |
| E       | E            | R       | R            |
| F       | F            | S       | 2            |
| G       | 2            | T       | T            |
| H       | H            | U       | U            |
| I       | I            | V       | V            |
| J       | 7            | W       | W            |
| K       | K            | X       | X            |
| L       | L            | Y (Y)   | A (Y)        |
| M       | M            | Z       | 3            |

## \* Mirror Images of Some Symbols

| Symbol | Mirror Image | Symbol | Mirror Image |
|--------|--------------|--------|--------------|
|        |              |        |              |
|        |              |        |              |
|        |              |        |              |
|        |              |        |              |
|        |              |        |              |
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|        |              |        |              |
|        |              |        |              |
|        |              |        |              |

Note  $\Rightarrow$  The objects given below are such objects, the image of which is identical to the object.

A, H, I, M, O, T, U, V, W, X, Y, 8, 0, , , , , , , , ,

(1) What is the mirror image of "PUNEET".

$\Rightarrow$  PUNEET  $\nparallel$  TEEVUT

(2) What is the mirror image of "SATISH".

$\Rightarrow$  SATISH  $\nparallel$  HZITA2

